

NERD

CNS Module - Part 2

Questions

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CNS Module - Part 2

Questions

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1. Answers of the MCQs
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3. Explanations of some questions



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PREFACE

Dear colleagues,

Presenting you with the third edition of **NERD | CNS Module (Questions)**.
This edition's content is made out to help you solve MCQs & Short Essay easily.

Hoping this book improves your studying experience to be as enjoyable and as fruitful as possible

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NERD | CNS Module - Part 2 (Questions)

By: Omar Elazab

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First Edition 2023

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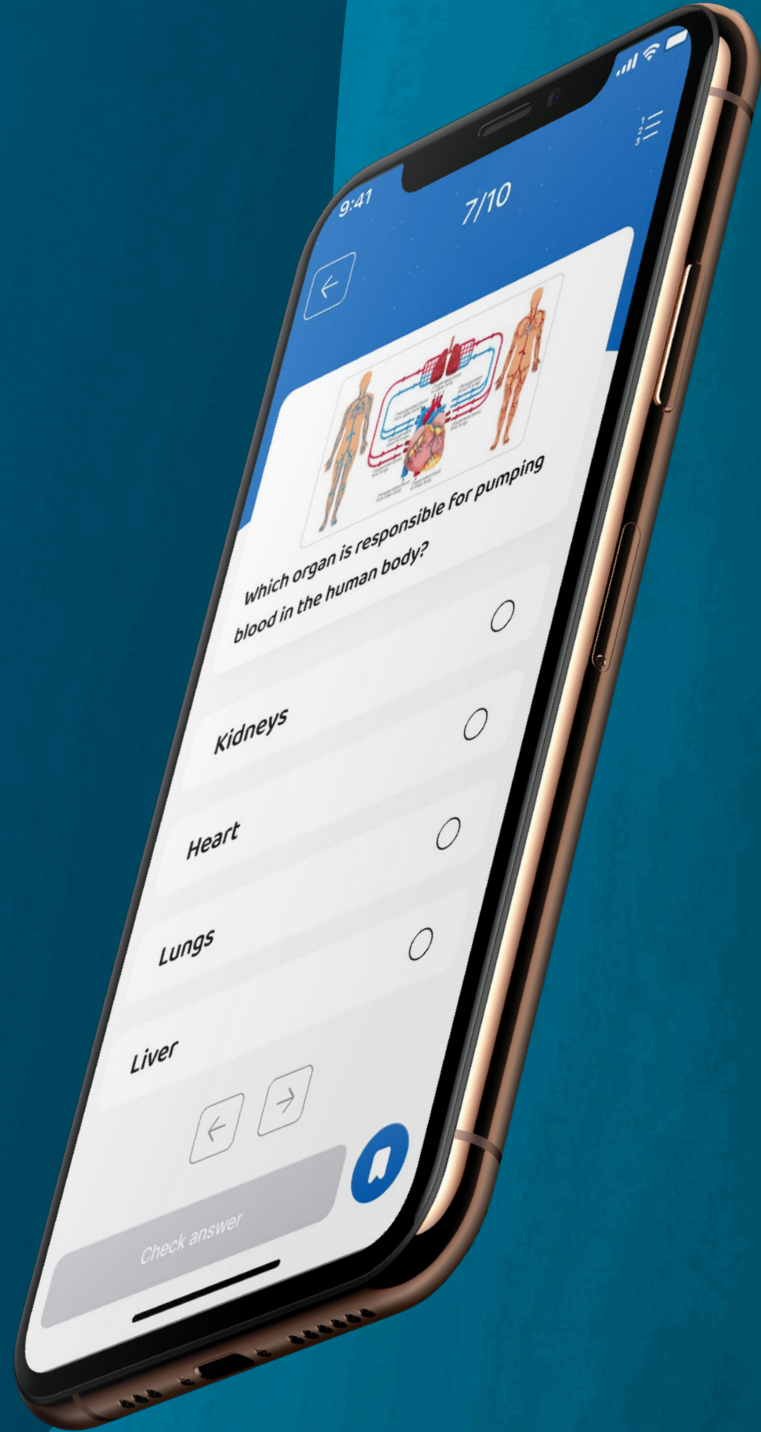


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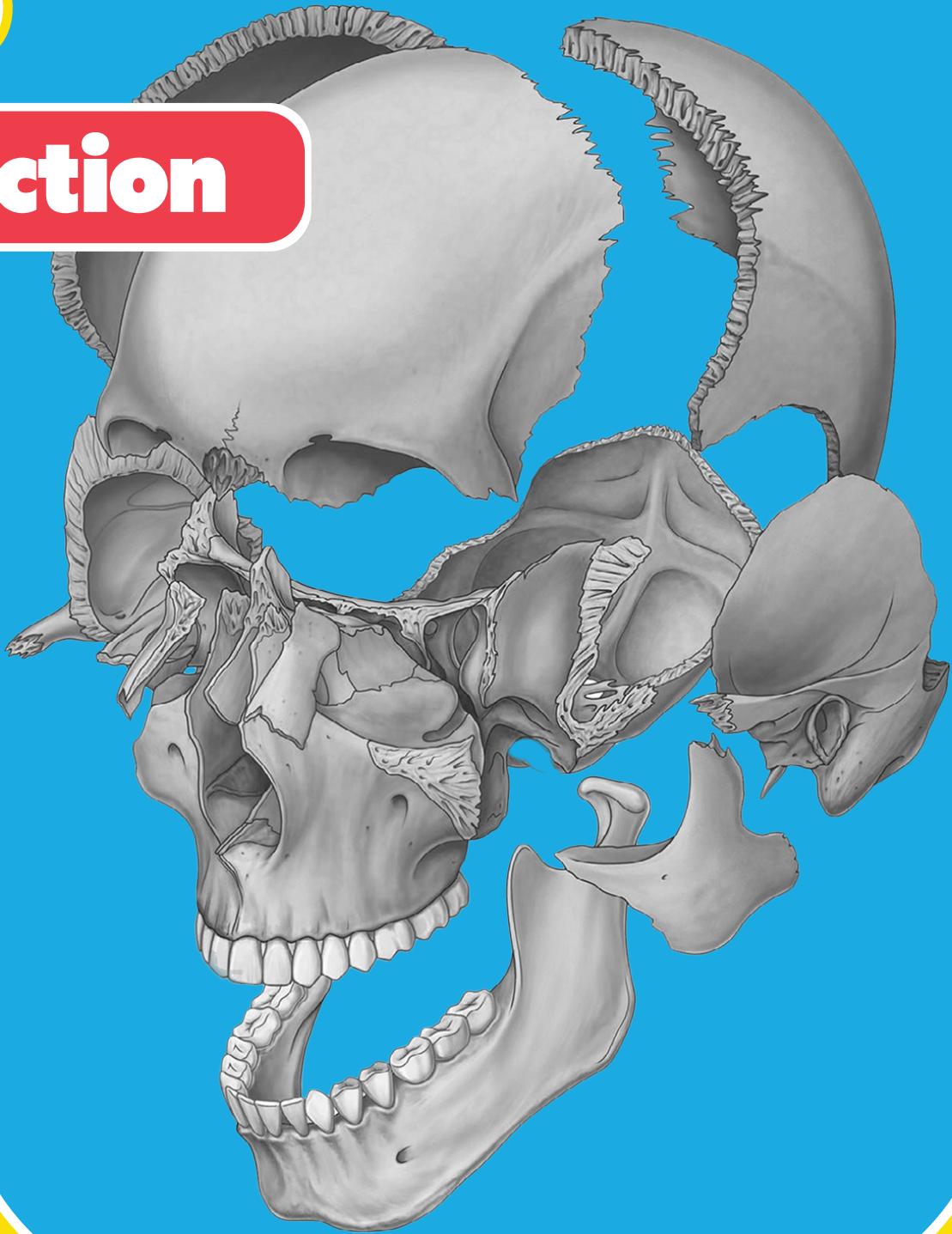
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MCQ

5

Section



Pathology



1. Which of the following is true about meningococcal meningitis:

- a. It most commonly affects adults around the sixth decade of life
- b. The Dura matter is mainly affected by suppuration
- c. The condition may be complicated by acute adrenal insufficiency
- d. All of the above

2. Rabies is transmitted to CNS via:

- a. Arteries
- b. Veins
- c. Peripheral nerves
- d. Direct implantation
- e. Lymphatics

3. A child with tuberculous meningitis his CSF examination will show an increased number of:

- a. Macrophages
- b. Lymphocytes
- c. Plasma cells
- d. Giant cells
- e. Epithelioid cells

4. Muscle paralysis in case of poliomyelitis is due to:

- a. Destruction of the muscle fibers by the toxins of the organisms
- b. Inflammation of the muscle
- c. Inflammation of the nerves supplying the affected muscles
- d. Degeneration of their anterior horn cells in the spinal cord
- e. Degeneration of the neuromuscular junctions

5. The following disease is transmitted by blood transfusion:

- a. Meningitis
- b. Poliomyelitis
- c. Autoimmune deficiency syndrome (AIDS)
- d. Brain abscess
- e. Rabies

6. Poliomyelitis is usually an infection of:

- a. Cerebral tissue
- b. Cerebellar tissue
- c. Anterior horn cells of the spinal cord
- d. Posterior horn cells of the spinal cord
- e. Peripheral nerves

7. CSF in acute septic meningitis doesn't show:

- a. Increased in amount
- b. Increased protein content
- c. Increased sugar content
- d. Polymorphonuclear cells
- e. Organisms

8. Gliosis means:

- a. Regeneration of the brain tissue
- b. Necrosis of the brain tissue
- c. Fibrosis of the brain tissue
- d. None of the above

9. CSF shows cobweb coagulum of fibrin in standing in case of:

- a. Chronic meningitis
- b. Brain abscess
- c. Encephalitis
- d. Cerebral infarction

10. Poliomyelitis is a:

- a. Bacterial infection
- b. Viral infection
- c. Fungal infection
- d. Parasitic infection
- e. None of the above

11. The cause of death in rabies is:

- a. Muscle paralysis
- b. Heart failure
- c. Paralysis of respiratory muscles
- d. Respiratory center failure
- e. Septicemia

12. Infection spreads from suppurative lung diseases to the brain via:

- a. Vertebral arteries
- b. Vertebral veins
- c. Local extension
- d. Peripheral nerves
- e. None of the above

13. Rabies is a:

- a. Bacterial infection
- b. Viral infection
- c. Fungal infection
- d. Parasitic infection
- e. Degenerative disease

14. Mode of infection in poliomyelitis is:

- a. Ingestion of contaminated food or drink
- b. Droplet infection
- c. Blood spread from a septic focus
- d. Direct spread from air sinuses

Chapter 2 ? Neurodegenerative & Demyelinating diseases

1. Alzheimer's disease:

- a. Congenital disease
- b. Inflammatory disease
- c. Degenerative disease
- d. Demyelinating disease
- e. Metabolic disease

2. A 85 year old man resident of a nursing home is found wandering the streets looking for his way to home, the patient's family tell the doctor that his condition has been worsening progressively over the last several years. He doesn't have a history of head trauma or cardiovascular disease in addition to the disorientation; he has demonstrated significant language impairment. Microscopical findings include:

- a. Rosenthal fibers
- b. Amyloid B proteins
- c. Lewy bodies
- d. Verocay bodies

3. Negri bodies are:

- a. Inflamed granulation tissue
- b. Epithelioid granuloma
- c. Aggregates of organizing fibrin
- d. Foreign body granulomas
- e. Intracytoplasmic inclusions

4. Multiple sclerosis is:

- a. Congenital disease
- b. Inflammatory disease
- c. Degenerative disease
- d. Demyelinating disease
- e. Metabolic disease

5. 68 year old man suffers from dementia for a long period, he died in a car accident. During autopsy silver stain was performed on sections of the frontal, parietal & temporal cortex it reveals intraneuronal inclusions. These inclusions are:

- a. Lewy bodies
- b. Tau proteins
- c. Negri bodies
- d. Schawmann bodies



1. Which of the following aneurysms can affect the cerebral vessels:
 - a. Mycotic aneurysms
 - b. Atherosclerotic aneurysms
 - c. Congenital aneurysms
 - d. All of the above
2. A previously healthy 31 year old woman experiences extensive subarachnoid hemorrhage at the base of the brain she is afebrile the CSF protein is slightly increased but the glucose is normal, which of the following is the most likely diagnosis:
 - a. Acute bacterial meningitis
 - b. Ruptured berry aneurysm
 - c. Cerebellopontine angle tumor
 - d. Pilocytic astrocytoma
 - e. Meningioma
3. Mycotic cerebral aneurysms is due to:
 - a. Congenital weakness of the wall of cerebral vessels
 - b. Septic emboli from subacute bacterial endocarditis
 - c. Atherosclerosis of the cerebral vessels
 - d. Increased intracranial tension
4. Subarachnoid hemorrhage is due to:
 - a. Trauma of meningeal artery
 - b. Trauma of cortical veins
 - c. Rupture of cerebral aneurysms
 - d. Sudden rise of blood pressure
 - e. Hemorrhagic blood disease
5. Congenital cerebral aneurysm affects:
 - a. Blood vessels of circle of Willis
 - b. Cerebral arteries
 - c. Meningeal arteries
 - d. Internal carotid artery
6. Hypertensive patients are at high risk of:
 - a. Extradural hemorrhage
 - b. Subarachnoid hemorrhage
 - c. Subdural hemorrhage
 - d. Intracerebral hemorrhage
7. Subdural hemorrhage occurs:
 - a. Between Dura & skull bones
 - b. Between Dura & arachnoid
 - c. In subarachnoid space
 - d. Intracerebral
8. Subdural hemorrhage is commonly related to:
 - a. Hypertension
 - b. Trauma
 - c. Encephalitis
 - d. Ruptured cerebral aneurysm
9. A female patient 40 years old presented with sudden onset of loss of memory brain MRI revealed multiple small secular aneurysms in the circle of Willis with rupture of one of them causing subarachnoid hemorrhage. The type of aneurism is due to:
 - a. Congenital arterial malformation
 - b. Inflammation of the vessels of the circle of Willis
 - c. Septic emboli in the vessels of the circle of Willis
 - d. Atheromatous emboli of the vessels of the circle of Willis
10. A 58 year old hypertensive patient presents with sudden onset of hemiplegia followed by symptoms of increased intracranial tension (headache & vomiting). ACT scan shows extensive intracerebral hemorrhage in the region of basal ganglia, most probable diagnosis is:
 - a. Ruptured arteriovenous malformation
 - b. Ruptured congenital berry aneurysm
 - c. Ruptured multiple microaneurysms (Charcot Bouchard microaneurysms)
 - d. Thrombotic occlusion of the middle cerebral artery

1. Which of the following tumors is a WHO grade IV:
 - a. Diffuse astrocytoma
 - b. Oligodendroglioma
 - c. Ependymoma
 - d. Medulloblastoma
2. Pilocytic astrocytoma is characterized by all EXCEPT:
 - a. Usually occurs in children
 - b. Usually occurs in the cerebellum
 - c. It's usually cystic
 - d. It's of bad prognosis
 - e. It shows Rosenthal fibers
3. Psammoma bodies are characteristic of:

a. Astrocytoma	d. Meningioma
b. Oligodendroglioma	e. Ependymoma
c. Medulloblastoma	
4. Pilocytic astrocytoma show the following EXCEPT:
 - a. Activation of the mitogen activated protein kinase (MAPK) pathway
 - b. Activation & alternations of BRAF
 - c. Inactivation of neurofibromatosis type 1 (NF1)
 - d. Mutations in IDH1 & IDH2
5. A 50 year old man has a 3cm diameter left frontal lobe mass with areas of calcification. Microscopically sheets of regular cells with spherical nuclei surrounded by clear halo of cytoplasm are seen. Which of the following diagnosis is most likely to be made on microscopic examination of this mass:
 - a. Thrombosed berry aneurysm
 - b. Oligodendroglioma
 - c. Meningioma
 - d. Schwannoma
 - e. Organizing abscess
6. The following tumor is always benign:
 - a. Pilocytic astrocytoma
 - b. Glioblastoma multiforme
 - c. Ependymoma
 - d. Medulloblastoma
 - e. Meningioma
7. The biopsy of a ring enhancing tumor in the left parietal lobe of a 60 year old patient reveals an astrocytic tumor that displays marked cellular atypia, mitotic activity, areas of pseudopalisading necrosis & multifocal microvascular hyperplasia. The diagnosis is:
 - a. Astrocytoma grade 1
 - b. Astrocytoma grade 2
 - c. Anaplastic astrocytoma
 - d. Astrocytoma grade 4 (Glioblastoma multiforme)
 - e. Pilocytic astrocytoma
8. Microscopic rosette formation is characteristic feature of:
 - a. Astrocytoma
 - b. Oligodendroglioma
 - c. Medulloblastoma
 - d. Meningioma
 - e. Schwannoma
9. A 10 year old girl develops ataxia & hydrocephalus. CT scan shows a midline cerebellar mass. Which of the following is the most likely diagnosis:
 - a. Astrocytoma
 - b. Meningioma
 - c. Neurofibroma
 - d. Medulloblastoma

- 10. A female 50 year old presented with brain tumor with focal erosion of the skull bones, the tumor was excised and examined histologically. It consisted of spindle shaped & syncytial cells arranged in whorly pattern. The most possible diagnosis is:**
- Astrocytoma
 - Ependymoma
 - Meningioma
 - Glioblastoma multiforme
 - Schwannoma
- 11. A 65 year old man develops new onset of seizures & headaches. A CT scan reveals a 6 cm left sided intracerebral mass which of the following is the most likely diagnosis:**
- Meningioma
 - Ependymoma
 - Schwannoma
 - Glioblastoma multiforme
- 12. Medulloblastoma is a:**
- Glial tumor
 - Neuronal tumor
 - Tumor of undifferentiated cells
 - Nerve sheath tumor
 - Malformative tumor
- 13. The following tumor shows whorly pattern of growth:**
- Astrocytoma
 - Oligodendroglioma
 - Ependymoma
 - Medulloblastoma
 - Meningioma
- 14. A 5 year old child suffered from headache & blurring of vision, radiological examination showed a cerebellar tumor. Microscopic examination of a biopsy from this tumor revealed diffuse infiltration by malignant small round cells with rosette formation, the most possible diagnosis is:**
- Anaplastic astrocytoma
 - Glioblastoma multiforme
 - Ependymoma
 - Neuroblastoma
 - Medulloblastoma
- 15. Astrocytoma is:**
- Glial tumor
 - Neuronal tumor
 - Tumor of undifferentiated cells
 - Nerve sheath tumor
 - Malformative tumor
- 16. Meningioma shows:**
- Rosenthal fibers
 - Vascular proliferation
 - Papillary formation
 - Rosette formation
 - Psammoma bodies
- 17. A 9 year old child has enlargement of the lateral ventricles with a 4 cm homogenous well circumscribed mass within the 4th ventricle. Microscopically it shows rosettes & pseudo rosettes, which of the following is the most likely diagnosis:**
- Ependymoma
 - Meningioma
 - Metastatic carcinoma
 - Schwannoma

18. Glioblastoma multiforme shows:

- a. Rosenthal fibers
- b. Vascular proliferation
- c. Papillary formation
- d. Rosette formation
- e. Psammoma bodies

19. The following histological picture is not characteristic of Glioblastoma multiforme:

- a. Highly anaplastic tumor cells
- b. Papillary formation
- c. Necrosis
- d. Tumor giant cells
- e. Vascular proliferation

20. The most common brain tumor in children:

- a. Astrocytoma
- b. Ependymoma
- c. Medulloblastoma
- d. Meningioma
- e. Schwannoma

21. The following is not a feature of meningioma:

- a. Usually occurs in adults
- b. Always a benign tumor
- c. Has a whorly cut surface
- d. Characterized by Psammoma bodies

22. Which of the following tumors is malignant:

- a. Meningioma
- b. Glioblastoma multiforme
- c. Schwannoma
- d. Neurofibroma

23. A male patient 40 year old presented with hemiplegia & deafness. Radiological examination showed a tumor at the Cerebellopontine angle, biopsy from this tumor showed a neurogenic tumor formed of hypercellular and hypocellular areas. The most possible diagnosis is:

- a. Neuroblastoma
- b. Schwannoma
- c. Ganglioneuroma
- d. Ganglioneuroblastoma
- e. Neurofibroma

24. An 18 year old male presents with a well circumscribed mass involving the 4th ventricle. Diagnostic microscopical findings in this condition include:

- a. Rosenthal fibers
- b. Chicken wire calcification
- c. Rosettes & pseudorosettes formation
- d. Verocay bodies

25. The following is not a type of glioma:

- a. Ganglioneuroma
- b. Astrocytoma
- c. Ependymoma
- d. Glioblastoma

26. A 33 year old male presents with seizures, MRI shows an ill-defined area of infiltrative tumor expanding the frontal lobe. Biopsy of this lesion shows glial tissue with slightly increased cellularity and mild nuclear pleomorphism. No mitotic activity, no necrosis & no vascular proliferation. What is the most likely diagnosis:

- a. Astrocytoma grade 1
- b. Astrocytoma grade 2
- c. Anaplastic astrocytoma
- d. Astrocytoma grade 4

27. A male patient 40 year old presented with left sided hemiplegia, brain MRI revealed a space occupying lesion in the right cerebral hemisphere. Biopsy examination showed a highly cellular tumor with malignant cells, areas of necrosis & vascular proliferation. The most possible diagnosis is:

- a. Glioblastoma multiforme
- b. Anaplastic astrocytoma
- c. Ependymoma
- d. Metastatic carcinoma
- e. Malignant meningioma

28. Oligodendroglioma is not a:

- a. Glial tumor
- b. Cerebral tumor
- c. High grade tumor
- d. Well circumscribed tumor

29. A 9 year old girl is evaluated for headache and ataxia over the last 2 months A CT scan reveals a midline partially cystic cerebellar mass, microscopic examination shows elongated bipolar astrocytes with fibrillar processes and Rosenthal fibers the diagnosis is:

- a. Astrocytoma grade 1
- b. Astrocytoma grade 2
- c. Anaplastic astrocytoma
- d. Astrocytoma grade 4

30. The following tumor arises from the lining of the ventricular system:

- a. Pilocytic astrocytoma
- b. Ependymoma
- c. Medulloblastoma
- d. Meningioma
- e. Schwannoma

31. Astrocytoma is a:

- a. Glial tumor
- b. Neuronal tumor
- c. Tumor of undifferentiated cells
- d. Nerve sheath tumor
- e. Malformative tumor

32. Oligodendrogliomas are not characterized by:

- a. Have IDH1 mutation
- b. Have IDH2 mutation
- c. Activation of the mitogen activated protein kinase (MAPK)
- d. Chromosomes 1P/19q codeletion

33. Medulloblastoma shows:

- a. Rosenthal fibers
- b. Vascular proliferation
- c. Papillary formation
- d. Rosette formation
- e. Psammoma bodies

34. Gliomas include all of the following EXCEPT:

- a. Astrocytoma
- b. Glioblastoma multiforme
- c. Oligodendroglioma
- d. Medulloblastoma
- e. Ependymoma

35. Pilocytic astrocytoma shows:

- a. Rosenthal fibers
- b. Vascular proliferation
- c. Papillary formation
- d. Rosette formation
- e. Psammoma bodies

36. The following tumor is always malignant:

- a. Pilocytic astrocytoma
- b. Ependymoma
- c. Medulloblastoma
- d. Meningioma
- e. Schwannoma

37. Astrocytoma is not:

- a. A glial tumor
- b. The commonest primary brain tumor in adults
- c. Always in the cerebral hemispheres
- d. Classified into grades according to the degree of nuclear atypia

6

Section



Pharmacology



1. The following is not common between pethidine & Buprenorphine:

- a. Analgesic effect in severe pain
- b. Respiratory depression is increased by increasing dose
- c. Nausea & vomiting
- d. Constipation

2. The following is true concerning opioids:

- a. Inhibit potassium outflux from neuronal membrane
- b. May be safely given in liver disease
- c. Naloxone effectively blocks effects of tramadol
- d. Pethidine is used in chronic moderate and severe pain
- e. None of the above

3. Choose the incorrect statement regarding pure opioid antagonists:

- a. Naloxone is taken IV in acute morphine toxicity & in opioid induced neonatal apnea (asphyxia neonatorum)
- b. Naltrexone should be given to opioid addict after full detoxification to avoid withdrawal symptoms
- c. Naltrexone is short duration and should be repeated in treatment of acute morphine toxicity
- d. Naltrexone is used as maintenance therapy to avoid relapse in opioid and alcohol addicts

4. The following analgesic is not recommended to be taken in renal failure or more than 48 hours for fear of convulsions:

- a. Morphine
- b. Tramadol
- c. Meperidine
- d. Codeine

5. Regarding Methadone, choose the incorrect statement:

- a. Used in detoxification and maintenance of opioid addicts
- b. Long acting & available orally with less withdrawal symptoms
- c. Less effective as analgesic than morphine
- d. Has risk of arrhythmia

6. Most serious adverse effect of morphine:

- a. Dependence
- b. Respiratory depression
- c. Hypertension
- d. A & B
- e. B & C

7. Which of the following is an opioid used in management of heroin withdrawal:

- a. Buspirone
- b. Bupropion
- c. Buprenorphine
- d. Bromocriptine

8. The following is wrong about opioid drugs:

- a. Methadone may prolong QT interval and precipitate torsade de pointe
- b. High doses of fentanyl may precipitate chest wall rigidity and decrease ventilation
- c. Pethidine is used in chronic moderate to severe pain inducing labor pain and biliary colic
- d. Buprenorphine is uses alternative to methadone in treatment of opioid addiction

9. Conscious sedation may be induced by:

- a. Fentanyl / Midazolam
- b. Midazolam / Droperidol
- c. Fentanyl / Domperidone
- d. None of the above

10. The following opioid drug should be avoided in patient taking MAOI:

- a. Morphine
- b. Fentanyl
- c. Meperidine
- d. Codeine

11. The following is not contraindication for morphine:

- a. Head injuries
- b. Acute undiagnosed abdominal pain
- c. Acute myocardial infarction
- d. Biliary colic

12. Long term use leads to tolerance for all actions of morphine EXCEPT:

- a. Analgesic effect
- b. Respiratory depression
- c. Constipation
- d. Hypotension

13. In moderate pain due to inflammation (Arthritis) the following analgesic is recommended:

- a. Morphine
- b. Methadone
- c. Tramadol
- d. Buprenorphine

14. One of the following is not strong pure opioid agonist:

- a. Morphine
- b. Pethidine
- c. Propoxyphene
- d. Fentanyl

15. The following is not common between Pethidine & Tramadol:

- a. Analgesics
- b. Action completely reversed by Naloxone
- c. Less respiratory depression than morphine
- d. Increase risk of convulsions with high doses

16. The following concerning Naloxone is not correct:

- a. Has a shorter half-life than its agonist
- b. Given orally
- c. Given in anesthesia to reverse CNS depression of its agonist
- d. It's a specific antagonist to opioid receptors

17. Neuroleptanalgesia may be induced by:

- a. Fentanyl & Midazolam
- b. Midazolam & Droperidol
- c. Fentanyl & Domperidone
- d. Fentanyl & Droperidol

18. Pethidine differs from morphine in all of the following EXCEPT:

- a. It's preferred for long treatment of chronic pain
- b. It's less liable to cause asphyxia neonatorum
- c. It's less liable to cause biliary colic
- d. It has atropine like action

19. All of the following are involved in mechanism of action of morphine EXCEPT:

- a. Most effects are mediated by activation of kappa receptors
- b. Inhibition of calcium influx resulting in inhibition of transmitter release
- c. Increase in K^+ outflux resulting in hyperpolarization of neuronal membrane
- d. Reduction of emotional response to pain

20. The following is wrong statement about Tramadol:

- a. Analgesic by inhibition uptake of 5HT (serotonin) and Norepinephrine
- b. Weak Mu agonist & partially antagonized by naloxone
- c. Less addiction, respiratory depression and convulsion than morphine
- d. Used as analgesic in postoperative pain as well as neuropathic pain

21. Choose the incorrect statement regarding Buprenorphine:

- a. Partial Mu receptor agonist
- b. Respiratory depression is not increased by increasing its dose
- c. Long acting and available sublingually in treatment of opioid addiction
- d. Can be given with Morphine to enhance its analgesic effect

22. Choose the incorrect statement regarding Heroin:

- a. Diacetyl morphine and converted to morphine in CNS
- b. Highly lipophilic with rapid onset of action
- c. Longer duration of action than morphine
- d. Not used clinically in most of countries

23. Indications of morphine don't include:

- a. Analgesic in cancer pain
- b. Acute pulmonary edema in acute left ventricular failure & myocardial infarction
- c. In anesthesia
- d. Severe pain following head injury

24. Choose the incorrect statement regarding Fentanyl:

- a. More potent with shorter duration and rapid onset than morphine
- b. Preferred in anesthesia
- c. Used with neuroleptic drugs (Droperidol) in conscious sedation
- d. Neurolept anesthesia is a combination neuroleptic drug, opioid drug & nitrous oxide in oxygen

• Cross match

25. Pethidine	a. Partial agonist used in management of opioid addict
26. Remifentanyl	b. Pure agonist used in management of opioid addict
27. Methadone	c. Not to be used for more than 48 hours to avoid risk of convulsions due to its toxic metabolite
28. Tramadol	d. Ultra short acting given by IV infusion
29. Buprenorphine	e. Mixed opioid non opioid, partially antagonized by naloxone





- 1. The advantage of adding Bromocriptine to L-dopa is:**
 - a. Activates D1 receptors improving the condition
 - b. Reduces the risk of psychosis of L-dopa
 - c. More effective than L-dopa
 - d. Longer $t_{1/2}$ & rapid absorption reducing fluctuations in L-dopa response
- 2. Nausea and vomiting of L-dopa may be reduced without interference with its antiparkinsonian effect by coadministration of:**
 - a. Metoclopramide
 - b. Domperidone
 - c. Carbidopa
 - d. B & C
- 3. A 51 year old patient with Parkinsonism is being maintained on levodopa-Carbidopa with adjunctive use of low doses of Entacapone but continues to have off-periods of akinesia. The most appropriate drug to rescue the patient but that will only provide temporary relief is:**
 - a. Apomorphine
 - b. Benztropine
 - c. Carbidopa
 - d. Ropinirole
 - e. Selegiline
- 4. The following adverse effect of L-dopa is not minimized by Carbidopa:**
 - a. Involuntary movements (Dyskinesia)
 - b. Nausea & vomiting
 - c. Arrhythmia
 - d. Postural hypotension
- 5. The following concerning antiparkinsonian drugs is/are correct:**
 - a. Apomorphine is given subcutaneously in emergency
 - b. L-dopa induced dyskinesia is reduced by Carbidopa
 - c. Benztropine reduces fluctuations in response to L-dopa
 - d. Amantadine may induce insomnia
 - e. A & D
- 6. Which of the drugs listed doesn't reduce fluctuations in response to L-dopa:**
 - a. Benztropine
 - b. Selegiline
 - c. Entacapone
 - d. Ropinirole
- 7. Effects of Selegiline in parkinsonism don't include:**
 - a. Sedative effective relieving insomnia of parkinsonism
 - b. Used with L-dopa/Carbidopa to prolong it's action
 - c. Inhibits MAO-B
 - d. Decreases fluctuations in L-dopa response
- 8. Which one of the following antiparkinson drug may cause peripheral vasospasm :**
 - a. Amantadine
 - b. Bromocriptine
 - c. Carbidopa
 - d. Entacapone
 - e. Ropinirole

1. Sedation is most marked with:

- a. Fluxotine
- b. Zolpidem
- c. Flumazenil
- d. Buspirone
- e. Valproate

2. The following concerning the use of benzodiazepines is not true:

- a. Abuse is associated with memory loss
- b. Toxic dose results in prolonged sleep
- c. Respiratory depression is reversed by Flumazenil
- d. Shorter acting agent are preferred as anxiolytics

3. The following is wrong statement about BZDs:

- a. Lorazepam is an intermediate duration and effective anxiolytic and used in status epilepticus
- b. Midazolam is long acting used in anesthesia as hypnotic and should be avoided as anxiolytic
- c. Alprazolam is effective in controlling the panic attacks
- d. Diazepam is long acting & useful during alcohol withdrawal

4. Abuse potential of Buspirone is low because:

- a. It has a weak anxiolytic effect
- b. It has no recognizable effect on brain neurotransmitters
- c. It has a delayed onset of action
- d. It's marked sedating effect is troublesome

5. The following drug is not matched with its mechanism of action:

- a. Diazepam increases frequency of opening of chloride channels
- b. Zolpidem binds with BZD1 subtype that increases GABA activity
- c. Barbiturates increase duration of opening of chloride channels
- d. Buspirone is selective 5HT1A antagonist

6. All of the following characteristics are shared by Flumazenil & Naloxone EXCEPT:

- a. Have shorter $t_{1/2}$ than their respective agonists
- b. Given orally
- c. Given in anesthesia to reverse CNS depression of their agonists
- d. Specific respiratory stimulants in BZD or opioid toxicity

7. The following statement is wrong regarding pharmacokinetics of benzodiazepines:

- a. Absorption of intramuscular injection except Lorazepam is slower than oral
- b. Metabolized in the liver by oxidation followed by glucuronidation except Lorazepam
- c. Clinical duration of action of long acting BZDs is shorter than the actual half-life as they form active metabolites
- d. Lorazepam has no active metabolites

8. Which of the following is preferred for treatment of anxiety with panic attacks:

- a. Buspirone
- b. Alprazolam
- c. Haloperidol
- d. Zolpidem

9. Which of the following doesn't have antidote :

- a. Clonazepam
- b. Thiopental
- c. Alprazolam
- d. Midazolam

10. The following concerning Zolpidem is not true:

- a. It's a short acting hypnotic causing less hangover than benzodiazepines
- b. It acts on benzodiazepine¹ receptor
- c. More liable to induce rebound insomnia & disturbance in REM sleep than benzodiazepines
- d. Its use is associated with less tolerance and dependence than benzodiazepines

11. BZDs don't induce the following effect:

- a. Anticonvulsant & spasmolytic
- b. Decrease stage II of NREM sleep
- c. Decrease REM sleep
- d. Decrease stage III & IV of NREM sleep

12. The following drug has no sedative hypnotic effect:

- a. Antihistamines
- b. Tricyclic antidepressants
- c. Flumazenil
- d. Clonidine

13. The following is a wrong statement about Barbiturates:

- a. Low therapeutic index
- b. Strong enzyme inhibitors
- c. Dose dependent CNS depression up to coma
- d. No specific antidote

14. The following drug is characterized by ultra-short duration of action:

- a. Clonazepam
- b. Lorazepam
- c. Midazolam
- d. Alprazolam

15. Flumazenil is an antidote for all the following EXCEPT:

- a. Clonazepam
- b. Buspirone
- c. Zolpidem
- d. Diazepam

16. The major limitation to the use of benzodiazepines in anxiety is:

- a. No antidote is available if overdose toxicity occurs
- b. Cardiovascular depression is common
- c. Dependence
- d. Increased risk of respiratory depression

17. An anxiolytic useful in a patient with history of drug dependence is:

- a. Buspirone
- b. Alprazolam
- c. Diazepam
- d. Lorazepam

18. Choose the incorrect statement:

- a. Buspirone is selective anxiolytic with delayed onset of action
- b. Ramelteon is selective melatonin receptor antagonist
- c. Flumazenil is short acting BZDs receptor antagonist
- d. Eszopiclone is intermediate acting non BZDs hypnotic

19. Buspirone is effective in case of:

- a. Panic attack
- b. Anxiety
- c. Insomnia
- d. All of the above

20. The following is wrong about Zolpidem:

- a. Short acting non BZDs hypnotic
- b. Less hangover and rebound insomnia than BZDs
- c. It's toxicity can be reversed by flumazenil
- d. Tolerance & dependence is more common than BZDs

21. The dose of benzodiazepines should be adjusted in:

- a. Elderly patients
- b. Concomitant administration with alcohol
- c. Hepatic dysfunction
- d. All of the above

22. Flumazenil can be used as antidote with which of the following:

- a. Zolpidem
- b. Thiopental
- c. Secobarbital
- d. Riluzole

23. All of the following drugs have slower onset when taken intra muscular EXCEPT:

- a. Alprazolam
- b. Midazolam
- c. Diazepam
- d. Lorazepam

24. The following anxiolytic drug don't act through GABA:

- a. Zolpidem
- b. Buspirone
- c. Lorazepam
- d. Phenobarbital

25. The following is not correctly matched to its use:

- a. Diazepam / Muscle spasticity
- b. Midazolam / Anesthesia
- c. Clonazepam / Epilepsy
- d. Buspirone / Panic attack

26. The following is not adverse effect of BZDs:

- a. Sedation & hangover
- b. Insomnia & decrease in REM sleep on sudden withdrawal
- c. Decrease in cognitive function of the brain
- d. Tolerance & dependence

27. The following statement is wrong:

- a. Chronic anxiety is added to antidepressant in the beginning of therapy till the effect of antidepressant appears
- b. Chronic insomnia should be treated with antidepressant such as Mirtazapine
- c. Short acting BZDs are preferred in insomnia with difficulty in falling a sleep
- d. Panic disorders require highly potent drugs such as Alprazolam & Buspirone
- e. Dose adjustment for most of BZDs is required in hepatic impairment

28. Conscious sedation may be induced by:

- a. Fentanyl / Midazolam
- b. Midazolam / Droperidol
- c. Fentanyl / Domperidone
- d. All of the above



1. The following about thiopental is wrong:

- a. Ultra short acting due to rapid metabolism
- b. Has a rapid onset of action due to rapid crossing of BBB
- c. Enzyme inducer and may precipitate porphyria
- d. May induce respiratory depression with no available antidote

2. Dentists avoid extraction in infection because of all the following EXCEPT:

- a. Fear of spread of infection
- b. Infection decreases extracellular unionized form of local anesthetics
- c. Infection reduces crossing of local anesthetics inside nerve cell
- d. Infection increases extracellular unionized form of local anesthetics

3. The following is not common adverse effect of nitrous oxide:

- a. Hypoxia if take pure without oxygen
- b. Increase pressure in the middle ear
- c. Abortion on prolonged exposure
- d. Hepatic toxicity

4. The following anesthetic agent is not acting through increase GABA activity:

- a. Halothane
- b. Propofol
- c. Isoflurane
- d. Ketamine

5. All of the following anesthetic drugs can cause respiratory irritation EXCEPT:

- a. Sevoflurane
- b. Isoflurane
- c. Desflurane

6. Abuse and expanding closed gas spaces are adverse effects of which of the following anesthetics :

- a. Ketamine
- b. Nitrous oxide
- c. Propofol
- d. Halothane

7. Which of the following is a wrong statement:

- a. Propranolol decrease metabolism of Lidocaine and may precipitate its toxicity
- b. Addition of epinephrine to local anesthetics may induce severe hypertension due to unopposed alpha action
- c. Lidocaine with epinephrine should be avoided in patient taking antidepressant drugs such as TCA & MAOI
- d. Felypressin is not suitable with local anesthetics in cardiac patients

8. One of the following anesthetics protocols is not correctly matched:

- a. Dissociative anesthesia / Ketamine
- b. Conscious sedation / Midazolam & Fentanyl
- c. Deep sedation / Propofol
- d. Neuroleptanalgesia / Propofol plus Fentanyl

9. The following is not true about cocaine:

- a. It's local anesthetic
- b. Acute toxicity may result in convulsions
- c. It's potent vasodilator
- d. Chronic abuse results in psychosis

10. Which of the following possesses a weak analgesic effect:

- a. Nitrous oxide
- b. Halothane
- c. Ketamine
- d. Fentanyl

11. Which of the following is ester local anesthetic:

- a. Lidocaine
- b. Bupivacaine
- c. Ropivacaine
- d. Benzocaine

12. Concerning nitrous oxide, the following is incorrect:

- a. Effective analgesic with low potency
- b. Safer than halothane with less respiratory depression and less hepatic toxicity
- c. Risk of abuse on prolonged exposure
- d. Iron deficiency anemia on chronic exposure

13. The following is not true about local anesthetics:

- a. Adding sodium bicarbonate increases activity of local anesthetics
- b. Bupivacaine is short acting
- c. Lidocaine is an amide
- d. Benzocaine is used as surface anesthetic

14. The following statement about halogenated anesthetics is wrong:

- a. Isoflurane may precipitate ischemia in cardiac patients
- b. Halothane is strong analgesic with medium rate of induction
- c. Desflurane suitable for one day surgery
- d. Sevoflurane is preferred as inhalation agent for children

15. The following is not true concerning ester type local anesthetics:

- a. Shorter acting than amides
- b. Less liable to induce allergy
- c. Less liable to systemic toxicity than amides
- d. Procaine is a member of this group

16. The following conditions decrease activity of local anesthetic EXCEPT:

- a. Repeated administration
- b. Infection
- c. Decrease intracellular acidity
- d. Addition of sodium bicarbonate

17. Which of the following is not a characteristic feature of amide local anesthetics:

- a. Longer duration of action than ester group
- b. Metabolized in the liver
- c. High risk of severe hypersensitivity reaction
- d. Duration of action is prolonged in patient with heart failure

18. Which of the following is wrong regarding Ketamine:

- a. Short acting IV agent producing dissociative anesthesia
- b. Has profound analgesic effect
- c. Useful in hypertension as it decreases blood pressure
- d. Hallucination may occur on recovery

19. Which of the following is not adverse effect of spinal anesthesia:

- a. Headache
- b. Spinal root injury
- c. CSF infection
- d. Hypertension

20. Which of the following statements concerning anesthetics is correct:

- a. Thiopental is ultra-short acting due to rapid metabolism
- b. Ketamine blocks NMDA receptors
- c. Propofol possesses a strong analgesic effect
- d. Halothane is the least likely inhalation agent to increase intracranial tension
- e. Dissociative anesthesia is induced by midazolam

21. Which of the following is wrong regarding Propofol:

- a. Rapid induction & rapid recovery due to rapid metabolism
- b. Used in conscious and deep sedation
- c. Respiratory depression and vomiting are common adverse effect
- d. Useful in one day surgery as IV anesthetic

22. A drug induced a GABA mediated sedative hypnotic effect in intensive care units:

- a. Fentanyl
- b. Propofol
- c. Ketamine
- d. B & C

23. Addition of epinephrine to local anesthetic don't:

- a. Decreases absorption
- b. Increases duration of action
- c. Decrease systemic adverse effects
- d. Increases potency

24. Lidocaine is preferred to bupivacaine in local anesthesia due to:

- a. Higher pka
- b. Longer duration
- c. Higher potency
- d. More rapid onset

25. Which of the following is wrong regarding local anesthetics:

- a. Binding to plasma protein affects duration of action
- b. Lipid solubility affects potency
- c. Addition of vasoconstrictor shorten duration of action
- d. Pka affects speed of onset

26. Which of the following is wrong statement about local anesthetics:

- a. All are vasodilators except cocaine
- b. Vasoconstrictors may be added to local anesthetic except cocaine
- c. Cocaine is used as local anesthetic in fingers and toes
- d. Vasoconstrictors added to local anesthetic include epinephrine & Felypressin

27. Which of the following method of administration of local anesthetic is more suitable for dentistry:

- a. Spinal anesthesia
- b. Epidural anesthesia
- c. Nerve block
- d. IV regional anesthesia

28. The following increase potency of anesthetics EXCEPT:

- a. Opioid drugs
- b. Benzodiazepines
- c. Alcohol abuse
- d. Elderly patient

29. The following is not adverse effect of halothane:

- a. Malignant hyperthermia
- b. Bradycardia & Arrhythmia
- c. Hypertension
- d. Hepatic toxicity
- e. Increase in intracranial pressure



1. Mild infrequent migraine attack is managed by:

- a. Triptan
- b. Ergot
- c. Diazepam
- d. Ibuprofen

2. Migraine prophylaxis of choice is:

- a. Timolol
- b. Topiramate
- c. Triptan
- d. None of the above

3. The following antiepileptic is not preferred during pregnancy:

- a. Lamotrigine
- b. Carbamazepine
- c. Levetiracetam
- d. Valproate

4. Ethosuximide useful in:

- a. Partial epilepsy
- b. Absence seizure
- c. Tonic clonic
- d. A & C

5. The following side effect is most likely to be seen with carbamazepine:

- a. Water retention
- b. Coarse facial features
- c. Hirsutism
- d. Gingival hyperplasia

6. All mentioned drugs useful in prophylaxis of migraine EXCEPT:

- a. Amitriptyline
- b. Timolol
- c. Diazepam
- d. Topiramate

7. Which of the following is not adverse effect of phenytoin:

- a. Osteomalacia
- b. Gingival hyperplasia
- c. Agranulocytosis
- d. Teratogenicity

8. Fatal dermatitis is a serious adverse effect of which of the following drugs:

- a. Levetiracetam
- b. Topiramate
- c. Lamotrigine
- d. Thiopental

9. All mentioned drugs useful in treatment of migraine EXCEPT:

- a. Nifedipine
- b. Ergotamine
- c. Metoclopramide
- d. Morphine

10. The following is/are true concerning antiepileptics:

- a. Females who get pregnant should discontinue the antiepileptic drug
- b. Lamotrigine may induce fatal dermatitis
- c. They are contraindicated in neuropathic pain
- d. Carbamazepine antagonizes the action of antidiuretic hormone
- e. B & D

11. Preferred therapy for a girl with combined absence and tonic clonic epilepsy is :

- a. Ethosuximide
- b. Carbamazepine
- c. Valproate
- d. Phenytoin plus carbamazepine

12. Uses of antiepileptics include all of the following EXCEPT:

- a. Lamotrigine / mood stabilizer in bipolar manic depression
- b. Phenytoin / digitalis arrhythmia
- c. Gabapentin / neuropathic pain
- d. Carbamazepine / dilutional hyponatremia

13. Valproate toxicity doesn't include:

- a. Hirsutism
- b. Hand tremor
- c. Thrombocytopenia
- d. Rare but fatal hepatitis

14. All the following are important guidelines in the management of epilepsy EXCEPT:

- a. All antiepileptics should be given after meals
- b. Pregnant females should discontinue the antiepileptic drug
- c. Gradual withdrawal of therapy to avoid status epilepticus
- d. Monotherapy is preferred to avoid drug interactions

15. A mood stabilizer safe in pregnant patients with bipolar depression:

- a. Carbamazepine
- b. Valproate
- c. Phenytoin
- d. Ethosuximide

16. An antiepileptic drug for absence seizures is :

- a. Diazepam
- b. Carbamazepine
- c. Ethosuximide
- d. Phenytoin

17. The following antiepileptic drug is not enzyme inducer:

- a. Phenytoin
- b. Carbamazepine
- c. Valproate
- d. Phenobarbital

18. Which of the following is not adverse effect of carbamazepine:

- a. Agranulocytosis
- b. Dilutional hyponatremia
- c. Megaloblastic anemia
- d. Enzyme inducer

19. An antiepileptic suitable for a pregnant woman with tonic clonic epilepsy is:

- a. Phenytoin
- b. Carbamazepine
- c. Valproate
- d. Ethosuximide

20. Hirsutism may be induced by:

- a. Valproate
- b. Interferon
- c. Heparin
- d. Phenytoin
- e. B & D

21. The following is not correctly matched to its mechanism of action:

- a. Levetiracetam / bind vesicle protein and modify glutamate release
- b. Lamotrigine / block Na^+ channels & presynaptic Ca^{++} channels
- c. Valproate / NMDA glutamate receptor antagonist
- d. Ethosuximide / Na^+ channel blocker

22. Choose the incorrect statement:

- a. Levetiracetam & valproate are broad spectrum used in all types of epilepsy
- b. Carbamazepine is 1st choice in partial epilepsy
- c. Valproate is 1st line in myoclonic epilepsy
- d. Clonazepam is 2nd line in partial epilepsy

23. Gingival hyperplasia is a common side effect of:

- a. Valproate
- b. Phenytoin
- c. Ethosuximide
- d. Carbamazepine

24. Teratogenic effects are more likely to be induced by:

- a. Phenytoin
- b. Valproate
- c. Carbamazepine
- d. A & B

25. The following antiepileptic is not used in tonic clonic epilepsy:

- a. Phenytoin
- b. Carbamazepine
- c. Valproate
- d. Ethosuximide

26. Regarding adverse effects of ergotamine, which is incorrect?

- a. Nausea & vomiting
- b. Chest pressure
- c. Vasospasm & gangrene, avoided in peripheral vascular disease
- d. Valvular heart disease with long term use
- e. Hypotension

27. Gingival hygiene and care is highly recommended when taking which of the following drugs:

- a. Phenytoin
- b. Valproate
- c. Carbamazepine
- d. Topiramate

28. The following drug(s) can be used in treatment of myoclonic epilepsy:

- a. Levetiracetam
- b. Valproate
- c. Topiramate
- d. All of the above

29. The following antiepileptic drug is not matched with its adverse effect:

- a. Lamotrigine / Stevens Jonson syndrome
- b. Gabapentin / Megaloblastic anemia
- c. Topiramate / renal stone & Glaucoma
- d. Levetiracetam / mood change & agitation

30. The following is not correctly matched to its mechanism of action:

- a. Phenytoin / sodium channel blockade
- b. Clonazepam / Cl^- channel blockade
- c. Valproate / inhibition of GABA transaminase
- d. Ethosuximide / Ca^{++} channel blockade

31. Which of the following is incorrect regarding antiepileptic drugs:

- a. Fosphenytoin is prodrug of phenytoin with less cardiac side effects
- b. Oxcarbazepine is analogue to carbamazepine with the same indications but less adverse effect
- c. Phenytoin & carbamazepine are contraindicated in absence & myoclonic seizures
- d. Clonazepam is 1st line in absence & myoclonic seizures

32. Regarding adverse effects & precautions with Sumatriptan, which is incorrect:

- a. Unpleasant taste with nasal spray
- b. Chest pressure
- c. Hypotension
- d. Serotonin syndrome with SSRIS
- e. Contraindicated in patients with ischemic heart disease

33. Sumatriptan precautions include all of the following except:

- a. Shouldn't be used for long term
- b. Diabetic patient
- c. Men over 40
- d. Postmenopausal

34. One of the following pairs of drugs used in treatment of migraine could be used intranasally:

- a. Sumatriptan & Diazepam
- b. Metoclopramide & Morphine
- c. Ergotamine & Triptan
- d. Triptan & Dihydroergotamine

35. The following antiepileptic drug acts by binding to VS_A to modify glutamate release:

- a. Valproate
- b. Levetiracetam
- c. Lamotrigine
- d. Gabapentin

36. Gangrene may be associated with:

- a. Acetaminophen
- b. Metoclopramide
- c. Opioid
- d. Dihydroergotamine

37. One of the following drugs could be used sublingually in treatment of migraine attacks:

- a. Ergotamine
- b. Triptan
- c. Topiramate
- d. Acetaminophen

• **Cross match**

38. Ergotamine	a. suppress firing of trigeminal nerve
39. Metoclopramide	b. long term use result in Valvular heart disease
40. Sumatriptan	c. added to ergotamine to increase its absorption
41. Topiramate	d. serotonin syndrome may occur with SSRI
42. Livido reticularis	a. Fentanyl
43. Fatal hepatotoxicity & hair thinning	b. Lamotrigine
44. Chest wall rigidity	c. Carbamazepine
	d. valproate
	e. Amantadine



1. **Reduction in sexual function is least seen with:**
 - a. Imipramine
 - b. Fluoxetine
 - c. Bupropion
 - d. Both B & C
2. **Fluoxetine doesn't induce:**
 - a. Serotonin syndrome if co administered with Tranylcypromine
 - b. Anxiety
 - c. Nausea
 - d. Marked sedation
3. **Which of the following is not adverse effect encountered with Fluoxetine:**
 - a. Nausea
 - b. Sexual dysfunction
 - c. Increase risk of suicide
 - d. Sedation & dry mouth
4. **The following don't induce sexual dysfunction:**
 - a. Tricyclic antidepressants
 - b. Mirtazapine
 - c. Fluoxetine
 - d. Bupropion
 - e. B & D
5. **Which of the following is wrong regarding antidepressants:**
 - a. Have delayed onset of action
 - b. MAOIs are reserved for atypical depression & as 2nd choice in refractory major depression
 - c. Fluoxetine & paroxetine are potent enzyme inducer leading to drug interaction with beta blocker & antipsychotics
 - d. TCA are reserved for resistant cases as toxicity limits its use
6. **Disadvantages of tricyclic antidepressants include:**
 - a. Narrow safety margin
 - b. Frequent side effects
 - c. Delayed therapeutic effect
 - d. All of the above
7. **An antidepressant useful for smoking cessation is:**
 - a. Venlafaxine
 - b. Bupropion
 - c. Imipramine
 - d. Mianserin
8. **Nocturnal enuresis is treated with imipramine but not fluoxetine because:**
 - a. Imipramine induces insomnia while fluoxetine induces sedation
 - b. Imipramine is safer than fluoxetine
 - c. Imipramine possess anticholinergic effect but fluoxetine doesn't
 - d. All of the above
9. **The following drug doesn't increase serotonin in the brain:**
 - a. Imipramine
 - b. Fluoxetine
 - c. Bupropion
 - d. Venlafaxine
 - e. Mirtazapine
10. **Severe hypertension is most likely to occur following intake of old cheese with:**
 - a. Tranylcypromine
 - b. Moclobemide
 - c. Selegiline
 - d. Bupropion

11. Moclobemide is preferred to phenelzine because:

- a. It inhibits MAO-A and MAO-B enzymes
- b. It's a selective MAO-A inhibitor only
- c. It can be safely used in combination with SSRIs
- d. All of the above

12. Indications of antidepressants don't include:

- a. Chronic neuropathic pain
- b. Convulsive disorders
- c. Eating disorders
- d. Obsessive-compulsive disorders

13. The following is not adverse effect of imipramine:

- a. Prolongation of QT interval leading to arrhythmia
- b. Confusion, dry mouth and increase IOP
- c. Insomnia & Hallucination
- d. Convulsion

14. The nonselective MAOIs have limited use as antidepressants because of their:

- a. Low efficacy
- b. High toxicity
- c. Potential to interact with many foods and drugs
- d. Both B & C

15. Patient intolerant to SSRIs due to their correlation with sexual disturbances can instead take:

- a. Venlafaxine
- b. Imipramine
- c. Amitriptyline
- d. Mirtazapine

16. Which of the following drug interaction with TCA is wrong:

- a. Alpha blocker / potential postural hypotensive effect
- b. TCA / potentiate the effect of sympathomimetic
- c. SSRIs / decrease toxicity of TCA
- d. MAOIs / serotonin syndrome

17. The following concerning venlafaxine is not true:

- a. Anticholinergic side effects are frequently encountered with its use
- b. It's a norepinephrine serotonin reuptake inhibitor
- c. Nausea is a troublesome side effect
- d. Hypertension may occur during therapy

18. The following is not correctly matched:

- a. Fluoxetine / selective serotonin reuptake inhibitor
- b. Mirtazapine / presynaptic α_2 receptor blocker
- c. Venlafaxine / norepinephrine serotonin reuptake inhibitor
- d. Moclobemide / selective monoamine oxidase - B inhibitor

19. The following is not true regarding tricyclic antidepressant toxicity:

- a. Arrhythmia is managed by NaHCO_3
- b. Convulsions is one of the features of toxicity
- c. Atropine toxicity is seldom seen
- d. Conduction defects are responsible for arrhythmia

• **Cross match**

20. Bupropion	a. sexual dysfunction & atropine side effects
21. Imipramine	b. hypertension
22. Venlafaxine	c. convulsions with no reduction in sexual dysfunction
23. Fluoxetine	d. sexual dysfunction & serotonin syndrome



1. Intrapartum group B streptococcus prophylaxis requirements:

- a. GBS carrier mother
- b. History of delivery of infant with GBS
- c. History of GBS bacteriuria at any time during pregnancy
- d. All of the above

2. Which of the following is wrong about meningitis due to streptococcal pneumoniae:

- a. Duration of treatment up to 14 days
- b. Ceftriaxone is chosen in ampicillin MIC < 1ug/ml
- c. Addition of vancomycin to ceftriaxone if MIC of both Ampicillin & ceftriaxone is high
- d. Rifampicin is added to vancomycin during using dexamethasone because it increases metabolism of vancomycin

3. Which of the following statement is wrong about using corticosteroids in meningitis:

- a. Taken for 2-4 days every 6 hours
- b. Dexamethasone is used not cortisone
- c. Should be started immediately after empiric antibiotic
- d. May decrease penetration of vancomycin to BBB

4. TB meningitis:

- a. Duration will take 9-12 month
- b. Ethambutol may be used
- c. Combination of 1st line therapy of anti TB drugs
- d. All of the above

5. Chemoprophylaxis is not recommended in meningitis due to:

- a. Neisseria meningitides
- b. Streptococcus pneumoniae
- c. Haemophiles influenza
- d. Listeria monocytogenes

6. Gray baby syndrome is associated with:

- a. Ceftriaxone
- b. Moxifloxacin
- c. Penicillin
- d. Chloramphenicol

7. Which of the following is 1st choice as empiric antibiotic in meningitis suspected to streptococcal agalactia in 55 year old patient:

- a. Ampicillin + ceftriaxone
- b. Vancomycin + ceftriaxone
- c. Vancomycin + ceftriaxone + Ampicillin
- d. Vancomycin + Cefepime

8. Dexamethasone is injected IV for:

- a. 2-4 days every 12 or 24 hours
- b. 10-14 days every 6 hours
- c. 2-4 days every 6 hours
- d. 10-14 days every 12 or 24 hours

9. Choose from the following pair of antibiotics not crossing BBB even if inflamed:

- a. Aminoglycosides & β -lactam
- b. Chloramphenicol & Aminoglycosides
- c. Rifampicin & Monobactam
- d. Amphotericin & Aminoglycosides

10. Prolonged QT interval is associated with:

- a. Moxifloxacin
- b. Chloramphenicol
- c. Ceftriaxone
- d. None of the above

11. Prophylaxis of close contact by rifampicin for 2 days used in case of:

- a. Haemophiles influenza meningitis
- b. Streptococcus agalactia meningitis
- c. Streptococcus pneumoniae meningitis
- d. Neisseria meningitides

12. Principles of antimicrobial therapy for meningitis:

- a. Antibiotics should be lipophilic ionized in physiological PH
- b. Steroids should be administrated after starting empiric AB
- c. Lumbar puncture should be done before choosing empiric AB
- d. Once organism is identified treatment should be continued for 7 to 21 days

13. Prophylaxis of close contact by rifampicin for 4 days used in case of:

- a. Haemophiles influenza meningitis
- b. Streptococcus agalactia meningitis
- c. Streptococcus pneumoniae meningitis
- d. None of the above

14. Psuedocholethiasis is associated with:

- a. Aminoglycosides
- b. Penicillin
- c. Chloramphenicol
- d. Ceftriaxone

15. Chloramphenicol is the last antibiotic of choice to be used due to:

- a. Anaphylactic shock
- b. Agranulocytosis
- c. Ototoxicity
- d. Pseudo cholelithiasis

16. Which of the following drugs is preferred in case of penicillin allergy in patient with streptococcus pneumoniae:

- a. Cefixime
- b. Rifampicin
- c. Gentamicin
- d. Vancomycin



1. Which of the following is correct about Fingolimod:

- a. Associated with 1st dose tachycardia
- b. 1st parenteral and oral disease modifying drug
- c. Used with caution in diabetic patient
- d. Increases T-lymphocyte penetration to BBB

2. The following is an incorrect statement:

- a. Natalizumab taken by IV infusion for control progressive multiple sclerosis who failed on interferon
- b. Fingolimod may induce bradycardia and macular edema
- c. Donepezil block central muscarinic receptors and used in Alzheimer disease
- d. Rivastigmine available as transdermal patch in management of dementia of Alzheimer and Parkinson disease

3. Increased risk of infection is mainly associated with:

- a. Prednisone
- b. Fingolimod
- c. Interferon
- d. A+B
- e. None of the above

4. Choose the wrong statement:

- a. Fingolimod is disease modifying agent taken orally to control relapse and decrease disability of multiple sclerosis
- b. Depression, flu like syndrome are adverse effects of interferon
- c. Natalizumab is monoclonal antibody taken orally against $\beta 1$ integrin
- d. Chemotherapeutic agents like cyclophosphamide can be used to control multiple sclerosis

5. Macular edema is mainly occurred with:

- a. Natalizumab
- b. Fingolimod
- c. Interferon
- d. Riluzole

6. Which of the following statement is wrong regarding management of multiple sclerosis:

- a. Methylprednisolone is used to control acute exacerbation
- b. The main target disease modifying drugs is to inhibit WBC mediated inflammatory processes
- c. Interferon $\beta 1A$ & $\beta 1B$ are the 1st line treatment
- d. Fingolimod is taken IV to alter lymphocyte migration to CNS

7. Choose the incorrect statement regarding mode of action of Riluzole:

- a. Block Na^+ channel
- b. Block GABA release
- c. Block NMDA receptor
- d. Block calcium channel

8. Which of the following drug is not increasing acetyl choline in the brain:

- a. Rivastigmine
- b. Donepezil
- c. Memantine
- d. Galantamine

9. Which of the following is incorrect regarding Natalizumab:

- a. Taken IV infusion monthly
- b. Decrease BBB invasion by T-lymphocyte
- c. Indicated in mild controlled multiple sclerosis
- d. Monoclonal antibody

10. NMDA receptor antagonists include:

- a. Galantamine
- b. Ketamine
- c. Memantine
- d. Rivastigmine
- e. B & C

11. Which of the following is correct regarding Fingolimod:

- a. Decrease BBB invasion by T- lymphocyte
- b. Prevent T lymphocyte migration from lymph node
- c. Used with caution in cardiac patients
- d. All of the above

12. Riluzole is prescribed in Amyotrophic lateral sclerosis to:

- a. Decrease onset of dementia
- b. Decrease confusion and headache
- c. Decrease need for ventilator
- d. All of the above

13. Choose the incorrect sentence regarding interferon:

- a. 1st line therapy for patient with MS
- b. Injected IV
- c. Prevent demyelination process
- d. Associated with increased hepatic enzymes

14. Fampridine drug in treatment of multiple sclerosis:

- a. Blocks Na⁺ channel
- b. Causes hyperpolarization
- c. Blocks K⁺ channel
- d. Decreases neuronal depolarization

15. Which of the following is the best choice for patients with dementia due to Parkinson's disease:

- a. Rivastigmine
- b. Memantine
- c. Donepezil
- d. Galantamine

16. Which of the following drugs is contraindicated in Alzheimer disease:

- a. Statins
- b. Bapineuzumab (monoclonal antibody against A β)
- c. Benztropine
- d. Galantamine

17. The following are adverse effect of corticosteroids EXCEPT:

- a. Hyperglycemia
- b. Muscle wasting
- c. Hypotension
- d. Osteoporosis

18. Depression is mainly associated with:

- a. Methylprednisolone
- b. Memantine
- c. Interferon
- d. A+C
- e. All of the above

19. To improve walking speed in patient with multiple sclerosis, you should prescribe:

- a. Fingolimod
- b. Fampridine
- c. Prednisone
- d. None of the above

20. The following is wrong regarding Riluzole:

- a. Inhibits glutamate release
- b. Block sodium channel
- c. NMDA receptor antagonist
- d. Used in management of multiple sclerosis

21. The following is wrong statement about Memantine:

- a. NMDA receptor antagonist leading to decrease glutamate excitotoxicity
- b. Used in moderate to severe cases of Alzheimer diseases
- c. Combined with donepezil in Namzaric
- d. Induced confusion and diarrhea



1. **A drug effective in reducing emotional bluntness in schizophrenics is:**
 - a. Carbamazepine
 - b. Clozapine
 - c. Chlorpromazine
 - d. Clonazepam
2. **Nicotine abuse may be managed by:**
 - a. Varenicline
 - b. Bupropion
 - c. Buprenorphine
 - d. Acamprosate
 - e. A & B
3. **The following is not adverse effect of lithium carbonate:**
 - a. Ataxia & Tremors
 - b. Polyuria & thirst
 - c. Hypertension
 - d. Renal tubular damage
4. **The following antipsychotic drug is preferred in children because it has less effect on cognitive function:**
 - a. Olanzapine
 - b. Aripiprazole
 - c. Quetiapine
 - d. Thioridazine
5. **The following is / are correctly matched with areas of dopamine blockade:**
 - a. Extrapyramidal effects / Basal ganglia
 - b. Antipsychotics / Mesolimbic
 - c. Antiemetic / chemoreceptor trigger zone
 - d. All of the above
6. **The correct statement regarding extrapyramidal effects of antipsychotics:**
 - a. Occur most commonly with low potency agents
 - b. Tardive dyskinesia is relieved by anticholinergics
 - c. Anticholinergics are useful in neuroleptic-induced parkinsonism
 - d. Both A & B
7. **The risk of inducing rigidity, tremors & tardive dyskinesia in schizophrenics is highest with:**
 - a. Haloperidol
 - b. Thioridazine
 - c. Clozapine
 - d. Olanzapine
8. **The following is not true about antipsychotics:**
 - a. Control symptoms of schizophrenia without affecting the basic disorder
 - b. Low potency drugs are preferred in elderly & cardiac patients
 - c. Thioridazine may block cardiac conduction & increase QT interval
 - d. They don't induce dependence
9. **A drug reserved for resistant cases of schizophrenia for fear of agranulocytosis:**
 - a. Haloperidol
 - b. Clozapine
 - c. Olanzapine
 - d. Risperidone
10. **The following abused drug is wrongly matched to replacement therapy:**
 - a. Pentobarbital / phenobarbital
 - b. Morphine / methadone
 - c. Alcohol / Disulfiram
 - d. Alprazolam / Diazepam
11. **Which of the following drugs could rapidly calm a violent patient:**
 - a. Haloperidol
 - b. Buspirone
 - c. Flumazenil
 - d. Fluoxetine

12. Which of the following improves negative symptoms in schizophrenia:

- a. Thioridazine
- b. Risperidone
- c. Haloperidol
- d. Chlorpromazine

13. The best drug to reduce excessive nicotine craving:

- a. Mirtazapine
- b. Bupropion
- c. Venlafaxine
- d. Fluoxetine

14. The following is wrong regarding management of alcohol abuse:

- a. Naltrexone blocks opioid receptors (↓ relapse)
- b. Acamprosate is weak NMDA antagonist & GABA receptor activator (↓ craving)
- c. Triazolam replaces alcohol because it has smoother withdrawal
- d. Disulfiram is used as psychotherapy

15. The following is incorrect regarding management of nicotine abuse:

- a. Nicotine replacement by nicotine gum, transdermal patch, oral inhaler & nasal spray
- b. Bupropion increase DA that will decrease craving
- c. Varenicline is partial agonist at CNS nicotinic Ach receptors
- d. Naltrexone decreases craving & relapse

16. Less extrapyramidal effects with risk of agranulocytosis may be seen with:

- a. Clozapine
- b. Thioridazine
- c. Haloperidol
- d. Risperidone
- e. Olanzapine

17. The following drug is 1st choice in treatment of psychosis in Parkinson disease:

- a. Haloperidol
- b. Risperidone
- c. Aripiprazole
- d. Quetiapine

18. The following is not used in alcohol abuse:

- a. Disulfiram
- b. Acamprosate
- c. Flumazenil
- d. Naltrexone
- e. Thiamine

19. The following antipsychotic drug has the least central side effects especially extrapyramidal side effects:

- a. Haloperidol
- b. Clozapine
- c. Aripiprazole
- d. Chlorpromazine

20. The following is not true about drug abuse:

- a. Short acting agents are less liable to be abused
- b. Death from cocaine overdose may result from severe hypertension
- c. A red eye is one of consequences of cannabis addiction
- d. Bupropion reduces craving in patient who quit smoking

21. Thiamine deficiency occurs most common with abuse of:

- a. Heroin
- b. Alcohol
- c. Cocaine
- d. Cannabis

22. The following is correctly matched to its adverse effect:

- a. Clozapine / marked extrapyramidal side effects
- b. Olanzapine / agranulocytosis more common compared to clozapine
- c. Haloperidol / conduction defects more common than Thioridazine
- d. Venlafaxine / hypertension more common
- e. Fluoxetine / prominent anticholinergic side effects

23. The following antipsychotic drug is preferred in children suffering from autism:

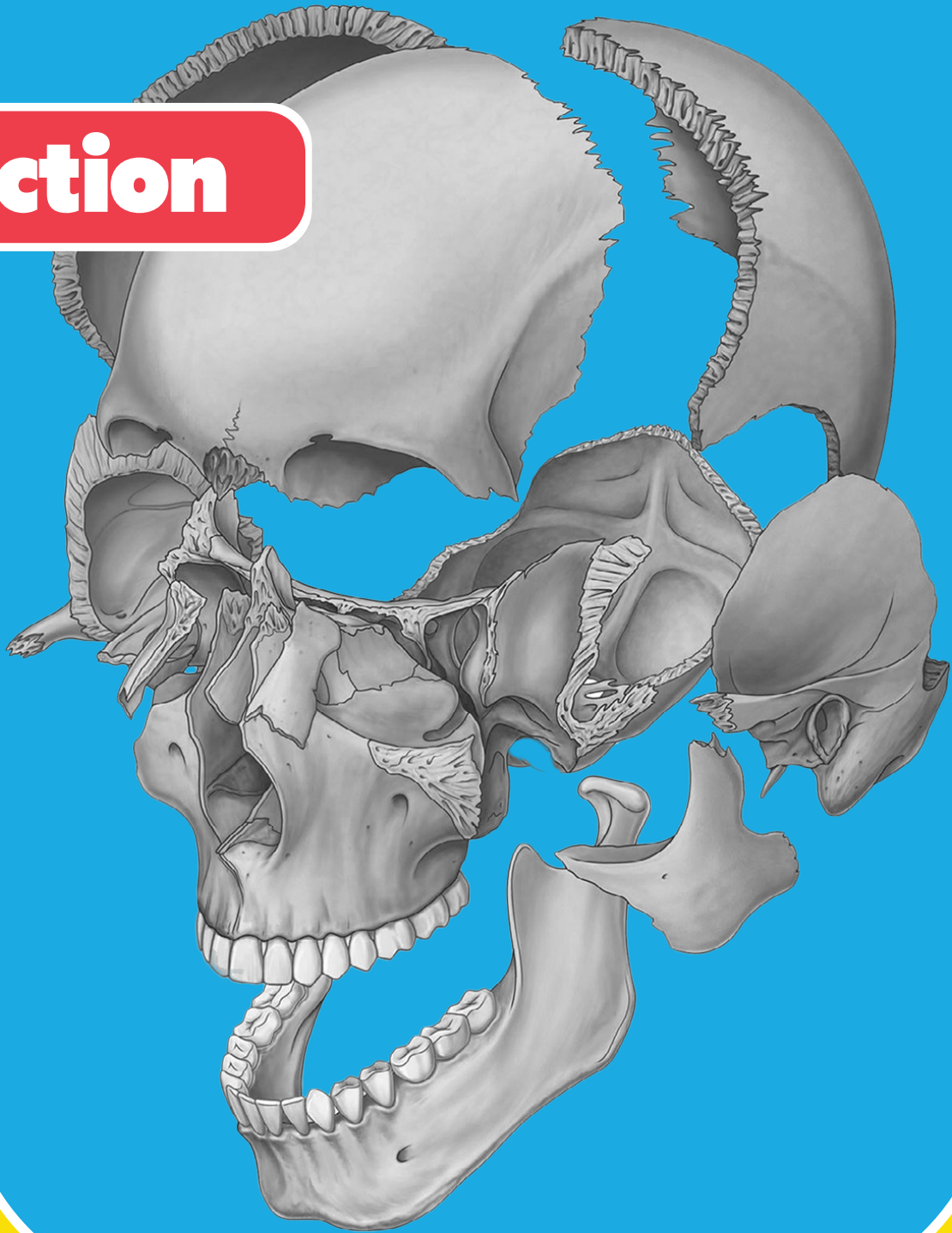
- a. Olanzapine
- b. Aripiprazole
- c. Quetiapine
- d. Thioridazine

Notes

[illegible]

7

Section



Microbiology

1. The drug of choice for prevention of epidemic cerebrospinal meningitis in children is:

- a. Penicillin G.
- b. Ciprofloxacin.
- c. Rifampicin.
- d. Cefotaxime.

2. Virulence of *N. meningitidis* is due to:

- a. Exotoxin production.
- b. Spreading factors production.
- c. Polysaccharide capsule.
- d. All of the above.

3. Which one of the following media is most suitable for isolating *Neisseria meningitidis* in nasopharyngeal swab?

- a. Sheep blood agar.
- b. Chocolate agar.
- c. MacConkey agar.
- d. Thayer-Martin medium.

4. Rifampicin is used as prophylaxis against epidemic cerebrospinal meningitis because:

- a. It has a broad-spectrum activity.
- b. Efficiently secreted into the saliva.
- c. Resistance to the drug developed rarely.
- d. All of the above.

5. Treatment of choice in epidemic cerebrospinal meningitis is:

- a. Tetracyclines.
- b. Streptomycin
- c. Chloramphenicol.
- d. Penicillin G.

6. *Neisseria* species are positive for the enzyme:

- a. Coagulase.
- b. DNase.
- c. Oxidase.
- d. Catalase.

7. *Neisseria meningitidis* Carriers harbor the organism:

- a. In the nasopharynx.
- b. In the blood.
- c. In the oral cavity.
- d. On the skin.

8. A Gram-negative diplococcus was isolated from a chocolate agar culture of a cerebrospinal fluid sample. Which one of the following tests would be most informative in identifying the organism as *Neisseria meningitidis*?

- a. Appearance of colonies.
- b. Growth on Thayer Martin medium.
- c. Negative oxidase test of spinal fluid.
- d. Ability of organism to ferment glucose and maltose.

9. Meningococcal vaccine:

- a. Provides protection against disease due to serogroup B.
- b. Is prepared from capsular polysaccharide of *N. meningitidis* A, C, Y, W135.
- c. Does not reduce carrier rate.
- d. Contains heat-inactivated *N. meningitidis*.

10. Each of the following statements concerning *Neisseria* is correct EXCEPT

- a. They are Gram- negative diplococci.
- b. They produce IgA protease as a virulence factor.
- c. They are oxidase positive.
- d. They grow best under anaerobic conditions.

11. Each of the following statements concerning *Neisseria meningitidis* is correct EXCEPT:

- a. it is an oxidase-positive, gram-negative diplococcus.
- b. It contains endotoxin in its cell wall.
- c. It produces an exotoxin that stimulates adenylate cyclase.
- d. It has a polysaccharide capsule that is anti-phagocytic.

12. which of the following false about the symptoms of meningitis

- a. Decrease appetite
- b. Unresponsiveness to light
- c. Seizures
- d. None of the above

13. most common of neonatal meningitis is

- a. listeria monocytogenes
- b. *E. coli*
- c. influenza b
- d. all of the above

14. chemical examination of CSF in case of bacterial meningitis include

- a. ↑ protein
- b. normal glucose
- c. CSF is clear.
- d. None of the above

15. physical examination of fungal meningitis of CSF is/are.

- a. CSF is turbid
- b. ↓ protein
- c. Normal glucose
- d. None of the above

16. positive biochemical reaction of *Cryptococcus neoformans* to

- a. oxidase
- b. catalase
- c. urease
- d. a & b

17. colony of *Cryptococcus neoformans* is..... while colony *N. meningitidis* is

- a. creamy mucoid, smooth
- b. smooth, creamy mucoid

18. All of the following are symptoms of meningitis except....

- a. irritability
- b. stiff neck
- c. increased appetite
- d. nausea and vomiting

19. For diagnosis of carrier of *N. Meningitis*, we take ...

- a. blood sample.
- b. CSF sample
- c. Nasopharyngeal swab
- d. Urine sample

20. In bacterial meningitis ...

- a. Proteins are elevated.
- b. CSF is clear.
- c. Glucose is reduced.
- d. a, c

21. used to demonstrate of encapsulated yeast cells of *Cryptococcus*.

- a. India ink
- b. Gram stain
- c. Seaboard's dextrose agar
- d. Giemsa stain

22. Cryptococcus neoformans is

- a. Yeast cells with gelatinous capsule
- b. Urease positive
- c. Pathogenic to mice
- d. All of the above

23. Pathogenic Neisseria are characterized by:

- a. Grow on ordinary media.
- b. Grow on chocolate agar.
- c. Grow at room temperature.
- d. Oxidase-Negative.

24. Epidemic cerebrospinal meningitis:

- a. Is transmitted only by cases.
- b. Is caused mainly by serotypes A, B, C.Y, W 135
- c. Penicillin G is used as chemoprophylaxis.
- d. Live at vaccine is used to prevent epidemics.

25. The most common route of meningitis transmission

- a. Hematogenous
- b. Direct spread from otitis media
- c. Nearby focus as sinusitis
- d. Trauma

26. Which of the Following organisms cause adult meningitis:

- a. Listeria monocytogens
- b. Escherichia coli
- c. Group B streptococci
- d. Neisseria meningitidis
- e. C & d

27. 2-year-old child has a fever, stiff neck & is irritable. Gram stain of the Spinal fluid reveals G-ve small pleomorphic coccobacillary organisms.**I. Which of the Following is the Most likely etiological micro-organism?**

- a. Hemophilus influenzae
- b. E. coli
- c. Streptococcus pneumoniae
- d. Staph. Golden

II. What is the most appropriate procedure to follow in order to reach an etiological diagnosis:

- a. Culture the Spinal fluid in mannitol salt agar
- b. Perform catalase test of the Isolated organism.
- c. Culture the Spinal fluid in chocolate agar
- d. Perform coagulase test.
- e. Perform latex agglutination to detect specific Ab in spinal fluid.

28. A 33-year-old HIV +ve man complains of: headache & blurred vision. CSF obtained by lumbar Puncture reveals a capsulated organism visible by india ink. Which of following is true Concerning the Organism morphology:

- a. Can be seen as spaghetti & meat ball on KOH.
- b. Consists of branching septate hyphae
- c. Exists as mycelial form at room temp & yeast at 37°C.
- d. Encapsulated non-dimorphic yeast.

29. Chronically ill young male spends most of his time raising & training pigeons. He develops mild Pulmonary infection & eventually presents to the physician with headache, mental status changes, fever. Clinical diagnosis of meningitis was confirmed with a latex agglutination test on CSF for Capsular polysaccharide of organism. What is the most likely causative organism:

- a. Aspergillus
- b. Candida albicans
- c. Cryptococcus neoformans
- d. Histoplasma capsulatum

30. 10 years old boy has a severe headache & vomiting that began yesterday. He is now confused. On examination, his temperature is 39°C & his neck is stiff. Spinal fluid reveals G+ve diplococci on Gram stain, High protein, low glucose. Which of the Following should be considered for his family members:

- a. No prophylaxis is necessary.
- b. They should be given vaccine.
- c. They should be given Rifampicin as prophylaxis.
- d. They should be given sulfonamide as prophylaxis.

31. 1 week newborn develops meningitis. Short G+ve rods are isolated. History reveals that the mother had eaten unpasteurized cheese during pregnancy & she recalled having flu like illness. Which of the Following is the Most likely etiological micro-organism:

- a. Corynebacterium diphtheriae
- b. Escherichia coli
- c. Group B streptococci
- d. Listeria monocytogens
- e. Streptococcus pneumoniae

32. The following is best used for treatment of cryptococcal meningitis:

- a. Amphotericin B
- b. Nystatin
- c. Fluconazole
- d. A & c in combination

33. A neonate was delivered 3 weeks prematurely. The Neonate had no signs of illness at birth, however developed meningitis 3 days later. The Causative organism isolated from the cerebrospinal fluid of the Neonate was oxidase-ve, lactose fermenting, indole +ve, G-ve bacilli. How did the neonate acquire this pathogen?

- a. In utero transmission early in pregnancy
- b. In utero transmission late in pregnancy
- c. During passage in birth canal
- d. Droplet transmission from anterior nares of health care providers
- e. Air born transmission from contaminated air in the delivery room.

34. An 18-year-old man in a military camp presented to your hospital with fever, severe headache, Drowsiness & neck rigidity. Gram -ve cocci were isolated from CSF the Reservoir of infection in this case could Be:

- a. A human carrier
- b. An infected animal
- c. Bird feces
- d. Food or water
- e. Soil

35. Quellung test is positive with:

- a. S. pneumoniae
- b. Enterobacter
- c. S. aureus
- d. Diphtheriae
- e. N. gonorrhoeae

36. Which of the following is the primary virulence factor of *S. pneumoniae*?

- a. Bile solubility.
- b. Optochin production.
- c. Pili.
- d. Polysaccharide capsule.

37. An eight-year-old boy presented with fever, severe headache and a stiff Neck. There was also projectile vomiting and the child entered into a coma.

I. The most important laboratory test to help diagnosis in this case is:

- a. Complete Blood Count
- b. Blood sugar determination
- c. Urine analysis
- d. Lumbar puncture
- e. Tuberculin test

II. Direct smear from the diagnostic specimen shows Gram-negative Intracellular diplococci. The organism is probably:

- a. *Haemophilus influenzae*
- b. *Neisseria meningitidis*
- c. *Streptococcus pneumoniae*
- d. *Neisseria gonorrhoeae*
- e. *Streptococcus pyogenes*

III. The treatment of choice is:

- a. Sulphonamides
- b. Penicillin + chloramphenicol
- c. Third generation cephalosporins
- d. Anti-toxin e- None of the above

38. Meningitis associated with *Neisseria meningitidis* usually occurs as what type of disease?

- a. Endemic.
- b. Epidemic.
- c. Pandemic.
- d. Isolated.

39. What is the reservoir of *Neisseria meningitidis*?

- a. Soil.
- b. Saltwater.
- c. Human carriers.
- d. Domesticated animals.

40. How is *Neisseria meningitidis* spread?

- a. Through respiratory secretions or droplets.
- b. Through contaminated fomites.
- c. Through consumption of contaminated food or water
- d. By insect vectors.

41. What is the major virulence factor of *Streptococcus pneumoniae*?

- a. DNase.
- b. Capsule.
- c. Pyogenic exotoxins.
- d. M protein.

1. Physical examination of HSV meningitis of CSF is\are

- a. CSF is turbid.
- b. ↓protein
- c. ↓glucose
- d. none of the above

2. In poliomyelitis neutralizing Abs are formed in:

- a. Subclinical stage
- b. Abortive stage
- c. Aseptic meningitis
- d. Paralytic stage

3. Polio virus can replicate in all following sites except:

- a. Tonsils
- b. Anterior horn cells
- c. Peyer's patches
- d. Voluntary muscles

4. Which is false about Salk vaccine:

- a. Prevent both GIT infection & paralysis.
- b. Produces systemic immunity by IgM & IgG
- c. More stable than Sabin vaccine
- d. Given by injection.

5. Rapid diagnosis of enteroviral meningitis is performed by:

- a. Viral isolation
- b. IgM detection
- c. RT-PCR
- d. E/M

6. Five-year-old boy had an episode of mild vesicular rash in his mouth, hands & feet. Approximately 6 days later, he developed vomiting, fever & nuchal rigidity. A CSF sample showed high cell count & normal protein & glucose levels. Which of the Following is the Most likely causative agent?

- a. Rotavirus
- b. Cryptococcus
- c. Measles virus
- d. Meningococci
- e. Coxsackie virus

7. One of the Advantages of Sabin vaccine over Salk vaccine that It:

- a. Contains the 3 serotypes of poliovirus.
- b. Can be given to feverish child.
- c. Can be given to immunodeficient child.
- d. Viruses are grown in monkey or human cell culture.
- e. Produce herd immunity.

8. Interference decreases the efficacy of:

- a. Sabin vaccine
- b. Salk vaccine
- c. Measles vaccine
- d. Influenza vaccine

9. Following isn't an advantage of Sabin vaccine:

- a. Produces local immunity.
- b. Given orally.
- c. No reversion to virulence
- d. Produce herd immunity.

10. When infection of poliovirus stop at pyer's patch .this infection called

- a. Inapparent infection
- b. Apparent infection
- c. Abortive infection
- d. None of the above

11. Abortive infection means.

- a. Infection stop at Peyer's patch.
- b. Infection stop at deep mesenteric LN.
- c. May cause viremia.
- d. Progress to paralysis
- e. b & c

12. Poliovirus may cause

- a. Flaccid paralysis
- b. Respiratory paralysis
- c. Poliomyelitis of bulbar
- d. All of the above

13. Which of the following used to diagnosis of poliovirus

- a. Isolation virus
- b. Elisa
- c. Immunofluorescence
- d. All of the above

14. All of the following statements are true regarding poliovirus except:

- a. It is transmitted by feco-oral route.
- b. Asymptomatic infections are common in children.
- c. There is a single serotype causing infection.
- d. Live attenuated vaccine produces herd immunity.

15. All of the following are true about inactivated polio vaccine (IPV) except:

- a. IPV is made from wild polio viruses.
- b. IPV must be given by injection.
- c. IPV is equivalent to oral polio vaccine (OPV) in inducing immunity in the oral cavity.
- d. IPV is equivalent to OPV in inducing immunity in the intestine.

16. which current inactivated vaccine of poliovirus give in 4 doses

- a. Salk vaccine
- b. Sabin vaccine
- c. Gamma globulin
- d. IPV

17. brain abscess caused by all of the following except

- a. Staphylococcus
- b. Candida
- c. Rabies
- d. All of the above
- e. Non of the above



1. *Cl. Tetani* is

- a. Gram +ve non motile
- b. Gram -ve rods motile
- c. Gram +ve rods bacilli
- d. Gram -ve cocci rods

2. Death occurs in tetanus due to

- a. Respiratory paralysis
- b. Brain coma
- c. Local muscle spasm
- d. Opisthotonos

3. Which type of blood agar used to diagnosis tetani

- a. Alfa hemolytic agar
- b. Beta hemolytic agar
- c. Gama hemolytic agar
- d. All of the above

4. Shape of *c. Tetani* is

- a. Bullet
- b. Drumstick
- c. Grape
- d. None of the above

5. Is a *c. Tetani* invasive organism?

- a. True
- b. False

6. Which of the following true about tetanospasmin

- a. Reach CNS through blood
- b. Produce generalized muscular spasm
- c. Block inhibitory Nt
- d. All of the above

7. *Cl. botulinum* produces.

- a. Heat stable neurotoxin.
- b. Heat labile neurotoxin.

8. *Cl. tetani* produces.....while *Cl. botulinum* produces

- a. Flaccid paralysis, spastic paralysis
- b. Spastic paralysis, flaccid paralysis

9. classic botulism caused by

- a. Eating honey
- b. Inhalation of toxin
- c. Eating canned food
- d. Skin popping

10. which of the following used as confirmatory diagnosis

- a. Used culture.
- b. Toxin detection
- c. PCR
- d. All of the above

11. Which of the following used to prevent intestinal botulism

- a. Sterilization of food before canned
- b. Avoid inhalation of aerosol
- c. Avoid eating honey
- d. All of the above

12. Food poisoning due to *Clostridium botulinum* is most often caused by:

- a. Contaminated poultry products
- b. Canned food.
- c. Dairy products and custards.
- d. Contaminated and poorly cooked seafood.

13. Food poisoning that NOT manifested by gastrointestinal symptoms is caused by:

- a. Staph. aureus.
- b. *Clostridium perfringens*
- c. *Clostridium botulinum*
- d. *Salmonella typhimurium*.

14. Death from clostridium botulinum food poisoning is due to:

- a. Respiratory and cardiac failure.
- b. Electrolyte imbalance.
- c. Marked fluid loss.
- d. All of the above.

15. The vaccine to protect against tetanus is:

- a. Tetanus toxoid.
- b. Live attenuated Clostridium tetani
- c. Heat killed Clostridium tetani.
- d. Formalin killed C. tetani.

16. The mechanism of action of C. botulinum toxin that cause food poisoning is:

- a. Acts on vomiting centers
- b. Interfere with the release of acetyl choline from nerve endings.
- c. Block the release of inhibitory mediators in spinal cord.
- d. Acts directly on intestinal mucosal cells

17. Treatment of clostridium botulinum food poisoning includes

- a. Fluid and electrolyte replacement
- b. Antitoxin
- c. Antibiotic
- d. All

18. The most important protective function of the antibody stimulated by tetanus immunization is:

- a. To prevent growth of the pathogen.
- b. To prevent adherence of the pathogen.
- c. To neutralize the toxin of the pathogen.
- d. All of the above

19. Neurotoxic Clostridia include:

- a. Clostridium perfringens.
- b. Clostridium botulinum.
- c. Clostridium tetani.
- d. b & c.
- e. All of the above.

20. Cl. tetani causes lockjaw paralysis because.

- a. The organism enter muscle cells and blocks contraction.
- b. The organism enters muscle cells and stimulates contraction.
- c. Its neurotoxin blocks the release of inhibitory mediator leading to generalized muscle spasm.
- d. Its neurotoxin stimulates the release of inhibitory mediator leads to localized muscle spasm.

21. Which of following types of Cl. Botulinum toxin is most significant cause of human illness:

- a. Types A, B, C, D
- b. Types A, B, E
- c. Types, C, D, H
- d. Types B, C, E

22. Pathogenesis of which of following organism include retrograde axonal transport:

- a. Cl. Botulinum
- b. Cl. Tetani
- c. C. diphtheria
- d. Bacillus Anthracis

23. Which of the following is NOT true regarding tetanus antitoxin?

- a. Antitoxin neutralizes accessible toxin when administered early.
- b. Antitoxin is ineffective against toxin already fixed in the CNS.
- c. Antitoxin will reverse existing manifestations of neurotoxicity.
- d. Tetanus immune globulin (TIG) is superior to equine antitoxin because the Risk of sensitivity is minimal.

24. Symptoms of botulism include double vision, inability to speak and Respiratory paralysis. These symptoms are consistent with:

- a. Invasion of the gut epithelium by *C. Botulinum*
- b. Secretion of an enterotoxin
- c. Ingestion of a neurotoxic
- d. Endotoxic shock
- e. Activation of cyclic AMP

25. Which of the Following is incorrect regarding active immunization against tetanus:

- a. DTaP vaccine is used for children.
- b. Tetanus toxoid is used.
- c. No need for booster dose
- d. Live attenuated vaccine is used.
- e. C & d



1. **65 years old woman reports that she has had several episodes of confusion & memory loss during the past few weeks. On examination, she is afebrile but has a staggering gait & myoclonus can be elicited. Over the next several months, her condition deteriorates & death ensues. On autopsy, microscopic examination of brain reveals many vacuoles but no viral inclusion bodies. Diagnosis is:**
 - a. Herpes encephalitis.
 - b. Creutzfeldt Jakob disease
 - c. Subacute sclerosing panencephalitis
 - d. Rabie
 - e. Progressive multifocal leukoencephalopathy
2. **Following are slow viral diseases except: -**
 - a. Herpes simplex encephalitis
 - b. Subacute sclerosing panencephalitis
 - c. Creutzfeldt Jacob disease
 - d. Rabies
 - e. a & d
3. **45 years old man was attacked by a street dog & he was never vaccinated before. How would you proceed:**
 - a. Give him rabies Ig only.
 - b. Give him rabies vaccine only.
 - c. Give him both vaccine & Ig.
 - d. Just clean the Wound + vaccine
 - e. Give him Ig after the Vaccine, only if adequate titers aren't detected.
4. **Prion is atypical virus being CC by having: -**
 - a. circular RNA
 - b. Ds DNA
 - c. Proteins only
5. **Progressive multifocal encephalopathy is caused by:**
 - a. Adenovirus
 - b. Cytomegalovirus
 - c. Herpes simplex virus
 - d. Polyoma virus (JC virus)
 - e. Cocksackie virus
6. **35 years old man is seen in the emergency room complaining of severe headache & vomiting. He is foaming at the mouth & can't drink any liquids. Analysis of his spinal fluid reveal no abnormality & no organisms are seen in gram stain. 2 days later despite supportive measures, he died. Pathologic examination of brain revealed eosinophilic inclusion bodies in the cytoplasm of neurons. the most probable causative agent is:**
 - a. Prion
 - b. JC virus
 - c. Rabies virus (Rhabdovirus)
 - d. Parvovirus B19
 - e. Herpes simplex virus type 1
7. **All the following are true regarding Negri bodies EXCEPT:**
 - a. They are intra-nuclear inclusion bodies.
 - b. They are intra-cytoplasmic inclusion bodies.
 - c. They may be detected in infected cases of rabies.
 - d. They can be demonstrated in the brain of infected dogs by direct immunofluorescence tests.
8. **Post exposure vaccination is successful in rabies because:**
 - a. virus is weak.
 - b. Incubation period is long.
 - c. vaccine is potent.
 - d. All of the Above

9. Rabies is a disease caused by:

- a. It is a severe form of encephalitis.
- b. It is a zoonotic disease.
- c. It has long I.P
- d. All of above

10. A 25-year-old man was bitten on his right arm by a stray dog. He didn't receive rabies vaccine before. He should be immediately managed by:

- a. Interferon alpha + rabies immunoglobulin
- b. Interferon alpha + rabies vaccine
- c. Interferon alpha + ribavirin
- d. Rabies immunoglobulin
- e. Rabies immunoglobulin + rabies vaccine

11. Best route of rabies virus to reach the spinal cord & CNS of the Host:

- a. Blood due to high level of viraemia
- b. Along the Afferent peripheral neurons
- c. Being carried by MQS
- d. All of above

12. Vaccine used in Rabies prevention in human is:

- a. Killed vaccines.
- b. Live attenuated
- c. Not efficient
- d. Efficient only in pre-exposure time

13. Which is true about prions: -

- a. Composed of DNA or RNA
- b. Have long I.P
- c. Usually affects CNS.
- d. Highly immunogenic
- e. Can be cultured.
- f. B&c

14. An immunocompromised person with a history of Personality changes, Hallucination, Seizures has MRI that reveals a temporal lobe lesion. Brain biopsy results show: MNGCs with INIBS. Which is the Most probable cause:

- a. VZV
- b. EBV
- c. HSV
- d. Listeria monocytogens

15. The best treatment for the previous case is:

- a. Acyclovir
- b. Ganciclovir
- c. Ribavirin
- d. Amantadine

16. subacute sclerosing panencephalitis is caused by which of the Following virus:

- a. Epstein bar virus
- b. HIV
- c. JC virus
- d. Measles virus
- e. Mumps virus

17. The presence of Negri bodies in host cells is a characteristic of:

- a. Mumps
- b. Infectious mononucleosis
- c. Congenital rubella
- d. Aseptic meningitis
- e. Rabies

18. Which of the following is the mode for West Nile fever virus transmission:

- a. Arthropod bite
- b. Blood transfusion
- c. Organ transplantation
- d. All of the above

19. Which of the following is the best to diagnose prion disease:

- a. Culture in vitro
- b. Brain biopsy
- c. Serology
- d. Antigen detection

20. A previously healthy 50-year-old woman died of west Nile encephalitis. What is the reservoir & vector Of This virus respectively?

- a. Cattle, mosquitoes
- b. Birds, mosquitoes
- c. Horses, mosquitoes
- d. Cattle, ticks
- e. Rodents, fleas

21. Does rabies virus cause viremia

- a. True
- b. False

22. Incubation period of rabies virus depend on

- a. Bite location.
- b. Amount virus.
- c. Severity
- d. All of the above

23. Fear of water is a symptom of a

- a. HSV
- b. Rabies virus
- c. Arenavirus
- d. None of the above

24. To management of rabies, which of the following must use

- a. Active immunization
- b. Passive immunization

25. spasm of throat muscle due to rabies lead to

- a. Hydrophobia
- b. Increased salivation
- c. Paralysis
- d. None of the above

26. The best route of rabies virus to reach the spinal cord and CNS of the host is

- a. Blood due to high level of viremia
- b. Along the afferent peripheral neurons
- c. Being carried by macrophages.

27. Post exposure Vaccination is successful in rabies because

- a. The virus is weak.
- b. The vaccine is potent.
- c. The incubatory period is long.
- d. All of the above

28. Diagnosis is of rabies in an infected animal can be done by

- a. Detection of specific antibodies in the serum
- b. Detection of Negri bodies in the brain tissue
- c. Both of them
- d. None of them

29. The best rabies vaccine used now in humans is

- a. Human diploid cell vaccine
- b. Simple vaccine
- c. Duck embryo vaccine
- d. Flurry vaccine

30. Treatment of rabies includes the followings EXCEPT

- a. Post-exposure vaccination
- b. Passive immunization
- c. Cleaning of the wound with tetanus prophylaxis
- d. Antiviral drugs

31. The presence of Negri bodies in host cell is characteristic of

- a. Mumps
- b. Rabies
- c. Infectious mononucleosis
- d. Aseptic meningitis.

32. A disease, muscle twitching and lack of coordination, occurs due to a prion, known as

- a. Marasmus
- b. Kuru disease
- c. Cancer
- d. Tetanus

33.is abnormal form of a cellular protein

- a. Cervical dystonia
- b. Prion
- c. Clear sensorium
- d. None of the above

34. All of the following true about prion disease except

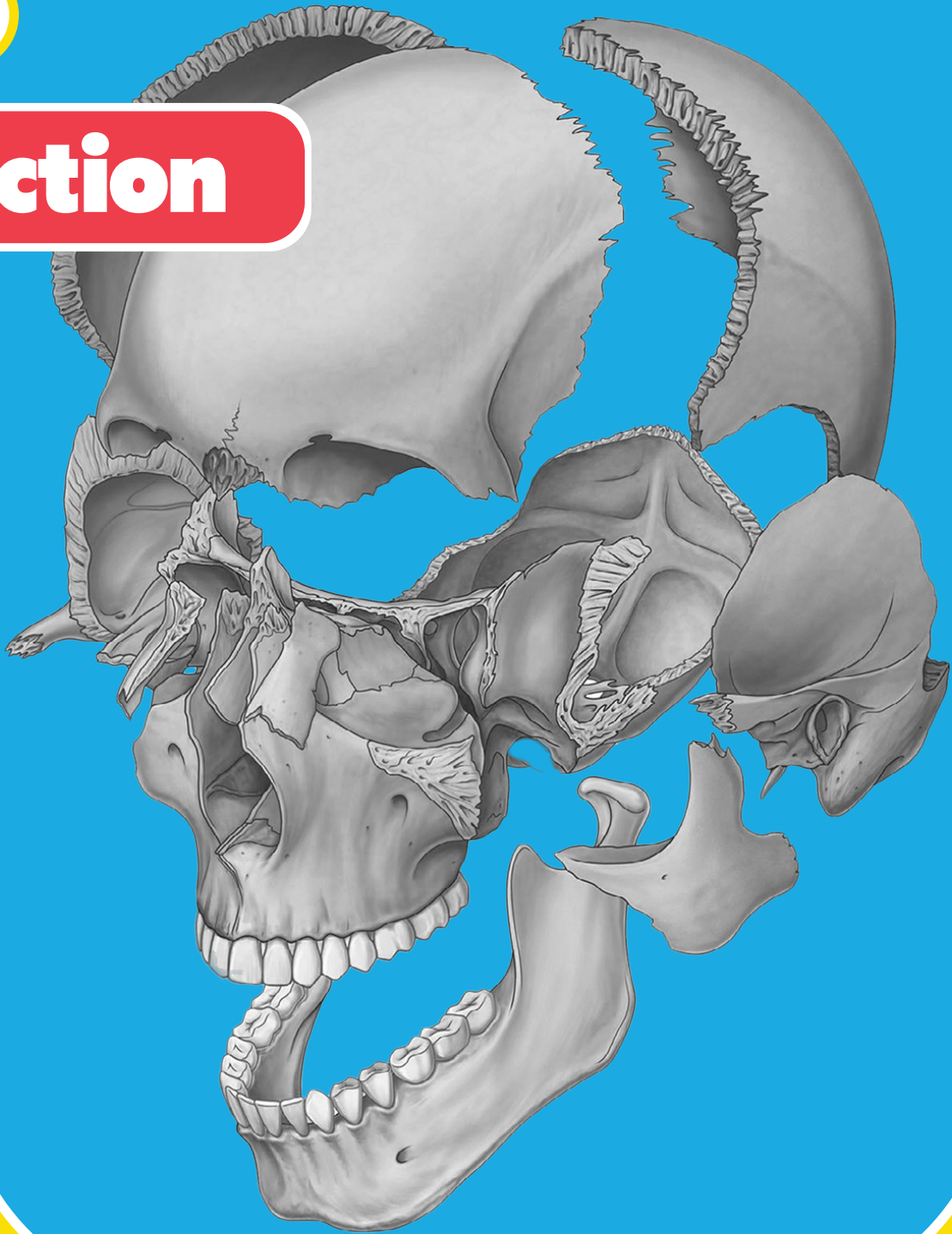
- a. Affect CNS
- b. Has long incubation period.
- c. Not has immune response.
- d. Not transmission by ingestion of inflected tissue



Notes

8

Section



Parasitology

Chapter 1 ? Protozoa of CNS (free-living amoebae)

1. Acanthamoeba can be a problem for people who:

- a. Inhale the trophozoites or cysts
- b. Wear contact lens
- c. Through skin abrasions
- d. All of the above

2. Naegleria fowleri infection is diagnosed by:

- a. Skin biopsy
- b. Stool examination
- c. CSF examination
- d. None of the above

3. The facultative parasites are:

- a. Acanthamoeba & Naegleria
- b. Toxoplasma gondii & sparganum
- c. Acanthamoeba & Toxoplasma gondii
- d. All of the above

4. A 20-year-old girl came to the ophthalmologist presenting by severe burning sensation, increased lacrimation and photophobia. On examination, keratitis of her left eye was observed ... she gave a history of prolonged wearing contact lenses

I. The parasitic cause of this case is:

- a. Sparganum
- b. Toxoplasma gondii
- c. Cysticercosis
- d. Acanthamoeba
- e. Naegleria

II. The mode of infection is:

- a. Autoinfection
- b. Congenital infection
- c. Through cornea
- d. Eating frog

5. The infective stage of Naegleria fowleri is:

- a. Cyst form
- b. Amoeboid trophozoite
- c. Flagellated trophozoite
- d. All of the above

6. The cyst that may be found in human tissue is:

- a. Acanthamoeba
- b. Naegleria fowleri
- c. Spirometra mansoni
- d. All of the above

7. 20% Biguanide in swimming pools is used as a control of:

- a. Naegleria fowleri
- b. Acanthamoeba
- c. Toxoplasma gondii
- d. Cysticercosis
- e. All of the above

8. Examination of corneal scraping to see cysts or trophozoites of:

- a. Acanthamoeba
- b. Naegleria fowleri
- c. Spirometra mansoni
- d. All of the above

9. Primary Amoebic Meningoencephalitis (PAM) is caused by:

- a. Entamoeba histolytica
- b. Naegleria fowleri
- c. Acanthamoeba
- d. All of the above

10. Acanthamoeba species cause:

- a. Granulomatous Amoebic Encephalitis (GAE)
- b. Keratitis
- c. Granuloma of internal organs
- d. Disseminated granulomatous lesions
- e. All of the above

11. Naegleria fowleri is:

- a. Obligatory parasite of man
- b. Free living amoeba accidentally pathogenic
- c. Non pathogenic amoeba
- d. All of the above

12. What of the following organisms have no definitive host, no reservoir host and no vector :

- a. Acanthamoeba & Naegleria fowleri
- b. Toxoplasma gondii & Sparganum
- c. Acanthamoeba & Toxoplasma gondii
- d. All of the above

13. The cyst of Acanthamoeba found in brain tissue has:

- a. 1 nucleus
- b. 2 nuclei
- c. 4 nuclei
- d. Contractile vacuole

14. Skin biopsy to diagnose disseminated skin lesions occurs in infection with:

- a. Naegleria Fowleri
- b. Toxoplasma gondii
- c. Sparganum
- d. Acanthamoeba
- e. Cysticercosis

15. The trophozoites of Naegleria Fowleri reach its habitat through:

- a. The blood
- b. The olfactory nerve
- c. The lymphatic system
- d. All of the above

16. Which is responsible for causing Chronic Granulomatous Amoebic Encephalitis:

- a. Naegleria Fowleri
- b. Sparganum
- c. Toxoplasma gondii
- d. Acanthamoeba

17. Keratitis may occurs with infection with:

- a. Naegleria Fowleri
- b. Acanthamoeba
- c. Sparganum
- d. Cysticercosis

18. CSF culture on non-nutrient agar seeded with gram -ve bacilli is used for diagnosis of:

- a. Naegleria Fowleri & Acanthamoeba
- b. Naegleria Fowleri
- c. Acanthamoeba
- d. Toxoplasma gondii
- e. All of the above

19. The cyst of this parasite has double wall (the outer is wrinkled & the inner is polyhedral):

- a. Acanthamoeba
- b. Naegleria Fowleri
- c. Toxoplasma gondii
- d. Sparganum

20. Amphotericin B is used to treat :

- a. Naegleria Fowleri
- b. Trichomonas vaginalis
- c. Acanthamoeba
- d. Balantidium coli

21. The opportunistic facultative protozoa is:

- a. Acanthamoeba
- b. cytosporidium parvum
- c. Entamoeba histolytica
- d. Entamoeba coli

22. Wearing contact lenses during swimming may lead to infection with:

- a. Trypanosome brucei
- b. Balantidium coli
- c. Acanthamoeba
- d. Naegleria Fowleri

23. Regarding to Acanthamoeba all these statements are correct EXCEPT:

- a. It infects immune-suppressed persons
- b. The infective stage are trophozoite & cyst
- c. It's facultative organisms
- d. Causing disseminated granulomatous amoebic disease
- e. It cause suppurative Meningoencephalitis

24. The trophozoite of this parasite can take flagellate form:

- a. Acanthamoeba
- b. Naegleria Fowleri
- c. Toxoplasma gondii
- d. Sparganum

25. Concerning Acanthamoeba the following is true :

- a. It causes diffuse hemorrhagic necrosis of the brain
- b. Culture of biopsy specimens shows cysts and trophozoites
- c. Trophozoites show amoeboid and flagellated forms
- d. Brain infection is usually secondary to intestinal infection

26. Culture medium used to isolate Naegleria Fowleri:

- a. NNN media
- b. Non-nutrient agar seeded with gram -ve bacilli
- c. Citrated blood
- d. Schneider's media

27. Regarding to Naegleria Fowleri , all these statements are correct EXCEPT:

- a. Can't transmit from person to person
- b. It infects healthy person
- c. The infective stage is amoeboid trophozoite
- d. Causing keratitis
- e. Causing Primary Amoebic Meningoencephalitis

28. As regards Acanthamoeba encephalitis:

- a. The invasive stage is amoeboflagellate
- b. Trophozoites only can be found in brain
- c. Chronic inflammation of the brain is seen
- d. The trophozoite reaches the brain through the cribriform plate
- e. It's usually fatal within 10 days

29. The habitat of Acanthamoeba in humans may be in:

- a. Brain and eyes
- b. Skin
- c. Lung
- d. Genito-urinary tract
- e. All of the above

30. Differentiation between Naegleria Fowleri and viral encephalitis infection occurs by :

- a. Complete blood film to detect polymorphonuclear leukocytosis (PMN or PNL) to differentiate it from viral encephalitis
- b. Complete blood film to detect eosinophilia in Naegleria Fowleri
- c. Complete blood film to detect monocytosis in viral encephalitis
- d. All of the above

31. A 25 year old patient presented with features of acute Meningoencephalitis with altered taste and smell ... His CSF on wet mount microscopy revealed motile trophozoites the most likely organism is:

- a. Naegleria Fowleri
- b. Acanthamoeba
- c. Entamoeba histolytica
- d. Plasmodium falciparum
- e. Toxoplasma gondii

32. The following protozoa can affect the brain:

- a. Naegleria Fowleri
- b. Acanthamoeba
- c. Plasmodium Falciparum
- d. All of the above

33. A male patient presented with fever, vomiting, altered taste and smell Few days later, he developed convulsions and coma , his parents gave a history of swimming in fresh water pools

I. To confirm this case we should do :

- a. CSF examination
- b. Blood examination
- c. Brain biopsy
- d. All of the above

II. The infective stage is:

- a. Cyst
- b. Amoeboid trophozoite
- c. Flagellate trophozoite
- d. All of the above



1. Definitive test for diagnosis of sparganosis is:

- a. Biopsy from the swelling to identify the sparganum larvae
- b. CT or MRI to confirm the site of lesion
- c. Blood picture to see eosinophilia
- d. All of the above

2. Mode of infection of Cysticercosis is/are:

- a. External autoinfection (contaminated hand to mouth)
- b. Internal autoinfection or retroinfection
- c. Ingestion of Taenia solium eggs through contaminated vegetables
- d. All of the above

3. A 20 year old female was referred to neurology clinic with a severe epileptic attack ... her family gave a history of previous attacks of intermittent drowsiness and convulsions. On examination tender palpable subcutaneous nodules were felt at different sites , CT scan showed cysts in brain ... Surgical removal of subcutaneous nodules revealed cystic structure about 1cm in size. She gave a history of eating pigs meat since 4 months

I. The preliminary diagnosis is :

- a. Cysticercosis
- b. Sparganosis
- c. Toxoplasmosis
- d. Naegleria Fowleri

II. The infective stage in this case:

- a. Cysticercus bovis
- b. Taenia solium egg
- c. Toxoplasma cyst
- d. Amoeboid trophozoite

4. Sparganosis could be prevented by:

- a. Avoid eating insufficiently cooked fish
- b. Avoid eating insufficiently cooked pork meat
- c. Avoid eating insufficiently cooked beef meat
- d. Boiling water before drinking

5. the most common parasitic infection of the CNS is:

- a. Echinococcosis
- b. Sparganosis
- c. Cerebral malaria
- d. Neurocysticercosis
- e. Amoebic brain abscess

6. Human act as blind intermediate host in:

- a. Cysticercosis & Sparganosis
- b. Cysticercosis & Toxoplasmosis
- c. Acanthamoeba & Naegleria
- d. Naegleria & sparganosis
- e. All of the above

7. Cysticercosis occurs as a complication in infection with:

- a. Taenia solium
- b. Taenia saginata
- c. Toxoplasma
- d. Naegleria

8. The infective stage of the parasite causing sparganosis is/are:

- a. Egg
- b. Coracidium
- c. Proceroid & sparganum larvae
- d. Sparganum larvae
- e. Coracidium & Sparganum larvae

9. Autoinfection is a mode of transmission of the following parasite:

- a. Cysticercus cellulosae
- b. Sparganum larvae
- c. Acanthamoeba
- d. Naegleria Fowleri

10. CSF analysis may be used in diagnosis of all these parasites EXCEPT:

- a. Naegleria Fowleri
- b. Acanthamoeba
- c. Cysticercus cellulosae
- d. Sparganosis

11. The most common presenting problem associated with the development of Neurocysticercosis:

- a. Facial palsy
- b. Decreased level of consciousness
- c. New onset seizures
- d. Blurred vision
- e. Nausea and vomiting

12. Confirmatory diagnosis of Cysticercosis by:

- a. Surgical removal of the nodules & examined microscopically to see Cysticercus cellulosae
- b. Immune-diagnosis by ELISA test
- c. X ray, CT & MRI of the skull
- d. Stool examination

13. Infective stage in Cysticercosis is:

- a. Egg of Taenia saginata
- b. Egg of Taenia solium
- c. Cysticercus bovis
- d. Cysticercus cellulosae

14. Taenia solium infection is serious as:

- a. It inhabits the small intestine
- b. Segments pass with stool
- c. Man can act as intermediate blind host
- d. Segments detach in groups

15. Elevated eosinophilia, high protein level and normal glucose may found in CSF sample indicate to infection with:

- a. Sparganosis
- b. Cysticercosis
- c. Toxoplasmosis
- d. Acanthamoeba
- e. Naegleria

16. A 25 year old male presents with headache, confusion and seizures. He gives history of multiple subcutaneous nodules and states that he frequently eats raw frog meat... what do you suspect :

- a. Granulomatous amoebic encephalitis
- b. Primary amoebic Meningoencephalitis
- c. Sleeping sickness
- d. Cerebral malaria
- e. Sparganosis

17. Mode of infection of sparganosis is/are:

- a. Drinking water containing Cyclops infected with Proceroid larva
- b. Eating undercooked frogs or bird containing sparganum larvae
- c. Applying the infected flesh of frogs as poultices containing sparganum larvae
- d. All of the above

18. The intermediate host of Taenia solium is/are:

- a. Pigs occasionally man
- b. Cattle, sheep, occasionally man
- c. Dogs
- d. Foxes & Wolves

Chapter 3 ?

Other important parasitic infections of the CNS

- 1. Congenital transmission may occur by the following parasite:**
 - a. Toxoplasmosis
 - b. Cysticercosis
 - c. Sparganosis
 - d. Acanthamoeba
 - e. Naegleria Fowleri
- 2. Pyrimethamine + sulfadiazine + folinic acid are used for treatment of:**
 - a. Toxoplasma gondii
 - b. Acanthamoeba
 - c. Cysticercosis
 - d. Sparganosis
- 3. The opportunistic protozoa is/are:**
 - a. Acanthamoeba
 - b. Toxoplasma gondii
 - c. All of the above
 - d. Naegleria Fowleri
- 4. Teratogenic protozoa is :**
 - a. Acanthamoeba
 - b. Naegleria Fowleri
 - c. Toxoplasma gondii
 - d. Cysticercosis
- 5. A typical triad of microcephaly or hydrocephalus + intracranial calcification & chorioretinitis occurs in infection with:**
 - a. Acanthamoeba
 - b. Naegleria Fowleri
 - c. Toxoplasma gondii
 - d. Cysticercosis
- 6. Single or multiple ring enhancing lesions in CT & MRI occurs in infection with:**
 - a. Acanthamoeba
 - b. Toxoplasma gondii
 - c. All of the above
 - d. Naegleria Fowleri
- 7. Treatment of toxoplasmosis in an AIDS patient includes:**
 - a. Albendazole
 - b. Mebendazole
 - c. Pyrimethamine
 - d. Praziquantel
 - e. Ketoconazole
- 8. Brain biopsy may used in diagnosis of :**
 - a. Acanthamoeba to detect cysts or trophozoites
 - b. Toxoplasma indicated in patient fail in response to treatment
 - c. All of the above
- 9. Chorioretinitis occurs in infection with:**
 - a. Congenital toxoplasmosis
 - b. Acanthamoeba
 - c. Naegleria Fowleri
 - d. Cysticercosis
 - e. Sparganosis
- 10. Blood transfusion is a source of infection with:**
 - a. Toxoplasma gondii
 - b. Acanthamoeba
 - c. Naegleria Fowleri
 - d. Sparganosis
 - e. All of the above

Notes

II



Short Essay

5

Section



Pathology



1. What are neurodegenerative diseases?

- Disorders characterized by the cellular degeneration of subsets of neurons that are typically related by function, rather than physical location in the brain.

2. How are neurodegenerative diseases classified?

- They can be classified based on:
 - Clinical presentation
 - Type of protein
 - Pathological processes
 - Genetic abnormalities.

3. What is dementia?

- The development of memory impairment and other cognitive deficits severe enough to decrease the affected person's capacity to function at the previous level despite a normal level of consciousness.

4. What is the most common cause of dementia in the elderly population?

- Alzheimer's Disease (AD)

5. What is the crucial event in the pathogenesis of Alzheimer's Disease (AD)?

- The accumulation of Beta Amyloid Peptide (A β) in the brain over time.

6. What is the role of Amyloid Precursor Protein (APP) in Alzheimer's Disease (AD)?

- APP is sequentially cleaved by enzymes to generate A β .
- Mutations in APP or in enzymes responsible for its degradation can lead to familial AD.

7. What is the protein responsible for the creation of A β in Alzheimer's Disease?

- Amyloid Precursor Protein (APP)

8. What is the major genetic risk factor for Alzheimer's Disease?

- Apolipoprotein E variant e4 (ApoE4)

9. What are the microscopic features of Alzheimer's Disease?

- Neuronal loss & Gliosis
- Extracellular neuritic plaques
- Intracellular Neurofibrillary tangles
- Cerebral amyloid angiopathy

10. What is the common coexisting neuropathology seen with Alzheimer's Disease?

- Cerebral Amyloid Angiopathy

11. What is the clinical feature that can assist in distinguishing Frontotemporal lobe degeneration from Alzheimer's Disease?

- Temporal deterioration of language and frontal lobes degeneration and atrophy

12. What are the causes of Parkinsonism?

- Parkinson's disease, Viral encephalitis, Vascular (post-stroke), Environmental Toxin, Drug-Induced

13. What gene mutations cause autosomal dominant Parkinson's Disease?

- Point mutations and duplications of the gene encoding α -synuclein

14. What are the clinical features of Frontotemporal lobe degeneration (FTLD)?

- Temporal deterioration of language and frontal lobe degeneration and atrophy.
- These symptoms precede memory disturbances.

15. Which type of transmission (inheritance) is common in FTLD?

- Up to 40% of FTLD cases show autosomal dominant transmission of the disease.

16. What genes are associated with FTLD?

- FTLD is associated with mutations in genes encoding for microtubule-associated protein tau.

17. What are the clinical features of Huntington Chorea?

- Movement disorder progression
- An increased risk of suicide
- Early cognitive symptoms that may progress to severe dementia.

18. Is Huntington Chorea an autosomal dominant disease?

- Yes, Huntington Chorea is an autosomal dominant disease.

19. What causes Parkinson's Disease?

- Parkinson's disease itself (85%),
- Viral encephalitis
- Vascular factors (post-stroke)
- Environmental toxins
- Drug-induced factors (e.g., antipsychotic drugs).

20. Are there different forms of Parkinson's Disease? And what is it?

- Yes
- besides the sporadic form, there are also autosomal dominant and recessive forms of Parkinson's Disease.





1. What is Multiple Sclerosis?

- An autoimmune demyelinating disorder characterized by distinct episodes of neurologic deficits.

2. True or False: Multiple Sclerosis is more common in men than women.

- False

3. What is the main pathology of Multiple Sclerosis?

- Autoimmune disease caused by a loss of tolerance to myelin antigens.

4. Which cells are involved in the pathogenesis of Multiple Sclerosis?

- CD4⁺ T cells
- CD8⁺ T cells
- B cells.

5. What are the gross characteristics of Multiple Sclerosis lesions?

- Multiple, well-circumscribed
- Gray-tan plaques near the ventricles and in various areas of the CNS.

6. What is the microscopic appearance of active Multiple Sclerosis plaques?

- Evidence of ongoing myelin breakdown with abundant macrophages containing myelin debris.

7. What is the clinical course of Multiple Sclerosis?

- Relapsing and remitting episodes of neurologic impairment with gradual accumulation of deficits over time.

8. What imaging studies are mandatory for Multiple Sclerosis diagnosis?

- MRI brain

9. True or False: Oligoclonal bands are usually identified in the CSF of patients with Multiple Sclerosis.

- True

10. What are the main symptoms of Multiple Sclerosis?

- Fatigue, visual impairment, cognitive impairment, weakness, spasms, pain, and bowel and urinary dysfunction.

11. What is the primary lipid of myelin?

- The primary lipid of myelin is a glycolipid called "galactocerebroside".

12. What are demyelinating disorders?

- Demyelinating disorders are diseases of the nervous system in which the myelin sheath of neurons is damaged, impairing the conduction of signals in the affected nerves.

13. What is Krabbe disease?

- Krabbe disease is an autosomal recessive disorder characterized by a deficiency of Galactosyl ceramidase, resulting in the accumulation of unmetabolized lipids, leading to demyelination and severe progressive degeneration of motor skills.

14. What are primary diseases of myelin?

- Primary diseases of myelin refer to diseases within the CNS where axons are tightly ensheathed by myelin, and myelin is not formed properly or has abnormal turnover kinetics.

15. What are demyelinating diseases of the CNS?

- Multiple sclerosis (MS)
- Acute disseminated encephalomyelitis (ADEM)
- Progressive multifocal leukoencephalopathy (PML)

16. What is the pathogenesis of multiple sclerosis (MS)?

- Multiple sclerosis (MS) is an autoimmune disease caused by a loss of tolerance to self-proteins (myelin antigens) due to a combination of environmental and genetic factors.

17. Which genetic factors are related to multiple sclerosis?

- HLA-DR and genes encoding receptors for the cytokines IL-2 and IL-7

18. What is the disease risk if multiple sclerosis is present in a first-degree relative?

- 15-fold higher

Chapter 3 ?

CNS Trauma

1. What is the clinical syndrome of altered mental status secondary to head injury?

- Concussion

2. What is a contusion in traumatic head injury? And what are its types?

- Direct parenchymal injury to the brain
- Coup injury at the point of impact
- Contrecoup injury opposite the point of impact.

3. What is diffuse axonal injury? And what are its common locations?

- Injury to white matter with axonal swelling due to acceleration and deceleration
- Commonly occurring in the corpus callosum, periventricular white matter, and hippocampus.

4. What is the location and symptoms of an epidural hematoma?

- Between the skull and dura
- Symptoms include lucid interval before loss of consciousness and possible tentorial herniation.

5. What is the location and symptoms of a subdural hematoma?

- Between the dura and arachnoid
- Symptoms include slow venous bleeding with delayed onset of headache and confusion.

6. Which vessel is most commonly involved in an epidural hematoma?

- Middle meningeal artery

- 7. What is the condition called when a patient experiences a lucid interval before losing consciousness?**
 - "Talk and die" syndrome
- 8. True or False: A trans-tentorial herniation may occur in an epidural hematoma.**
 - True
- 9. What is the appearance of an epidural hematoma on a head CT scan?**
 - Biconvex-shaped hemorrhage that does not cross suture lines
- 10. What is the most common cause of a subdural hematoma?**
 - Tear of bridging veins drain the brain to Dural sinuses
- 11. True or False: Brain atrophy is a risk factor for subdural hematoma.**
 - True
- 12. True or False: A subdural hematoma can cross suture lines.**
 - False

Chapter 4 ?

Microvascular diseases of CNS

- 1. What is the most common cause of cerebral infarction?**
 - Arterial occlusion of the cerebral arteries by either thrombi or emboli
- 2. What is the term for small infarcts in deep non-cortical white matter or the brainstem that result from the occlusion of penetrating branches of cerebral arteries?**
 - Lacunar Infarcts
- 3. What causes lacunar infarcts?**
 - Occlusion of penetrating branches of cerebral arteries
- 4. What is the most common cause of thrombotic occlusion of the carotid or cerebral arteries?**
 - Atherosclerosis
- 5. What is the term for fluid-filled cavity demarcated by gliotic scar?**
 - Old cyst surrounded by gliotic scar
- 6. What causes cerebral infarction in herniation?**
 - Compression of cerebral arteries from outside
- 7. What is the microscopic change seen in cerebral infarction after years?**
 - Astrogliosis surrounding a cyst
- 8. What are the types of aneurysms?**
 - Congenital (Berry Aneurysm)
 - Mycotic
 - Atherosclerotic
 - Charcot-Bouchard Microaneurysms
- 9. What is the most common site for Congenital (Berry) aneurysms?**
 - The blood vessels around the circle of Willis
- 10. What is the etiology of Congenital (Berry) aneurysms?**
 - Congenital defects of the media at the arterial bifurcations
 - Abnormality in specific collagen subsets
 - Focal destruction of the internal elastic membrane resulting from hemodynamic alterations

11. What is the shape of an Atherosclerotic aneurysm?

- Fusiform

12. What causes Charcot-Bouchard Microaneurysms?

- Hypertension causes hyaline arteriolar sclerosis of the deep penetrating arteries

13. What are the effects of cerebral aneurysms?

- Increased intracranial tension
- Rupture with subsequent hemorrhage
- Hydrocephalus
- Thrombosis and calcification

14. What is the most common site for arteriovenous malformation (AVM)?

- Middle cerebral artery distribution

15. What is the most common type of cavernous malformation?

- Cavernous angioma

16. What is the type of vascular malformation characterized by capillary-sized vessels separated by normal neural parenchyma?

- Capillary telangiectasia

17. What is cerebral edema?

- The accumulation of excess fluid within the brain parenchyma

18. What are the two underlying mechanisms for the development of cerebral edema?

- Vasogenic edema
- Cytotoxic edema

19. What is the cause of vasogenic edema?

- Disruption of the normal blood-brain barrier

20. What is the cause of cytotoxic edema?

- Increase in intracellular fluid due to neuronal and glial cell membrane injury

21. True or False: Cerebral edema can be localized or generalized.

- True

22. What are the gross morphological features of an edematous brain?

- Softer than normal - Appears to "overfill" the cranial vault
- Flattened gyri
- Narrowed sulci
- Compressed ventricular cavities

23. What are the two general mechanisms by which the brain may be deprived of oxygen?

- Functional hypoxia
- Ischemia

24. What is a transient ischemic attack (TIA)?

- Ischemia due to tissue hypoperfusion, characterized by symptoms lasting less than 24 hours

25. What is the clinical term for acute-onset neurologic deficits resulting from hemorrhagic or obstructive vascular lesions?

- Stroke

26. What is multi-infarct dementia?

- A slow, stepwise decrease in mental function caused by cerebral atherosclerosis

27. What causes global cerebral ischemia?

- Severe systemic hypotension

28. True or False: Neurons are more susceptible to hypoxic injury than glial cells.

- True

29. What are the gross morphological changes seen in the brain in the setting of global ischemia?

- Swollen brain
- Wide gyri
- Narrowed sulci
- Poor demarcation between gray & white matter
- Pseudo laminar necrosis
- Border zone (watershed) infarcts

30. What are border zone infarcts?

- Wedge-shaped areas of infarction that occur in regions of the brain that lie at the most distal portions of arterial territories in global cerebral ischemia

31. True or False: Focal cerebral ischemia is caused by cerebral arterial occlusion.

- True

32. What is the most common cause of cerebral arterial occlusion?

- Thrombi or emboli

33. What are lacunar infarcts?

- Small infarcts in deep non-cortical white matter or the brainstem that result from the occlusion of penetrating branches of cerebral arteries

34. True or False: Venous occlusion is a common cause of cerebral infarction.

- False

Chapter 5**Pathology of CNS infections****1. What are the causes of meningitis?**

- Infection (bacterial, viral, tuberculous, spirochetal, or cryptococcal)
- Certain drugs and chemicals
- SLE, Meningeal carcinomatosis

2. What are the clinical features of meningitis?

- Fever, headache, neck stiffness,
- Photophobia
- Altered mental state
- Cranial nerve deficit

3. How is acute meningitis characterized?

- Inflammation of the leptomeninges with cerebrospinal fluid (CSF) in the subarachnoid space

4. What are the etiologies of acute meningitis?

- Pyogenic (meningococci, pneumococci, E. coli, Beta Strept)
- Aseptic (enteroviruses, measles, mumps, influenza)

5. What are the morphological changes in acute meningitis?

- Pus and fibrin in subarachnoid space
- Dilated vessels, neutrophils infiltrate
- Excess proteins, reduced glucose

6. What are the complications of acute meningitis?

- Spread to encephalitis, pyemia & septicemia
- Adrenal necrosis (Waterhouse-Friedrichsen syndrome)
- Fibrosis leading to cranial nerve compression and hydrocephalus

7. How is aseptic meningitis characterized?

- Normal glucose and protein levels in CSF
- Usually self-limiting with no complications

8. What are the etiologies of chronic meningitis?

- Tuberculosis
- Syphilis

9. What is a brain abscess?

- A localized collection of pus within the brain tissue

10. What are the etiologies of brain abscess?

- Pyogenic bacterial infections (sinusitis, otitis media, pyemia, suppurative lung diseases)

11. What are the morphological changes in brain abscess?

- Irregular cavitation (liquefaction) with surrounding edematous brain tissue, ring-enhancing lesions on CT/MRI
- Microscopic presence of liquefied tissue and neutrophils
- Necrotic wall in acute abscess and gliosis in chronic abscess

12. What are the complications of brain abscess?

- Pressure effects according to the site (could be fatal)
- Spread to other areas

13. What is viral encephalitis?

- Infection of the brain associated with meningeal inflammation (meningoencephalitis)

14. What are the morphological features of viral encephalitis?

- Perivascular and parenchymal mononuclear cell infiltrates
- Microglial nodules
- Neuronophagia
- Inclusion bodies in certain viruses (e.g., Negri bodies in rabies)

15. What is the direct effect of HIV-1 on the brain?

- HIV-1 can cause opportunistic infections or tumors (primary CNS lymphoma)
- It can also directly injure the brain, leading to aseptic meningitis, meningoencephalitis (AIDS dementia), and other CNS injuries.

16. What is the cause and results of progressive multifocal leukoencephalopathy (PML)?

- It is caused by the JC virus
- Resulting in demyelination of the white matter.

17. Where in the brain does HSV encephalitis commonly affect?

- Inferior and medial regions of the temporal lobes
- Orbital gyri of the frontal lobes.

18. Which fungi commonly cause encephalitis?

- Candida albicans
- Mucormycosis
- Aspergillus fumigatus
- Cryptococcus neoformans

19. How do most fungi invade the brain?

- By hematogenous dissemination
- By direct extension from sinuses

1. Which tumors are classified as primary tumors of the CNS?

- Gliomas
- Neuroepithelial tumors
- Meningiomas
- Tumors of the pineal region

2. Where do approximately 70% of brain tumors occur in adults?

- Supratentorially (within the cerebral hemispheres or coverings)

3. Where do approximately 70% of brain tumors occur in children?

- Infratentorially (within the cerebellum or brainstem)

4. What are the three most common neuroepithelial brain tumors in children?

- Pilocytic astrocytoma
- Medulloblastoma
- Ependymoma

5. What are the three most common types of spinal cord tumors in decreasing order of frequency?

- Schwannomas
- Meningiomas
- Gliomas (ependymoma, astrocytoma)

6. What is the key characteristic of diffuse glioma?

- Single-cell infiltration of normal brain tissue by tumor cells

7. What is a common acquired mutation associated with diffuse gliomas?

- Isocitrate dehydrogenase (IDH1 and IDH2)

8. True or False: Diffuse glioma is resectable.

- False

9. How diffuse glioma is diagnosed?

- Immunostaining for the mutated form of IDH1 is positive in diffuse glioma.

10. Where are diffuse astrocytic tumors usually found?

- Cerebral hemispheres

11. What are the three grades of diffuse astrocytic tumors?

- Diffuse Astrocytoma, WHO Grade II
- Anaplastic Astrocytoma, WHO Grade III
- Glioblastoma IV

12. What are the common clinical manifestations of diffuse astrocytic tumors?

- Seizures, headaches, and focal neurologic deficits

13. What is the prognosis of diffuse astrocytic tumors?

- Can be static for several years
- But can progress to rapid tumor growth with progression to anaplastic astrocytoma and glioblastoma and its prognosis is poor

14. What is the gross appearance of diffuse astrocytic tumors?

- Poorly defined, gray, infiltrative tumors
- Firm or soft and gelatinous; cystic degeneration may be seen.
- In glioblastoma, variation in the gross appearance of the tumor. some areas are firm and white, others are soft and yellow (the result of tissue necrosis)

15. What are the key microscopic features of diffuse astrocytic tumors?

- Increase in the number of glial cell nuclei
- Nuclear pleomorphism
- Glial fibrillary acidic protein (GFAP)-positive astrocytic cell processes

16. True or False: Anaplastic astrocytoma have regions that are more densely cellular and have greater nuclear pleomorphism.

- True

17. What are the histologic criteria for differentiating grades of diffuse astrocytoma?

- Diffuse astrocytoma: No appreciable mitotic activity
- Anaplastic astrocytoma: Mitoses present
- Glioblastoma: Necrosis and/or microvascular proliferation

18. What are the common clinical manifestations of oligodendrogliomas?

- Seizures

19. Where are oligodendrogliomas mostly located?

- Cerebral hemispheres, mainly in the frontal or temporal lobes

20. True or False: Oligodendrogliomas have a better prognosis than that of diffuse astrocytic tumors of similar grade.

- True

21. What is the key genetic finding in oligodendrogliomas?

- Codeletion of chromosomes 1p and 19q

22. What are the key microscopic features of oligodendrogliomas WHO grade II?

- Sheets of regular cells with rounded nuclei containing finely granular rounded chromatin surrounded by a clear halo of cytoplasm and anastomosing capillaries

23. What are the key microscopic features of oligodendrogliomas WHO grade III?

- More aggressive with higher cell density, nuclear anaplasia, and prominent mitotic activity

24. Where are pilocytic astrocytomas most commonly located?

- Cerebellum
- They can also involve the third ventricle
- The optic pathways
- Spinal cord
- Occasionally the cerebral hemispheres

25. What genetic alterations are associated with pilocytic astrocytomas?

- Activation of the mitogen-activated protein kinase (MAPK) pathway, including activating alterations of BRAF and inactivation of neurofibromatosis type 1 (NF1)

26. What is the gross appearance of pilocytic astrocytomas?

- Often cystic, with a mural nodule in the wall of the cyst

27. What are the key microscopic features of pilocytic astrocytomas?

- Low to moderate cellularity often with a biphasic pattern
- Bipolar cells with Rosenthal fibers
- Multipolar cells with microcysts
- Eosinophilic granular bodies in loose regions

28. What is plastic ependymomas?

- Ependymoma In the foramina of Luschka and Magendie to surround cranial nerves and vessels

29. Where do Ependymomas typically occur in children?

- They typically occur near the fourth ventricle.

30. What is the most common location for Ependymomas in adults?

- The spinal cord is the most common location for Ependymomas in adults.

31. Describe the gross pathology of Ependymomas.

- Ependymomas are well-circumscribed tumors that are soft and spongy
- Occasionally with gritty calcium deposits.

32. What are the microscopic features of Ependymomas?

- Uniform small cells, round to oval nuclei with granular chromatin
- Fibrillary matrix, and perivascular anucleate zones (pseudo rosettes).

33. What are the characteristics of Embryonal Neoplasms?

- Highly cellular
- Poorly differentiated
- Mitotically active.
- They tend to disseminate along cerebrospinal fluid.

34. Name some examples of Embryonal Neoplasms.

- Medulloblastoma
- Atypical teratoid/ rhabdoid tumor (AT/RT)
- Embryonal tumor with multilayer rosettes (PNETs).

35. What are the different types of meningiomas WHO grade I?

- Syncytial
- Fibroblastic
- Transitional
- Psammomatous

36. What are the features of atypical meningiomas (WHO grade II)?

- Increased mitotic activity
- Brain invasion
- At least three of the following features: increased cellularity, small cells with an increased nuclear- to-cytoplasmic ratio, prominent nucleoli, sheeting and foci of spontaneous necrosis.

37. What are the characteristics of anaplastic (malignant) meningiomas (WHO grade III)?

- Overtly malignant cytology and/or markedly elevated mitotic activity

38. What is the most common type of primary CNS lymphoma?

- Diffuse large B cell lymphoma

39. Which population is most commonly affected by primary CNS lymphoma?

- Immunosuppressed persons

40. What's the relation between primary CNS lymphoma and Epstein-Barr virus?

- Primary CNS lymphoma is positive for the oncogenic Epstein-Barr virus

41. What is the response of primary CNS lymphoma to chemotherapy?

- Relatively poor

42. Where are germ cell tumors most commonly found in the brain?

- Along the midline, particularly in the pineal and suprasellar regions

43. Is there a gender predominance for germ cell tumors in the pineal region?

- Yes, there is a strong male predominance

44. What is the most common primary CNS germ cell tumor?

- Germinoma

45. How do metastatic tumors reach the CNS?

- Via hematogenous route
- Direct invasion from adjacent anatomical structures

46. Where are the majority of brain metastases located?

- Cerebral hemispheres (80%)
- Cerebellum (15%)
- Brainstem (5-9%)

47. Which primary sites are most common for metastatic tumors to the CNS?

- Lung
- Breast
- Skin (melanoma)
- Kidney
- Gastrointestinal tract

48. What is the gross appearance of metastatic tumors in the CNS?

- Sharply demarcated masses with surrounding edema

49. What are the two familial tumor syndromes associated with an increased risk of CNS cancers?

- Tuberous Sclerosis
- Von Hippel-Lindau Disease

50. What is the underlying cause of von Hippel-Lindau Disease?

- Germline mutations of the VHL tumor suppressor gene

51. What is the key role of the VHL protein in von Hippel-Lindau Disease?

- Cellular oxygen sensing

52. What is the most common non-neoplastic lesion associated with von Hippel-Lindau Disease?

- Hemangioblastoma

53. What is the most common embryonal tumor of the CNS?

- Medulloblastoma

54. Where does medulloblastoma arise in the brain?

- Cerebellum or dorsal brain stem

55. What are the morphological variants of medulloblastoma?

- Classic
- Desmoplastic/nodular
- Medulloblastoma with extensive nodularity
- Large cell/anaplastic

56. What is the most common cytogenetic aberration in medulloblastomas?

- Loss of the short arm of chromosome 17 (17p)

57. What is the most favorable genetic subtype of medulloblastoma?

- WNT-activated, recognized by nuclear β -catenin

58. Where do hemangioblastomas commonly occur?

- Cerebellum (most common)
- Brainstem
- Spinal cord

59. Are hemangioblastomas usually sporadic or associated with von Hippel-Lindau Disease?

- Both, with sporadic cases accounting for 70% and VHL-associated cases accounting for 30%

60. What is the appearance of a hemangioblastoma?

- Well circumscribed, partly cystic and highly vascular lesion or completely solid
- Occasionally, it's yellow due to lipid
- Its classic appearance is Cyst with a solid vascular nodule

61. What is the classic microscopic appearance of a hemangioblastoma?

- Large, vacuolated neoplastic stroma cells with vascular cells
- Numerous thin-walled vessels
- Hemorrhage may occur

62. What is the origin of hemangioblastoma stromal cells?

- Unknown



6

Section



Pharmacology

1. Which drugs are classified as opioid antitussives?

- Codeine
- Dextromethorphan

2. Name some strong pure agonists opioids

- Morphine
- Pethidine
- Fentanyl
- Heroin
- Methadone

3. What are the partial agonists opioids

- Buprenorphine

4. What is the classification of codeine? And what's its uses?

- Moderate analgesic.
- Combined with paracetamol or aspirin in moderate pain.
- Also an antitussive.

5. Which drugs are classified as opioid antidiarrheals?

- Loperamide
- Diphenoxylate

6. What is the classification of propoxyphene? And what's its uses?

- Weak analgesic.
- Combined with paracetamol or aspirin in moderate pain.

7. What are the sources of opioid analgesics?

- Plant: morphine - codeine
- Synthetic: fentanyl - methadone - heroin - pethidine
- Endogenous: endorphins - enkephalins - dynorphins (opio-peptides)

8. How do opioids work?

- Opioids (directly or through release of opio-peptides) activate opioid receptors (Mu, kappa & delta) present in the CNS and periphery, producing morphine-like effects.

9. Which areas of the body are affected by opioid receptors?

- Afferent pain-conducting fibers (peripheral analgesia)
- Spinal cord (spinal analgesia)
- Brain stem, thalamus & cerebral cortex (supraspinal analgesia)
- Limbic system (euphoria & emotional response to pain)

10. What is the mechanism of euphoria caused by opioids?

- Euphoria is due to increased dopamine in the nucleus accumbens, which is formed in the ventral tegmental area (VTA).

11. What is the cellular mechanism of opioid action?

- Activation of opioid receptors inhibits neuronal activity through:
 - inhibition of Ca^{2+} entry
 - stimulation of K^{+} outflux, leading to neuronal inhibition.

12. What are the pharmacological actions of morphine?

- Analgesia, Sedation, Euphoria, Vasodilation, Respiratory depression, Cough suppression, Uterine tone, Histamine release.

13. What are the adverse effects of morphine?

- Convulsion, Chest wall rigidity,
- GI upset, Vomiting, Constipation
- Tolerance
- Urinary retention
- Uterine relaxation, Asphyxia neonatorum, Delayed labor,
- Hypotension.

14. What are the main effects of morphine?

- Analgesia
- Sedation
- Euphoria

15. What inhibitory effects does morphine have?

- Hypotension
- Respiratory depression
- Cough suppression
- Decreased uterine tone

16. What stimulatory effects does morphine have?

- Miosis
- Vomiting
- Increased biliary pressure
- Urine retention (also ↑ADH secretion)
- Constipation
- Hypotension, itching & bronchospasm

17. In which four types of severe pain is morphine contraindicated or not recommended?

- Acute undiagnosed abdominal pain
- normal labor pain
- closed head injury
- biliary colic

18. What are the differences between pethidine (meperidine) and morphine?

- Less constipation and neonatal respiratory depression
- does not delay labor
- less spasm in sphincter of Oddi
- less risk of convulsions

19. What are the advantages of partial agonists over pure agonists in opioid treatment?

- Less addiction
- alternative treatment for opioid addicts

20. What are the therapeutic uses of pure antagonists?

- Reversal of opioid overdose

21. What are the therapeutic uses of naloxone?

- Acute opioid toxicity
- Asphyxia neonatorum

22. What are the therapeutic uses of naltrexone?

- Opioid abuse
- Alcohol abuse

23. When should naltrexone be given to opioid addicts?

- After full detoxification, to avoid withdrawal syndrome

24. What are the drug interactions of opioids with other CNS depressants?

- Additive depressant effect, especially in old patients

25. What is the interaction between pethidine or tramadol and MAOI?

- Serotonin syndrome (Hyperthermia + convulsion)

26. How should acute morphine toxicity be treated?

- Support respiration
- Naloxone (IV) repeated when necessary

27. What are the therapeutic uses of naltrexone?

- Opioid abuse
- Alcohol abuse

28. How does naltrexone prevent relapse in addicts?

- Blocks euphoria of opioids that leads to Loss of desire to take drug

29. True or False: Naltrexone can be given to opioid addicts before full detoxification. And why?

- False
- To avoid withdrawal syndrome

Chapter 2 ? Drug Therapy of Parkinson's Disease

1. What is the cause of iatrogenic Parkinson's disease?

- Long-term use of D2 antagonists that cross the blood-brain barrier such as antipsychotics

2. What is the aim of drug therapy in Parkinson's disease?

- To restore balance between dopamine and acetylcholine activity in the basal ganglia

3. What is L-Dopa?

- L-Dopa is a dopamine precursor that can cross the blood-brain barrier (BBB).

4. How is L-Dopa converted to dopamine?

- L-Dopa is decarboxylated by the central dopa decarboxylase enzyme to dopamine (DA) in the dopaminergic neurons.

5. What is the purpose of carbidopa in L-Dopa therapy?

- Carbidopa, a peripheral decarboxylase enzyme inhibitor, is given with L-Dopa to prevent its conversion to dopamine peripherally.

6. What causes fluctuations in response of L-dopa?

- Short half-life
- fluctuation in L-dopa level

7. How can on-off effect be managed?

- Add Entacapone
- or Use Apomorphine (SC)

8. What is end-of-dose akinesia?

- Gradual loss of the useful effect (wearing off) before the time of the next dose

9. How can end-of-dose akinesia be managed?

- Add Selegiline
- Use dopamine agonists

10. True or False: L-Dopa can cross the blood-brain barrier (BBB).

- False. L-Dopa can cross the BBB, while dopamine cannot.

11. What is the mainstay of therapy for Parkinson's disease?

- L-Dopa is the most effective antiparkinson's drug and is considered the mainstay of therapy.

12. What are the peripheral adverse effects of L-Dopa?

- Nausea, vomiting, peptic ulcer
- postural hypotension, and arrhythmias are common adverse effects of L-Dopa.

13. What is dyskinesia?

- Dyskinesia refers to abnormal involuntary head, lip, and tongue movements associated with L-Dopa therapy.

14. What are the management options for fluctuations in response of L-Dopa therapy?

- adding entacapone or using apomorphine (SC) for on-off effect
- adding selegiline or dopamine agonists for end-of-dose akinesia.

15. What is the advantage of using dopamine agonists compared to L-Dopa?

- Dopamine agonists have less fluctuation in response, longer half-life
- can be taken transdermally or orally with higher bioavailability.

16. What is the alternative drug used in emergency situations for Parkinson's disease? And why is it used?

- Apomorphine is an alternative drug used in emergency situations
- to manage off time and akinetic episodes.

17. What is the mechanism of action of selegiline?

- Selegiline is a selective MAO-B inhibitor that delays the breakdown of dopaminergic neurons
- prolongs the action of L-Dopa.

18. What can cause orange urine?

- Entacapone

19. Which drugs belong to the Anticholinergic class?

- Benztropine
- Trihexyphenidyl

20. What is the mechanism of anticholinergic drugs?

- Block central cholinergic activity and restore DA/Ach balance
- Improve tremors (mainly) more than rigidity & bradykinesia.

21. What are the therapeutic uses of anticholinergic drugs?

- Drug-induced Parkinsonism
- Parkinson's disease (especially in the young)

22. What are the side effects of anticholinergic drugs?

- Similar to atropine but less severe (dry mouth, constipation, confusion)
- Used Cautiously in old patients (avoid in patients over 70)

23. What are the first choice agents in early Parkinson's disease?

- L-dopa/carbidopa
- Non-ergot dopamine agonists (Rotigotine)
- MAO-B inhibitors

1. What are the uses of Benzodiazepines? (Just 3)

- Generalized anxiety disorders
- Secondary anxiety due to psychiatric disease
- Panic attacks
- Insomnia, nightmares and night terrors
- Epilepsy
- Muscle spasticity
- Balanced anesthesia
- Conscious sedation
- Pre-anesthesia
- Alcohol withdrawal

2. How do Benzodiazepines affect sleep?

- Shorten sleep latency
- Increase stage 2 NREM sleep
- Decrease stage 3 & 4 NREM sleep
- Decrease REM sleep

3. What is the classification of Benzodiazepines based on duration of action?

- **Ultra-short acting:** Midazolam
- **Intermediate acting:** Alprazolam, Lorazepam
- **Long acting:** Diazepam, Clonazepam

4. What are the pharmacokinetics of Benzodiazepines?

- Hepatic oxidation with cytochrome P450 (CYP3A4), except Lorazepam

5. What are the selective drugs for anxiety and insomnia?

- Zolpidem
- Ramelteon
- Buspirone

6. What are the advantages of zolpidem over Benzodiazepines?

- Rapid onset
- Less tolerance & dependence
- Less adverse effects on cognitive function, hangover
- REM sleep suppression

7. What are the disadvantages of buspirone?

- Delayed onset (2 weeks)
- Not effective in insomnia or panic attacks

8. What are the general guidelines for therapy with sedative-hypnotics?

- Drug therapy is indicated only in severe acute anxiety or short-term treatment of acute insomnia
- Start with a small dose for a short time to avoid abuse and dependence
- Terminate slowly to avoid withdrawal syndrome
- Adjust dose in hepatic dysfunction
- Decrease doses in the elderly to avoid risk of falling

9. How do benzodiazepines facilitate the action of GABA?

- Binding to the BZD receptor site of GABA-A receptor complex and affinity of GABA to bind to its receptors facilitating the inhibitory effects of GABA

10. What is the mechanism of action of zolpidem?

- Binds to the BZD1 receptor subtype (involved in sedation & hypnosis) facilitating the inhibitory effects of GABA

11. What are the uses of benzodiazepines?

- GAD (all except ultra-short)
- 2ry anxiety due to psychiatric disease
- Panic attack: Alprazolam
- Insomnia
- Epilepsy (NOT 1st choice)

12. What are the adverse effects of benzodiazepines when it's used as anxiolytic?

- Abuse, tolerance, dependence
- paradoxical excitement

13. What are the effects of benzodiazepine withdrawal if it's used as hypnotic and sedative?

- Insomnia, nightmares

14. What are the motor-related side effects of benzodiazepines?

- Ataxia and motor incoordination

15. What are the two main effects of benzodiazepines when combined with alcohol or other CNS depressants?

- Additive CNS depression

16. What is the primary use of benzodiazepines in panic attacks?

- Anxiolytic and has antidepressant effect

17. What is the best benzodiazepine for treating epilepsy?

- Clonazepam

18. What is the mechanism of action of Flumazenil?

- Competitive antagonist to Benzodiazepine receptors

19. What are the therapeutic uses of Flumazenil?

- Benzodiazepine toxicity
- Termination of Benzodiazepine effect after anesthesia
- Hepatic encephalopathy



1. What are the classifications of local anesthetics?

- Ester L.A.
- Amide L.A.

2. What are the members of Ester L.A.?

- Procaine
- Benzocaine
- Cocaine

3. What are the members of Amide L.A.?

- Lidocaine
- Bupivacaine
- Ropivacaine

4. How does the rate of firing affect the effectiveness of local anesthetics?

- Blockade of Na^+ channels by LA is use-dependent, making it is more effective in rapid-firing nerves (sensory nerves) than slow firing nerves (motor nerves).

5. Which nerve fibers are more sensitive to local anesthetics?

- Small nerve fibers conducting pain, temperature, and autonomic activity are more sensitive to LA than large fibers conducting pressure and motor impulses.

6. What is the mechanism of action of local anesthetics?

- They block Na^+ channels in nerve membranes from inside, preventing the generation and propagation of action potentials.

7. Why are vasoconstrictors added to local anesthetics?

- Vasoconstrictors are added to prolong the duration of action and reduce systemic adverse effects and bleeding tendency

8. What factors affect the activity of local anesthetics?

- Plasma protein binding
- lipid solubility
- pKa of the drug
- addition of vasoconstrictors
- extracellular and intracellular pH.

9. What are the therapeutic uses and methods of administration of local anesthetics?

- Surface anesthesia, Minor surgeries
- Infiltration, Minor surgeries
- nerve block, for dentistry
- IV regional anesthesia
- spinal anesthesia, leg and pelvis surgery
- Epidural anesthesia, for painless labor

10. What are the characteristics of local anesthetics with ester L.A. classification?

- Short duration of action
- Less systemic toxicity
- More allergy (PABA metabolite)
- Duration of action is increased in genetic enzymedeficiency (Pseudo CHE)

11. Name a local anesthetic with the classification of lipophilic aromatic I. Amide L.A. and has an intermediate duration.

- Lidocaine 4%

1. What is the purpose of preanesthetic medications?

- To induce amnesia, reduce anxiety, and facilitate the induction of anesthesia.
- Suppress autonomic reflexes
- Potentiate weak anesthetics

2. What is the role of IV anesthetics and succinylcholine in general anesthesia?

- IV anesthetics are used for rapid induction
- Succinylcholine is used for intubation.

3. What are the two main types of inhalation anesthetics used for maintenance?

- Gas: Nitrous oxide (N_2O)
- Volatile liquids (halogenated drugs such as Halothane, Flurane).

4. What are the main types of intravenous anesthetics used for induction?

- Barbiturates (Thiopental Na^+)
- Non-barbiturates (Propofol, Ketamine, opioid analgesics, and BZDs).

5. What are general anesthetics used for?

- Inducing reversible loss of consciousness, analgesia, amnesia, and skeletal muscle relaxation.

6. What is the purpose of balanced anesthesia?

- To achieve all requirements for general anesthesia and minimize adverse effects of individual drugs.

7. List the components of balanced anesthesia.

- Preanesthetic medication
- IV anesthetics plus succinylcholine
- Inhalation anesthetics and NMBs (if needed) during operation

8. How do general anesthetics inhibit neuronal activity?

- They facilitate inhibitory effects of GABA at GABA receptors
- Inhibit excitatory effects of glutamate at NMDA receptors.

9. What is MAC (Minimal Alveolar Concentration)?

- The minimal concentration of anesthetic gas that produces immobility in 50% of patients when exposed to a painful stimulus.

10. What factors can increase MAC?

- Adjuvants (Opioids, sedative hypnotics, clonidine)
- Elderly
- Drugs that increase sympathetic activity
- Alcohol abuse

11. What is dissociative anesthesia?

- A hypnotic state with dissociation, amnesia, and analgesia, where patients are irresponsive to commands. It is induced by ketamine.

12. What is conscious sedation?

- A state of amnesia, sedation, and analgesia without complete loss of consciousness. Patients can respond to commands. It is induced by Propofol alone or midazolam + fentanyl.

13. What is deep sedation?

- A state of unconsciousness that is indistinguishable from anesthesia.

14. What are the disadvantages of halothane as an inhalation anesthetic?

- Hepatotoxicity
- Heart-related adverse effects
- Hypotension
- Malignant hyperthermia
- Uterine relaxation leading to bleeding
- Increased intracranial pressure.

15. What are the advantages and disadvantages of nitrous oxide (N_2O) as an inhalation anesthetic?

- **Advantages:**
 - » Strong analgesic
 - » Safe inhalation
- **Disadvantages:**
 - » Prolonged use can lead to hypoxia
 - » Euphoria
 - » Chronic exposure can cause megaloblastic anemia and abortion.

16. What are the advantages and disadvantages of ketamine as an intravenous anesthetic?

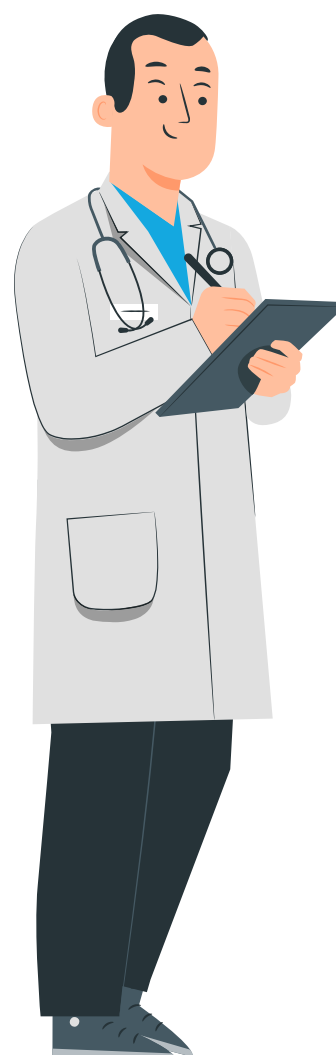
- **Advantages:**
 - » Strong analgesic
 - » Cardiovascular stimulant
 - » bronchodilator
- **Disadvantages:**
 - » Emergence phenomenon (hallucination and dissociation upon recovery)
 - » Increases blood pressure
 - » Increased intracranial pressure

17. What are the advantages and disadvantages of desflurane as an inhalation anesthetic?

- **Advantages:** Rapid onset and recovery
- **Disadvantages:** Respiratory irritation

18. What are the advantages and disadvantages of sevoflurane as an inhalation anesthetic?

- **Advantages:** Rapid onset and recovery, no respiratory irritation Preferred inhalation agent for induction
- **Disadvantages:** nephrotoxicity



1. What are the classic antiepileptic drugs?

- Phenytoin
- Carbamazepine
- Valproate
- Ethosuximide

2. What are the newer antiepileptic drugs?

- Lamotrigine
- Levetiracetam
- Oxcarbazepine
- Topiramate
- Gabapentin
- Pregabalin

3. What are the antiepileptic drugs that block presynaptic Ca^{++} channels?

- Gabapentin
- Pregabalin
- Lamotrigine

4. What are the antiepileptic drugs that block Na^{+} channels?

- Classic drugs
- Lamotrigine
- Oxcarbazepine
- Topiramate
- Levetiracetam

5. What are the antiepileptic drugs that block T type of Ca^{++} channels in thalamus?

- Ethosuximide
- Valproate

6. What is the effect of antiepileptic drugs on GABA?

- **Facilitate GABA action:** Benzodiazepines, Topiramate, Barbiturates
- **Increase GABA Level:** Valproate - Decrease GABA Breakdown: Vigabatrin

7. Which antiepileptic drugs can act with more than one mechanism?

- Valproate
- Topiramate
- Lamotrigine

8. What are the first-line antiepileptic drugs for different types of seizures?

- **Partial (Tonic/Clonic):** Carbamazepine, Lamotrigine
- **Absence:** Ethosuximide
- **Myoclonic:** Valproate

9. What are the second-line antiepileptic drugs for different types of seizures?

- **Partial (Tonic/Clonic):** Levetiracetam
- **Absence:** Carbamazepine, Levetiracetam, Lamotrigine
- **Myoclonic:** Levetiracetam, Topiramate

10. What are the adverse effects of carbamazepine?

- Less learning ability in children
- Enzyme inducer (Metabolism of antiepileptics, warfarin & other drugs)
- Agranulocytosis - Leukopenia
- Teratogenicity
- Water intoxication & Dilutional hyponatremia (Avoid with vasopressin)

11. What are the adverse effects of valproate?

- Hypersensitivity (Skin rash)
- GIT Disturbances (Start with small doses after meal)
- Fine hand tremors, Ataxia, Less sedative
- Enzyme inhibitor (Metabolism of antiepileptics & other drugs)
- Thrombocytopenia
- Teratogenicity (Add folic acid supplement)
- Others (Hair loss, polycystic ovary, increased appetite, weight gain)

12. Which antiepileptic drug is known to cause megaloblastic anemia?

- Valproate

13. True or False: Unpredictable serum levels should be monitored for Phenytoin.

- True

14. What is a specific adverse effect of Lamotrigine?

- Fatal dermatitis (Stevens-Johnson Syndrome)

15. What are the other uses of Antiepileptic drugs?

- Mood stabilizers in bipolar depression
- Antiarrhythmic
- Prophylactic in migraine
- Neuropathic pain

16. Which drugs are used for rapid control of status epilepticus?

- BZDS (Midazolam or Diazepam)

17. What is the first-line treatment for maintaining control in status epilepticus?

- Fosphenytoin

18. Which antiepileptic drugs are preferred in the elderly?

- Carbamazepine
- Lamotrigine
- Gabapentin

Chapter 5 ?

Migraine

1. What drugs are used to terminate acute migraine attacks?

- Mild analgesics (NSAIDs & acetaminophen)
- Triptans
- Ergot
- Antiemetic
- Benzodiazepine and opioids

2. How does specific antimigraine therapy prevent stretching of pain nerve endings?

- Vasoconstriction of dural vessels

3. What are the preparations of triptans?

- Sumatriptan: prototype (oral, SC, nasal spray)

4. What are the adverse effects of specific antimigraine therapy related to routes?

- Allergy in Injection site (with SC)
- unpleasant taste (with intranasal)

5. What are the adverse effects of triptan (specific antimigraine therapy) related to vasoconstriction?

- Chest pressure & pain (resolves spontaneously in 30 minutes)

6. What are the adverse effects of triptan (specific antimigraine therapy) related to sensory nerve transmission?

- Paresthesia, warmth, drowsiness, malaise

7. What precautions should be taken with specific antimigraine therapy?

- 1st dose given cautiously in diabetics, hypertensive
- Pregnancy
- With serotonergic drugs

8. What is the risk of combining triptan with serotonergic drugs?

- Serotonin syndrome

9. What is the mechanism of action of triptans?

- Activates 5HT 1B/1D receptors on presynaptic trigeminal nerve endings, leading to the release of vasodilatation
- Vasoconstriction of dural vessels

10. What is the mechanism of action of ergots?

- 5HT 1B/1D agonist (similar to triptans)

11. What are the adverse effects of ergots?

- GIT symptoms (nausea, vomiting, diarrhea)
- Chest pressure & pain
- Vasospasm leading to gangrene

12. What are the contraindications of ergots?

- Contraindicated in pregnancy
- Peripheral vascular disease
- Should not be used for a long time (can cause valvular heart disease)

13. Which antiemetic is used as monotherapy in emergency migraine cases?

- Parenteral metoclopramide (D_2 antagonist)

14. How is oral metoclopramide used in migraine treatment?

- It is added to other antimigraine drugs (ergotamine) to increase their absorption by its prokinetic effect

15. When are benzodiazepines & opioids used in migraine treatment?

- In resistant cases, but they may lead to habituation & rebound headache

16. What are the drugs used for migraine prophylaxis?

- Beta blockers (propranolol and timolol)
- Antidepressants (amitriptyline)
- Anticonvulsants (valproate & topiramate)
- Ca^{++} channel blockers (nifedipine, verapamil)

17. What is the choice of therapy in acute migraine attacks based on severity?

- Mild/Infrequent: Mild analgesics (NSAIDs)
- Outpatient with moderate to severe: Triptans
- Prolonged/frequent: Ergotamine
- Inpatients (emergency) with severe attack: IV antiemetic (metoclopramide). If not effective, add IV dihydroergotamine



1. What is the mechanism of action of antidepressants?

- Increase brain biogenic amines (norepinephrine and serotonin) within 12 hours, leading to downregulation of brain β and 5-HT₂ receptors (delayed for 2-4 weeks)
- increase brain-derived neurotrophic factor essential for survival and function of neurons

2. What is the mechanism of action of MAO inhibitors?

- Inhibit metabolism of biogenic amines inside cytoplasm of nerve endings, resulting in increased stores available for release by exocytosis

3. What is the mechanism of action of amine pump inhibitors?

- Inhibit reuptake of biogenic amines into neurons, resulting in their accumulation in the synaptic cleft and enhancing their effect at post-synaptic receptors

4. What are the different classes of amine pump inhibitor?

- Tricyclics (TCA)
- selective serotonin reuptake inhibitors (SSRI)
- serotonin norepinephrine reuptake inhibitors (SNRI)
- dopamine and norepinephrine reuptake inhibitors (NDRI)

5. What are the drug interactions with tricyclic antidepressants?

- Alcohol and anesthetics (sedation)
- hypotensive drugs (potentiate postural hypotensive effects of TCAs)
- sympathomimetics (TCAs inhibit uptake of biogenic amines)
- MAOIs (may induce acute serotonin syndrome)
- SSRIs (inhibits metabolism of TCA, leading to increased TCA toxicity)

6. What is the treatment for TCA toxicity?

- NaHCO₃
- Diazepam

7. Is hemodialysis recommended for TCA toxicity treatment?

- No

8. What is the mechanism of action of TCAs?

- They block the reuptake of NE and 5-HT, as well as autonomic receptors (H₁, M, α).

9. What are the adverse effects of MAOIs?

- Postural hypotension and atropine-like adverse effects
- Insomnia, shift to mania, and weight gain
- Serious food and drug interactions

10. What are the precautions with MAOIs?

- Warn patients about serious food (rich in tyramine) and drug interactions
- Avoid using TCAs or SSRIs within 2 weeks of stopping MAOIs (washing out period)
- Avoid MAOIs in the elderly due to their postural hypotension effect and atropine-like actions

11. What are the therapeutic uses of antidepressants? (Just 3)

- Major depression
- Urinary incontinence
- Nocturnal enuresis
- Post-traumatic stress disorders
- Anxiety disorders
- Eating disorders
- Chronic neuropathic pain
- Smoking cessation

12. What is the mechanism of action of bupropion?

- It is a melatonin MT₁ and MT₂ receptor agonist and a 5HT_{2c} receptor antagonist, leading to increased NE and DA release.

13. Which class of antidepressants is reserved for atypical depression associated with anxiety and phobia?

- MAOIs

14. Which class of antidepressants is the first choice for initial therapy?

- SSRIs

15. Which class of antidepressants is reserved for resistant cases?

- TCAs

16. In which conditions are MAOIs the second choice?

- Refractory major depression

17. Which antidepressants are potent enzyme inhibitors?

- Fluoxetine
- paroxetine

18. Which drugs can be added in resistant cases?

- Atypical antipsychotics
- Lithium
- T3

19. Which antidepressants are considered atypical?

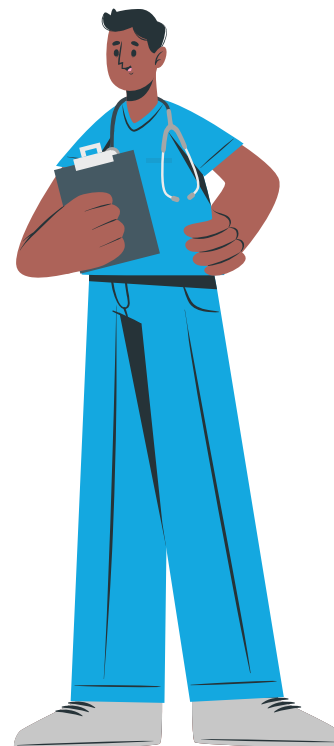
- Mirtazapine
- Bupropion

20. What is the sedating antidepressant?

- Paroxetine

21. What is the activating antidepressant?

- Fluoxetine



Chapter 7 ? Antimicrobial therapy for meningitis

1. How is Drug Induced Aseptic Meningitis treated?

- Immediate cessation of suspected drugs

2. Which antibiotics achieve therapeutic CSF level even without meningeal inflammation?

- Sulfonamides, Trimethoprim, Chloramphenicol, Rifampicin

3. Which antibiotics achieve therapeutic CSF level only with meningeal inflammation?

- β -lactam, Monobactam, Vancomycin, quinolones, Acyclovir

4. Which antibiotics do not achieve therapeutic CSF level even with inflammation?

- Aminoglycosides, 1st & 2nd Generation of cephalosporines, Amphotericin B

5. What is the initial management for bacterial meningitis?

- Immediate dexamethasone + Empiric antibiotic

6. What is the aim of dexamethasone therapy in meningitis?

- Improve outcome by reducing inflammation-induced brain edema and complications

7. What antibiotics are used for empirical therapy in suspected bacterial meningitis in a one month case?

- Ampicillin + cefotaxime or Ampicillin + Gentamicin

8. When should empirical therapy be continued until?

- Definitive diagnosis by lumbar puncture (2-3 days)

9. What antibiotics are used in intrathecal injection for nosocomial meningitis?

- Aminoglycosides, Polymyxin B, and Colistin

10. What antibiotics should be avoided in intrathecal injection for meningitis?

- Penicillins & cephalosporins (beta lactams)

11. What is the choice of antibiotic in bacterial meningitis for patients <1 month?

- Vancomycin

12. What is the triple therapy for cases with patients > 50 years or immunocompromised?

- Vancomycin + Ceftriaxone +Ampicillin

13. What is the alternative therapy for cases except in newborn (< 1 month)?

- Vancomycin + Meropenem

14. Which antibiotic is used in all cases of bacterial meningitis except in children < 1 month?

- Vancomycin

15. What is the empirical therapy for bacterial meningitis in children < 1 month?

- Ampicillin + cefotaxime or Ampicillin + Gentamicin

16. Which antibiotic is used in all cases of bacterial meningitis except in surgical interference in the brain?

- Ceftriaxone (3rd generation cephalosporines)

17. Which pathogen-specific antibiotic therapy is used for *Neisseria meningitidis*?

- Penicillin G (or Ampicillin) if MIC is low (< 0.1 ug/ml)
- Ceftriaxone if MIC is high (> 0.1 ug/ml)

18. What are the alternative antibiotics for *Neisseria meningitidis*?

- Moxifloxacin, Meropenem, Chloramphenicol (LAST choice)

19. What is the prophylactic antibiotic for close contact with *Neisseria meningitidis*?

- Rifampicin (for 2 days) or Ceftriaxone (single dose) or Ciprofloxacin (single dose)

20. Which antibiotic is used for Methicillin susceptible *Staphylococcus Aureus*?

- Nafcillin

21. Which antibiotic is used for *Streptococcus Pneumoniae* if MIC is low (<0.1 ug/ml)?

- Penicillin G (or Ampicillin)

22. Which antibiotic therapy is used for *Streptococcus Pneumoniae* with both penicillin and ceftriaxone MIC being very high?

- Ceftriaxone + Vancomycin (Add Rifampicin in the beginning of therapy)

23. Which antibiotic is used for Methicillin resistant *Staphylococcus Aureus* (MRSA)?

- Vancomycin

24. What is the alternative antibiotic for MRSA?

- Daptomycin or Linezolid

25. Which antibiotic is used for *Haemophiles Influenzae*?

- Ceftriaxone (if β -lactamase producing)
- Ampicillin (if β -lactamase -ve)

26. What is the prophylactic antibiotic for close contact with *Haemophiles Influenzae*?

- Rifampicin (for 4 days)

27. Which antibiotic is used for *Listeria monocytogenes* in meningitis?

- Ampicillin + Gentamicin (combination therapy for first 10 days, then monotherapy with ampicillin)

28. What is the treatment for Group B streptococcus (GBS) meningitis?

- Ampicillin (or Penicillin G)

29. What are the adverse effects of Ceftriaxone?

- Pseudo cholelithiasis (drug complexes with calcium and mimics gallstones)

30. What are the adverse effects of Moxifloxacin?

- QT interval prolongation & CNS adverse effects (headache, dizziness)

31. What are the adverse effects of Chloramphenicol?

- Agranulocytosis (bone marrow depression), Neurotoxicity & grey baby syndrome

32. What are the adverse effects of Penicillin?

- Hypersensitivity reaction

33. What are the adverse effects of Aminoglycosides?

- Nephrotoxicity & ototoxicity

34. What is the management of Non-Bacterial Meningitis caused by Fungal Infection?

- Candida meningitis: Liposomal Amphotericin-B + 5-flucytosine
- Aspergillus meningitis: Voriconazole

35. Which drugs are used in the first line therapy for Tuberculous Meningitis?

- Rifampicin, Isoniazid, Pyrazinamide, Ethambutol & Streptomycin

36. What is the recommended treatment for Herpes Simplex meningitis?

- Acyclovir

37. Should dexamethasone be taken before or with antimicrobial therapy in meningitis?

- Before or with antimicrobial therapy

38. What should be done if there is a delay in lumbar puncture in meningitis?

- Do not postpone the first dose of antimicrobial therapy

Chapter 8**Drug therapy for neurodegenerative and myelene disease****1. What is the aim of management for Alzheimer disease?**

- Increase cholinergic activity in the brain (by inhibiting Cholinesterase enzyme)
- Reduce excitotoxicity action of overstimulation of NMDA receptors (by using NMDA receptor antagonists)

2. What are the cholinesterase enzyme inhibitors used in the treatment of Alzheimer disease?

- Donepezil
- Rivastigmine
- Galantamine

3. What is the mechanism of action of cholinesterase enzyme inhibitors?

- Inhibits Acetylcholine esterase > increases cholinergic transmission in the brain

4. What is the mechanism of action of memantine?

- Noncompetitive NMDA receptor antagonist and protects neurons from glutamate-induced excitotoxicity

5. What is Namzaric?

- Combination of Memantine ER + Donepezil, taken once daily

6. What is the management approach for Multiple Sclerosis?

- Treatment of acute episodes (exacerbations) with corticosteroids
- Disease Modifying Drugs (DMDs)
- Symptomatic treatment

7. What is the treatment for an acute episode (exacerbation) of MS?

- Methyl prednisolone: 1 gm/day for 5 days IV (Large dose)
- followed by oral prednisone with gradual decrease

8. What are the first-line therapy drugs used in the treatment of MS?

- Interferon β 1a & Interferon β 1b

9. What are the adverse effects of Interferon β 1a and Interferon β 1b?

- Flu-like syndrome
- local injection site reaction
- depression
- bone marrow depression

10. What is the mechanism of action of interferon in the treatment of MS?

- Immunomodulatory effects, reduce inflammation that leads to demyelination of axon sheaths

11. What is the route of administration for Fingolimod?

- Oral daily

12. Which receptor does Fingolimod bind with on the surface of T lymphocytes?

- S1P receptors

13. What is the mechanism of action of Fingolimod?

- T-cell sequestration in lymph nodes
- Decrease T lymphocyte proliferation and migration to CNS
- Decrease neural inflammation and demyelination

14. What is the adverse effect of Fingolimod related to patients with diabetes?

- First dose bradycardia
- Macular edema
- Increase risk of infarctions

15. Which type of drugs are indicated to reduce relapse rate and accumulation of disability in multiple sclerosis?

- Disease Modifying Drugs (DMDs)

16. What is the mechanism of action of Natalizumab?

- Blocks the adhesion of T-lymphocytes to endothelial cells of BBB

17. What are the adverse effects of Natalizumab?

- Allergic reaction
- anti-natalizumab antibodies
- progressive multifocal leukoencephalopathy (PML)

18. Which drugs are used in Amyotrophic lateral sclerosis?

- Riluzole

19. What is the mechanism of action of Riluzole?

- Blocks presynaptic voltage gated Na^+ channels and decrease release of glutamate
- Increase reuptake of glutamate in glial cells
- Weak NMDA receptor antagonist





1. What is the mechanism of action of typical antipsychotics?

- Blocking D_2 receptors in the mesolimbic system improves positive symptoms

2. What receptors do atypical antipsychotics block in addition to D_2 receptors?

- 5HT_{2A} receptors

3. What are the typical antipsychotics?

- Chlorpromazine
- Thioridazine
- Haloperidol
- Fluphenazine

4. What are the atypical antipsychotics?

- Risperidone
- Aripiprazole
- Clozapine
- Olanzapine
- Quetiapine

5. What are the extrapyramidal adverse effects of antipsychotics?

- Akathisia
- Dystonia
- Parkinsonism
- Tardive Dyskinesia
- Neuroleptic malignant syndrome

6. What are the therapeutic uses of antipsychotics?

- Antipsychotics for violent patients, 1st psychosis and 2nd psychosis
- Antiemetic
- Antipruritic

7. What is the treatment for akathisia?

- Propranolol
- anticholinergic drugs

8. What is the prophylactic treatment for chronic extrapyramidal side effects?

- Minimal dose and duration
- Avoid routine use of anticholinergics

9. What is the treatment for tardive dyskinesia?

- Switch to atypical agent
- Anti-dyskinesia drugs

10. What are the low potency typical antipsychotics?

- Chlorpromazine
- Thioridazine

11. What are the high potency typical antipsychotics?

- Haloperidol
- Fluphenazine

12. What are the advantages of typical antipsychotics?

- Cheaper than atypical antipsychotics
- Fast-acting for rapid control of agitation
- Long-acting injectable depot formulation

13. What are the advantages of atypical antipsychotics over typical antipsychotics?

- Less central side effects
- More efficacy

14. What are some special features of Risperidone?

- Less extrapyramidal & autonomic side effects, less weight gain.
- May induce galactorrhea & gynecomastia and QT interval prolongation

15. What are some special features of Olanzapine?

- Increase in body weight.
- Autonomic side effects, risk of convulsions, sedation
- Less central side effects (extrapyramidal)

16. What is the therapeutic use of Haloperidol?

- Tranquilizer in agitation

17. What is the therapeutic use of Chlorpromazine?

- Antiemetic in severe vomiting
- Intractable hiccups

18. What is the therapeutic use of Risperidone?

- Autism

19. How is bipolar manic depression managed during the euthymic period?

- Lithium and/or antipsychotics

20. How is bipolar manic depression managed during the manic period?

- Lithium and/or Valproates with antipsychotics

21. How is bipolar manic depression managed during the depressive period?

- Lithium and Antidepressants with antipsychotics

22. What are the indications of lithium?

- Bipolar depression
- Recurrent endogenous depression
- Adjunct to antidepressants in refractory depression

23. What are the adverse effects of lithium?

- **CNS:** Fine tremors
- **GIT:** Nausea, Vomiting, Diarrhea
- **Renal:** Polyuria, Nephrogenic diabetes insipidus, Renal tubular damage
- **Thyroid:** Benign enlargement with or without hypothyroidism
- Toxic drug with a narrow therapeutic index

24. What's the manifestations of lithium toxicity?

- Coarse tremors and GIT upset
- Drowsiness, ataxia and cardiovascular collapse
- Confusion and convulsions
- Coma and death

25. What is the treatment for lithium toxicity?

- Stop lithium. Gastric lavage and activated charcoal in case of acute ingestion
- Fluid intake
- Diuresis (osmotic)
- Dialysis: hemodialysis (or peritoneal dialysis)

26. What is the most effective method for treating Li⁺ toxicity?

- Hemodialysis

27. When is hemodialysis recommended for Li⁺ toxicity?

- If Li⁺ level > 3 mEq/L
- Severe signs and symptoms of toxicity
- Contraindications for fluid intake as heart failure
- Renal failure

1. What are the risks of chronic abuse of opioids?

- Fatal overdose, insomnia, anxiety & tremors, increased risk of infection

2. How is opioid abuse managed?

- Methadone (oral) or buprenorphine (SL)
- Naltrexone
- Symptomatic treatment of withdrawal symptoms

3. What is the mechanism of nicotine addiction?

- Stimulates nicotinic Ach receptors in VTA
- leading to dopamine release in reward pathway

4. What are the symptoms of nicotine withdrawal syndrome?

- Insomnia, anxiety, aggressiveness, and increased appetite

5. What is the mechanism of action of Varenicline?

- Block the ability of nicotine to stimulate dopamine system in VTA
- Decrease craving & withdrawal syndrome

6. How is nicotine abuse managed?

- Psychotherapy
- Nicotine replacement
- Bupropion
- Varenicline

7. What are the acute effects of cocaine abuse? And what does it depend on?

- Euphoria
- Route-dependent intensity (IV -> the rush, Intranasal -> the high, Smoking -> less euphoria)

8. What are the risks associated with chronic cocaine abuse?

- Psychosis (hallucinations & delusions)

9. How is cocaine abuse managed?

- Bupropion
- antipsychotics

10. What is the psychoactive substance in cannabis?

- Tetrahydrocannabinol (THC)

11. What are the risks of chronic cannabis abuse?

- Risk of psychosis, red eyes, apathy, sexual dysfunction, gateway drug

12. What is the mechanism of action of Bupropion in cocaine abuse management?

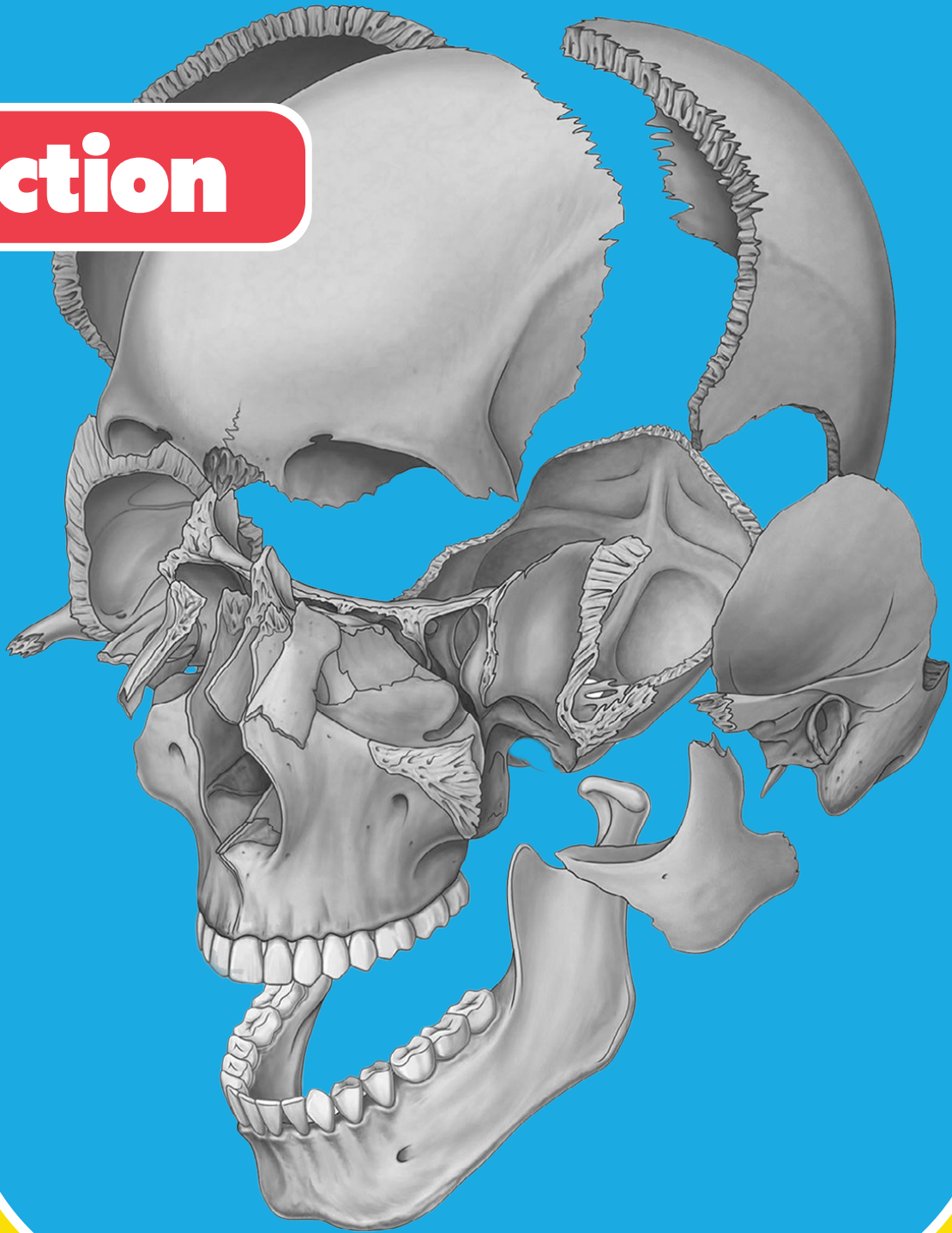
- Decreases craving, also has antidepressant effect

13. How is cannabis abuse managed?

- Naltrexone
- Buspirone
- Fluoxetine
- Dronabinol

7

Section



Microbiology



1. What are the characteristics of proteolytic strains?

- Glucose and maltose fermentation
- gelatin liquefaction
- H₂S production.

2. How can the presence of toxin be detected in botulism?

- In patient blood or food remnants using PCR or ELISA.

3. What is the treatment for infections affecting muscle tone causing flaccid paralysis ?

- Trivalent antitoxin (types A, B, and E)
- Respiratory support.

4. What is Tetanus?

- A bacterial infection caused by *Clostridium tetani* that produces a toxin causing painful muscle contractions.

5. What is the etiology of Tetanus? And what are its morphological features?

- *Clostridium tetani*
- Gram +ve bacilli, Anaerobic, spore forming & motile.

6. What are the virulence factors of *Clostridium tetani*?

- Exotoxin production 'Tetanospasmin', no invasion.

7. How does Tetanospasmin cause muscle contractions?

- Tetanospasmin binds to ganglioside receptor inhibiting release of inhibitory neurotransmitters such as GABA and glycine
- Causing continuous excitation of motor neurons.

8. What are the clinical manifestations of Tetanus?

- Lockjaw or trismus
- Exaggerated reflexes
- Respiratory failure leading to death.

9. How is Tetanus diagnosed?

- Strictly clinical
- Lab diagnosis is only for epidemiological purposes. Gram +ve spore forming bacilli 'Drum stick' seen under microscopy.

10. What is the prevention for Tetanus?

- Tetanus toxoid vaccination:
 - DTap for children
 - Tdap as a booster dose for adolescents.

11. What is the treatment for Tetanus?

- General measures to maintain airway and place patient in a dark quiet area
- Administration of antitoxin (Tetanus immunoglobulin)
- Antibiotics such as Metronidazole or Penicillin G.

12. What is Botulism?

- Food poisoning caused by ingestion of Botulinum toxin preformed in food contaminated with *Clostridium botulinum*.

13. What is the etiology of Botulism? And what are its morphological features?

- *Clostridium botulinum*
- Gram +ve bacilli, Anaerobic, spore forming & motile.

14. How does Botulinum toxin cause botulism?

- Botulinum toxin is absorbed from the gut and inhibits the release of acetylcholine (ACh) at peripheral nerve synapses.

15. What are the clinical manifestations of Botulism?

- Descending weakness and paralysis
- Diplopia & dysphagia
- Respiratory muscle failure
- No fever.

16. What are the laboratory diagnosis methods for Botulism?

- Isolation from food remnants on anaerobic media
- Detection of toxin in patient blood or food remnants.

17. What is the prevention for Botulism?

- Proper canning techniques
- Adequate heating of canned food.

Chapter 2 ?

Bacterial Meningitis

1. What is meningitis?

- Meningitis is an inflammatory process involving the meninges.

2. What are the three types of meningitis based on duration?

- Acute meningitis
- Sub-Acute Meningitis
- Chronic meningitis.

3. What are the causes of acute meningitis?

- Pyogenic bacteria and viruses.

4. What are the causes of sub-acute meningitis?

- Mycobacteria & Fungi.

5. What are the causes of chronic meningitis?

- Complication of sub-acute meningitis.

6. Which bacteria commonly cause meningitis in adults?

- Meningococci
- Pneumococci
- H. influenza
- Other pyogenic bacteria.

7. Which bacteria commonly cause meningitis in infants?

- E. coli
- Strept. agalactae.

8. Which fungus causes meningitis in immunocompetent individuals?

- Cryptococcus neoformans.

9. Which fungi cause meningitis in immunocompromised individuals?

- Aspergillosis
- Mucormycosis.

10. Which viruses commonly cause viral meningitis?

- Enteroviruses and Herpes viruses.

11. What are the routes of entry to the CNS for meningitis-causing organisms?

- Hematogenous (following bacteremia)
- Direct implantation of organism (traumatic or iatrogenic)
- Local extension from infected bone or spine.

12. What are the classic symptoms of meningeal irritation?

- Headache
- Neck stiffness
- Projectile vomiting
- Photophobia.

13. What is the diagnostic test for meningitis?

- CSF analysis using lumbar puncture.

14. What are the components examined in CSF analysis for meningitis diagnosis?

- General look (turbidity)
- Supernatant examination (leucocytic count, glucose levels, and protein levels + antigen detection)
- Deposit examination (Gram stain, cultures, PCR).

15. What is the etiology of meningococcal meningitis? And what are its morphological features?

- *Neisseria meningitidis*
- Gram-negative kidney-shaped diplococci, capsulated, oxidase positive.

16. What are the virulence factors of *Neisseria meningitidis*?

- Polysaccharide capsule
- Pili
- IgA protease
- Factor H binding protein.

17. How does *Neisseria meningitidis* colonize the nasopharynx?

- By becoming part of the transient flora as an asymptomatic carrier.

18. What are the clinical symptoms of meningococcal meningitis?

- Classic meningitis symptoms (headache, neck stiffness)
- Fever
- Septicemia (fulminant meningococemia).

19. What is the diagnostic method for meningococcal meningitis?

- CSF examination (general look, supernatant examination, and deposit examination)
- Blood culture
- PCR

20. What is the treatment for meningococcal meningitis carriers?

- Chemoprophylaxis with Ciprofloxacin or Rifampin.

21. What is the preferred chemoprophylactic drug for meningococcal meningitis carriers? And why?

- Rifampin
- As it is secreted into saliva and eradicates carriers.

22. What is the vaccination strategy for preventing meningitis epidemics?

- Polysaccharide vaccines against groups A, C, Y, W-135
- Factor H binding protein as an immunogen for group B.

23. What are the types of polysaccharide vaccines for meningitis?

- Conjugate vaccines (contain polysaccharides conjugated to carrier proteins)
- Unconjugated vaccines (contain only polysaccharides).

24. What are the virulence factors of *Streptococcus pneumoniae*?

- Capsule
- protease
- autolysin
- pneumolysin

25. How does *Streptococcus pneumoniae* cause meningitis?

- It spreads from the mastoid to the meninges
- Invades the bloodstream following lobar pneumonia.

26. How is pneumococcal meningitis diagnosed?

- CSF and blood specimens are examined. Direct microscopy shows Gram-positive diplococci surrounded by a clear unstained halo. Isolation is done on enriched media (blood agar).
- Latex agglutination test

27. What are the prevention measures for pneumococcal meningitis?

- 13-valent pneumococcal conjugate vaccine for immunocompromised patients and children < 5 years
- 23-valent pneumococcal polysaccharide vaccine for healthy individuals age 65 years or older.

28. What are the important virulence factors of *Haemophilus influenzae*?

- Polysaccharide capsule with a polyribosylribitol phosphate (PRP)
- IgA protease
- Lipopolysaccharide endotoxin
- pilli

29. How is *Haemophilus influenzae* diagnosed?

- CSF and blood specimens are examined by direct microscopic examination
- Isolation on chocolate agar with CO₂ and Factor V & X.
- SRapid capsular antigen detection
- PCR

30. What are the prevention measures for *Haemophilus influenzae*?

- Conjugated polysaccharide vaccine containing the Hi-b PRP capsule is given between 2 and 15 months of age.

31. What is the etiology of neonatal meningitis?

- *Streptococcus agalactiae* (Group B *Streptococcus*)

32. How is neonatal meningitis diagnosed?

- Vaginal and rectal swabs are cultured on blood agar.
- Beta hemolytic on blood agar
- Positive hippurate hydrolysis test
- Positive CAMP test

33. What are the prevention measures for neonatal meningitis?

- Screening for all pregnant women at 35 to 37 weeks' gestation by vaginal and rectal cultures.
- Intrapartum penicillin G (or ampicillin) is given IV if cultures are positive or in certain risk factors.

34. How is *Listeria monocytogenes* transmitted?

- Food-borne zoonotic disease associated with eating unpasteurized cheese or contaminated vegetables with animal feces.

35. How is *Listeria monocytogenes* isolated?

- On blood agar
- Cold enrichment technique

36. What type of immunity is primarily involved in defense against *Listeria*?

- Cell mediated immunity

37. What is the most important virulence factor of *Listeria*?

- Lysteriolysin O

38. What are the clinical presentations of neonatal *Listeriosis*?

- Early onset (granulomatosis infantiseptica)
- late onset (meningitis or meningo-encephalitis with sepsis)

39. What is the leading cause of bacterial meningitis in cancer and transplantation patients?

- *Listeria monocytogenes*

40. How does *Listeria* appear under direct microscopic examination?

- Intracellular Gram positive bacilli

41. What is the catalase status of *Listeria*?

- Catalase positive

42. What does *Listeria* look like under the microscope? And what is the difference between them?

- Diphtheroids
- It's motile and catalase positive

Chapter 3 ?**Fungal Meningitis****1. What are the modes of transmission for *Cryptococcal meningitis*?**

- Inhalation of dust contaminated with bird droppings

2. True or False: *Cryptococcal meningitis* can be transmitted from human to human.

- False

3. Which individuals are affected by *Cryptococcal meningitis*?

- Both healthy people and immunocompromised individuals

4. What is the primary portal of entry for *Aspergillosis* organisms?

- Respiratory tract

5. What are the etiological agents of *rhinocerebral mucormycosis*?

- *Mucor*, *Rhizopus* & *Absidia*

6. What are the risk factors for *rhinocerebral mucormycosis*?

- Diabetic ketoacidosis
- hematologic malignancies (Leukemia & lymphoma)

7. How do fungal spores of *mucormycosis* enter the body?

- Inhaled fungal spores germinate inside nasal cavity

8. What do fungal hyphae of *mucormycosis* invade? And what's its results?

- Blood vessels
- Resulting in thrombosis and infarctions

9. What are the clinical presentations of *rhinocerebral mucormycosis*?

- Bloody nasal discharge
- red and swollen facial area (orbital cellulitis)

10. What are the neurological manifestations of rhinocerebral mucormycosis?

- Loss of vision, diplopia, altered mental state

11. How can mucormycosis be diagnosed?

- Biopsy from CNS lesion or direct microscopy showing broad aseptate hyphae
- Isolation on lab media producing abundant cottony colonies

12. What is the treatment for rhinocerebral mucormycosis?

- Surgical debridement
- Antifungal therapy (Amphotericin B)

Chapter 4 ?

Viral Meningitis

1. What is the Salk vaccine 'IPV'?

- Formalin killed vaccine containing the 3 poliovirus strains, grown in human or monkey cell cultures.

2. How is the Salk vaccine administered?

- IM in 3 doses at 2, 4, 6 months.

3. What are the advantages of IPV?

- Cannot undergo genetic virulence.
- Safe administration in immunodeficient patients.

4. What are the disadvantages of IPV?

- Requires parenteral administration.
- Need for booster vaccine for lifelong immunity.
- Confers immunity only after 4 boosters.
- No local intestinal immunity (no IgA).
- Expensive.

5. What is the structure of Cocksackie virus?

- Enterovirus.

6. What are the characteristics of enteroviruses?

- Picornaviruses, small non-enveloped ssRNA +ve sense with icosahedral capsid

7. What is the mode of transmission for enteroviruses?

- Feco-oral route
- Can be respiratory for Cocksackie

8. What is the most common cause of viral meningitis?

- Enteroviruses

9. What are the members of the Herpesviridae family that can cause viral meningitis?

- Herpes simplex
- Varicella zoster virus

10. What are the members of the Paramyxoviruses family that can cause viral meningitis?

- Mumps virus

11. What are the members of the Flaviviruses family that can cause viral meningitis?

- West Nile virus

12. What are the members of the Retroviruses family that can cause viral meningitis?

- HIV

13. What are the characteristics of poliovirus? And how many antigenic types it has?

- Small non-enveloped ssRNA +ve sense with icosahedral capsid
- 3 antigenic types

14. How is poliovirus inactivated?

- Heating to 55°C for 30 minutes
- Pasteurization
- Chlorination of water

15. What is the reservoir and mode of transmission for poliovirus?

- Human is the reservoir
- Transmission by feco-oral route

16. What are the clinical presentations of poliomyelitis?

- Asymptomatic or subclinical infection
- Abortive poliomyelitis
- Non-paralytic poliomyelitis
- Paralytic poliomyelitis

17. How is poliomyelitis diagnosed?

- Isolation of the virus from throat swabs, rectal swabs, and stools
- Serology with paired serum specimens

18. What are the general preventive measures for poliomyelitis?

- Prevent fecal contamination of drinking water and food
- Avoid operations in children with fever
- Gamma globulin can protect against paralytic disease if given before infection or before nervous lesion

19. What is the name of the vaccine that provides immunity to poliovirus serotypes?

- Salk "IPV"
- Sabin "OPV"

20. What are the advantages of the Sabin vaccine (OPV)?

- Administered easily by oral drops
- No need for repeated booster doses
- Provides lifelong immunity
- Protects the gastrointestinal tract and central nervous system
- Provide herd immunity

21. What are the disadvantages of the Sabin vaccine (OPV)?

- Can mutate to a more virulent strain
- Unsafe for immunodeficient patients
- Not given to feverish children
- Requires refrigeration to avoid heat inactivation

22. What is the name of the killed vaccine for poliovirus serotypes?

- Salk "IPV"

1. What is the reservoir for West Nile Virus encephalitis?

- Wild birds

2. Which mosquito is the vector for West Nile Virus encephalitis?

- Culex mosquito

3. How can the West Nile Virus be transmitted?

- Organ transplants
- blood transfusion

4. What are the clinical symptoms of West Nile Virus encephalitis?

- Fever
- headache
- with or without signs of meningitis

5. How can West Nile Virus encephalitis be diagnosed?

- Detection of antibodies (Anti WNV IgM) in the spinal fluid
- PCR
- isolation from brain tissue, blood, or spinal fluid

6. What are the general prevention measures for West Nile Virus?

- Screening of donated blood
- Vector control
- Personal protection

7. What are the modes of transmission for rabies encephalitis?

- Bite of a rabid animal
- Exposure to aerosols of bat secretions containing rabies virus

8. How does the rabies virus spread within the body?

- The virus multiplies in the CNS
- And migrates through peripheral nerves to salivary glands and other organs

9. What are the clinical forms of rabies encephalitis?

- Furious rabies
- paralytic (dumb) rabies

10. How can rabies encephalitis be diagnosed?

- Direct methods (DIF, RT-PCR, viral isolation)
- Serology (immunofluorescence or neutralization tests)

11. What are the prevention and control measures for rabies?

- Eliminate stray animals
- Vaccinate domestic dogs and cats
- Follow pre-exposure and post-exposure recommendations

12. What is the recommended prophylaxis for rabies?

- Pre-exposure prophylaxis (vaccination) for high-risk individuals and travelers to high-risk areas
- Post-exposure prophylaxis (vaccination and immunoglobulin) for unvaccinated individuals

13. What are the slow CNS infections associated with muscle tone and nervous system?

- Measles virus (SSPE)
- JC virus (PML)
- prions (Kuru and vCJD)

14. How is progressive multifocal leukoencephalopathy (PML) caused?

- JC virus infection in immunocompromised patients

15. How are prion diseases transmitted?

- They are transmitted by inoculation or ingestion of prion-containing beef.
- Corneal transplant and transplant of dura mater of an infected cadaver can also transmit the infection.

16. What causes prion diseases?

- The disease is caused by a misfolded form of a normal prion protein
- Thereby altering the function of the protein, and leading to death of neurons.

17. Where does prion diseases primarily affect?

- The disease is confined to the CNS where death of neurons causes vacuolation and spongiform changes in the brain.

18. What are the clinical presentations of prion diseases?

- Gradual onset, slowly progressive dementia, and myoclonus.
- Severe disease resulting in coma and death.

19. How can prion diseases be diagnosed?

- Brain biopsy shows spongiform changes.
- No serologic tests are available, and Prions cannot be grown in culture.

20. What is the prevention and treatment for prion diseases?

- Infected animals should be slaughtered, and their carcasses destroyed.
- No available treatment or vaccine is present.

21. What are the etiologies of Viral encephalitis?

- Herpesviruses (HSV, Varicella Zoster virus, Epstein-Barr virus, Cytomegalovirus)
- Arboviruses (West Nile Virus, St. Louis encephalitis virus, Lacrosse strain of California encephalitis virus, Eastern and Western equine encephalitis viruses)
- HIV Encephalitis
- Post-infectious encephalitis

22. What is the most common cause of HSV encephalitis?

- HSV-1 in immunocompromised
- HSV-2 in neonatal encephalitis.

23. How does HIV gain entrance to the CNS?

- Within monocytes which easily traverse the BBB.

24. What are the symptoms of viral encephalitis?

- Altered mental status, seizures, and focal neurologic deficits.
- Personality change, hallucinations
- aphasia suggest temporal lobe localization in HSV encephalitis.

25. What are the complications and outcomes of Viral encephalitis?

- Cognitive disability, seizures, and motor deficits are common complications.
- Mortality of HSV encephalitis before acyclovir was 60-70%, and with treatment ~30-40%.
- WNV is associated with significant morbidity and mortality.

26. How is HSV encephalitis diagnosed?

- CSF analysis (mild elevation in CSF lymphocytes, elevated protein, normal glucose)
- PCR-based testing of CSF
- Radiology (temporal lobe abnormalities)
- Detection of antibodies in the spinal fluid or blood.

27. What is the prevention for HSV-2 transmission to the neonate?

- Pregnant women with active lesions late in pregnancy should receive acyclovir and should be considered for Cesarean section.

28. What is the treatment for HSV-1, HSV-2, and VZV encephalitis?

- Intravenous acyclovir.

29. What is the epidemiology and mode of transmission of West Nile Virus encephalitis?

- Endemic in Africa but has caused encephalitis in areas of Europe and Asia as well.
- Transmission can happen via organ transplants or blood transfusion.

30. How is West Nile Virus encephalitis diagnosed?

- Detection of antibodies (Anti WNV IgM) in the spinal fluid
- PCR
- Isolation from brain tissue, blood, or spinal fluid.

31. What is the prevention for West Nile Virus encephalitis?

- Screening of donated blood.
- General measures for vector control and personal protection.
- No vaccine is available for WNV.

32. What is the treatment for West Nile Virus encephalitis?

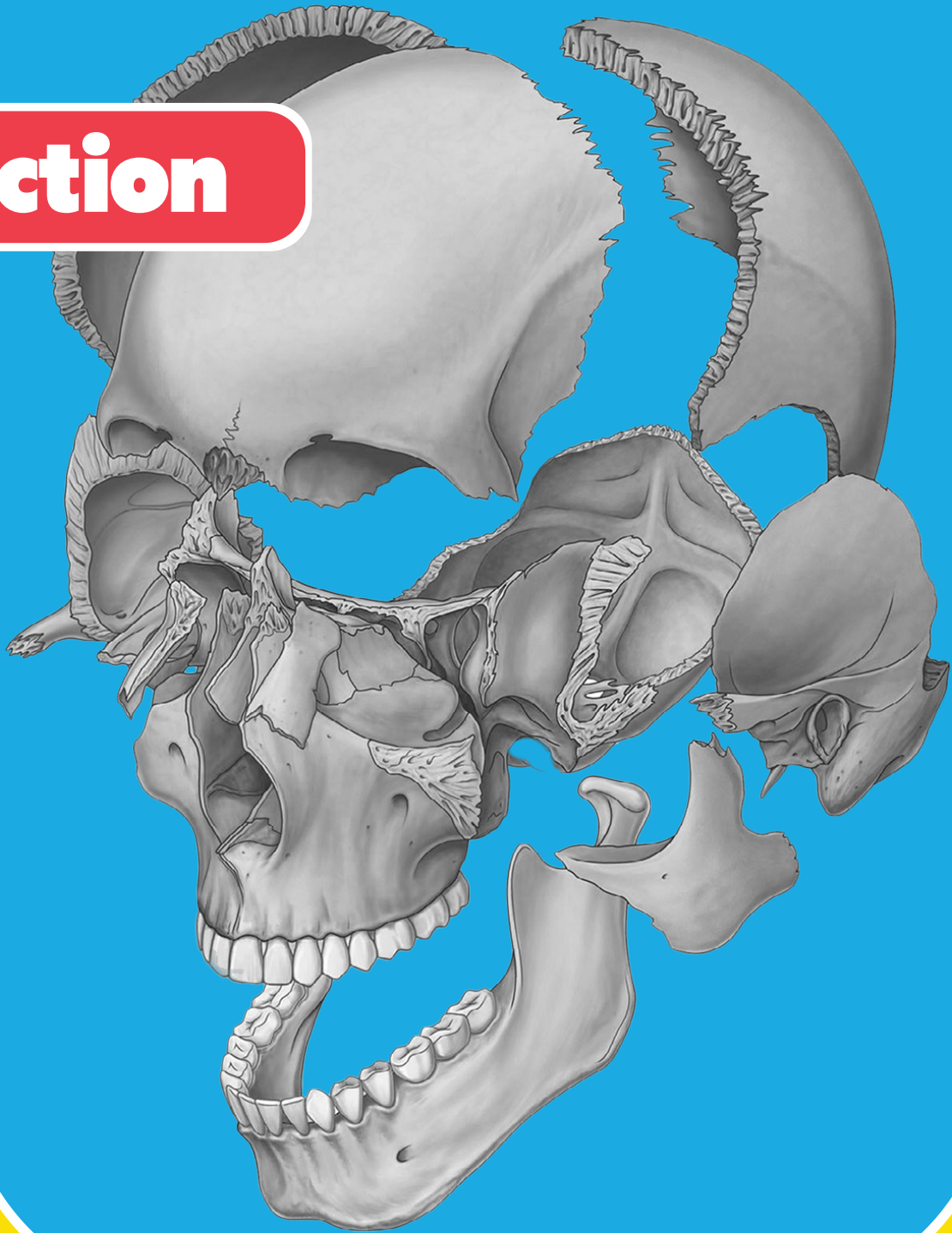
- **Supportive therapy:** hospitalization, respiratory support, fluids & electrolytes.
- No available antiviral therapy.



Notes

7

Section



Parasitology



1. Name two of the parasites affecting CNS.

- Naegleria fowleri
- Acanthamoeba

2. Numerate the stages of Naegleria fowleri.

- Trophozoite
- Cyst

3. Numerate the forms of Naegleria fowleri.

- Amoeboid trophozoite
- Flagellated trophozoite

4. What's the mode of transmission of Naegleria fowleri?

- Swimming in contaminated water
- Inhalation of contaminated air

5. Amoeboid trophozoite of Naegleria fowleri moves by..... while flagellated trophozoite moves by.....

- Broad pseudopodium (lobopodium)
- Flagella

6. What's the infective and diagnostic stages of Naegleria fowleri?

- Infective: amoeboid trophozoite
- Diagnostic: amoeboid trophozoite

7. Name four symptoms and signs of primary amoebic Meningoencephalitis.

- Alterations in smell and test
- High fever with sensory and motor changes
- Increased intracranial tension
- Coma

8. List three way of the lab diagnosis of Naegleria fowleri.

- Complete blood picture
- CSF examination
- PCR

9. Name two findings of CSF examination revealing Naegleria fowleri infection.

- Cloudy, purulent, or blood and under tension
- Increased protein content and decreased glucose content
- Bacteriologically free with increased lymphocytes

10. What's the treatment of Naegleria fowleri?

- Amphotericin B



1. Describe Acanthamoeba cyst in two points.

- Rounded
- With double wall with wrinkled outer cyst wall and polyhedral inner wall

2. What's the infective stage of Acanthamoeba?

- Trophozoite
- Cyst

3. What's the mode of transmission of Acanthamoeba?

- Inhalation
- Enter through skin or GIT lesion
- Through corneal abrasion

4. What's the pathogenic stage of Acanthamoeba?

- Trophozoite

5. Name the diagnostic stages of Acanthamoeba.

- Trophozoite
- Cyst

6. Name three diseases caused by Acanthamoeba.

- Granulomatous amoebic encephalitis
- Amoebic keratitis
- Disseminated granulomatous amoebic disease

7. List four of symptoms and signs of granulomatous amoebic encephalitis.

- Severe frontal headache
- High fever, nausea and vomiting
- Neck rigidity
- Sensory and motor changes and mental confusion
- Skin lesions

8. Amoebae produce injuries in GAE by.... (two points)

- Contact mediated lysis
- Ingesting cytoplasm of the cells

9. Name two ways to diagnose GAE.

- CSF examination
- Brain biopsy
- Culture

10. Biopsy to diagnose acanthamoeba keratitis is..... while it's..... in disseminated granulomatous amoebic disease.

- Corneal scraping
- Skin biopsy

11. List the ways to prevent infection with Acanthamoeba.

- Proper cleaning of contacts lenses
- Don't wear of lenses while swimming

12. Compare between Naegleria fowleri and Acanthamoeba regarding route of infection, course & diseases.

	N. Fowleri	Acanthamoeba
Route of infection	Nasal mucosa	Skin, eye, GIT
Course	Acute, fatal	Chronic, non fatal
Diseases	Primary amoebic Meningeoen- cephalitis	Granulomatous amoebic encephalitis

Chapter 3 ?

Sparganosis

1. Name two intermediate hosts of *Spirometra mansoni*.

- Cyclops
- Frogs, snakes, bird
- Mammals

2. What's the mode of infection of sparganosis?

- Drinking contaminated water
- Eating raw or undercooked frogs or birds
- Applying infected frogs as poultices

3. Sparganosis generally involves..... and lobes and may extend to

- Frontal
- Parietal
- Cerebellum

4. What's the gold standard to diagnose sparganosis?

- Biopsy

5. What's the treatment of sparganosis?

- Surgical removal

6. How to prevent infection with sparganum mansoni larvae?

- Destruction of cyclops by filtering and boiling water before drinking
- Don't use raw frogs as poultices
- Cook frogs and birds properly

7. What's the infective stage of Cysticercosis?

- Proceroid larva
- Sparganum larva

Chapter 4 ?

Cysticercosis

1. Name modes of infection with Cysticercosis.

- Heteroinfection
- Autoinfection

2. What's the types of auto infection of *Taenia solium*?

- External autoinfection
- Internal autoinfection

3. Name three organs cysticerci are located in and cause Cysticercosis.

- Muscles
- Eye
- CNS

4. Name two symptoms of Neurocysticercosis.

- Headache, vomiting and vertigo
- Convulsions and epilepsy
- Disequilibrium, abnormal behavior and meningoencephalitis

5. Name three ways to diagnose Cysticercosis.

- Stool examination
- ELISA
- CSF analysis
- Biopsy
- Imaging technique

6. List two lines of treatment of Cysticercosis.

- Albendazole
- Praziquantel
- Antiepileptic
- Surgical treatment

7. How to prevent Cysticercosis?

- Early detection and treatment of *taeniae solium*
- Protecting food of contamination with infected human's feces
- Personal hygiene
- Environmental sanitization



1. Name three parasites don't inhabit the brain but have neurological complications.

- Toxoplasma gondii
- Echinococcus granulosum
- Plasmodium falciparum
- African trypanosoma
- Entamoeba histolytica

2. What's the types of toxoplasma gondii infection?

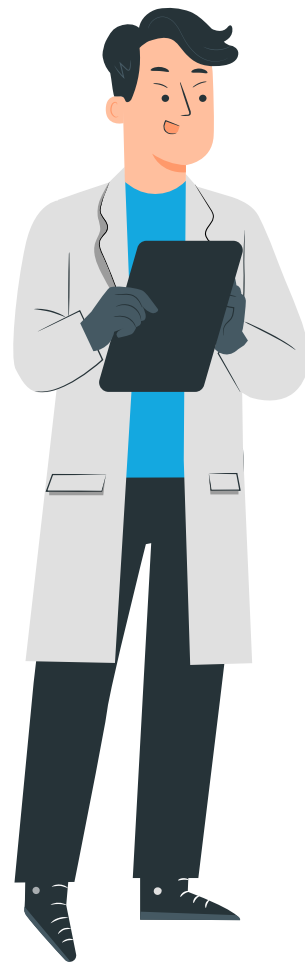
- Infection in immunocompromised hosts
- Congenital toxoplasmosis

3. List four symptoms and signs of toxoplasma gondii in immunocompromised hosts.

- Fever, headache, vomiting
- Speech difficulty, behavioral changes, seizure
- Ataxia, visual field defect, altered mental status
- Focal neurological deficit and cranial nerve deficit

4. Name two investigations to detect toxoplasma gondii.

- CD4⁺ count
- CT and MRI
- Serology
- PCR
- Brain biopsy
- CSF examination



Notes

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